

Techno-Economic Assessment of Soft-Wing Airborne Wind Energy System

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by

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Cover: Manual launch of a traction kite for power generation, adapted
from TU Delft [1].

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Preface

This work marks the end of my study time here in Delft, and it has been a great journey. I did not choose the bachelor's in Aerospace Engineering because of a particular fascination with aeroplanes or spacecraft, but rather because of my interest in wind, partly driven by the many watersports I enjoy. This interest eventually led me to pursue the master's in Wind Energy and Aerodynamics. During my master's, I took the airborne wind energy course taught by Roland Schmehl. Seeing one of my main hobbies, kitesurfing, combined with my interest in technology, immediately caught my attention. It seemed like a great opportunity to work on a concept that genuinely interested me. Through this thesis, I was further introduced to the field of airborne wind energy and had the opportunity to learn more about both the technology itself and the challenges of bringing it to market.

I would therefore first like to thank my supervisor, Roland Schmehl, not only for introducing me to this topic, but also for his enthusiasm, ideas, and guidance throughout the process. I would also like to thank Hanna Yousef for joining my supervision team so early in her own PhD trajectory and for providing valuable feedback throughout the project. The many smaller discussions we had along the way were especially valuable. I would also like to thank the teams at Kitepower and Beyond the Sea for sharing their data, knowledge, and practical experience. I am particularly grateful to Claudio Vergara and Tom Lollies from Kitepower for taking the time to meet with me and provide valuable insights. I would also like to thank Joren Bredael for sharing the AWESPA toolchain with me and explaining its implementation. Finally, I am grateful to everyone else who shared their knowledge, feedback, and time and helped make this work possible.

I would also like to thank my brothers, sister, and especially my parents for their continuous support, encouragement, and confidence in me throughout my studies. Their support has meant more to me than they probably realise. Another huge thank you goes to my friends, who have made not only this period but also my entire time in Delft unforgettable through all the fun moments we shared. They supported me in many ways and were an important part of these years.

Writing this thesis was both challenging and rewarding. I not only developed knowledge through research and applied theory, but the process itself also gave me valuable personal insights and helped me develop a more critical mindset. This has prepared me well for the challenges ahead, and I am excited for the new period to come.

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Abstract

Ground-generation airborne wind energy systems (AWES) use tethered devices flying crosswind to drive ground-based generators, potentially accessing stronger winds at higher altitudes with less structural material above ground. Leading-edge inflatable (LEI) kites have demonstrated this principle, yet their design still relies on expert judgement and in-house testing, while their scale-dependent technical and economic behaviour remains insufficiently understood. Scaling can increase annual energy production but also affects system design, aerodynamic performance, component service life, and recurring replacement expenditure. Existing coupled techno-economic frameworks were developed for fixed-wing concepts and omit the finite service life and recurring replacement of soft-wing components. It therefore remains unclear how these effects jointly determine the levelised cost of energy (LCoE) across system scales.

This thesis develops and applies a coupled techno-economic framework that extends an existing modular performance-assessment tool with sizing and economic models linking design and operation to annual energy production, service life, replacement, and lifetime cost. Component-based relations determine airborne mass and tether size; reel-out hours and simulated loads set canopy and tether replacement intervals. The framework is applied to TU Delft V3-derived systems at Bangor Erris, Ireland. Flat wing area and allowable tether stress are varied for prescribed generator mechanical power limits from 40 kW to 250 kW, defined as upper bounds on instantaneous reel-out power rather than average pumping-cycle power. An inner optimisation and outer grid search identify the lowest evaluated LCoE at each limit.

The component-based model, which scales the LEI wing and kite control unit separately, produced a more-than-proportional increase in airborne mass with wing area. The nominal LEI wing mass scaled with $S^{1.22}$, an exponent applicable only to the wing; alternative relations produced only minor LCoE changes. For the fixed TU Delft V3 reference system under the mid-maturity assumptions, the calculated LCoE was approximately 2992 €/MWh and should be interpreted as indicative. Recurring operating expenditure exceeded annualised capital expenditure, with ground-crew labour as the largest contribution, followed by canopy and tether replacement, while the ground station dominated the initial investment. Maturity scenarios with fewer ground-crew hours and longer component lives substantially reduced LCoE, suggesting operational autonomy and component durability as promising directions without establishing their relative importance.

Across the prescribed generator limits, annual energy production increased and LCoE decreased at a diminishing rate, with no intermediate system-scale optimum found within the evaluated range. The LCoE reduction arose primarily from spreading ground-crew expenditure over greater energy production, while rising component replacement expenditure increasingly offset this benefit.

This contrasts with the fixed-wing reference formulation, which showed an intermediate optimum. The different model boundaries prevent interpreting this contrast as an inherent distinction between fixed-wing and soft-wing systems. The principal trends persisted across sensitivity cases, but the numerical LCoE values and exact design selections remain indicative because uncertain-input interactions were not assessed, several relations rely on limited data or extrapolation, and the complete framework lacks validation against measured lifetime expenditure. Flat LCoE surfaces and numerical variability further limit exact design selection, making the relative trends and mechanisms the more reliable basis for interpretation. By resolving these coupled effects, the framework extends LCoE-driven conceptual design assessment from fixed-wing formulations to soft-wing AWE systems and identifies component service life and operational labour as principal lifetime-cost drivers. It enables consistent assessment of system configurations and operating strategies, providing early-stage insight into the trade-offs that distinguish a technically favourable design from an economically favourable one.

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Nomenclature

Abbreviations

Abbreviation	Definition
AEP	Annual energy production
AWE	Airborne wind energy
AWES	Airborne wind energy system
AWESPA	Modular toolchain for assessing airborne wind energy systems and farms
BoP	Balance of plant
BoS	Balance of system
CAD	Computer aided design
CAPEX	Capital expenditure
CF	Capacity factor
CRF	Capital recovery factor
EcoMo	Economic model used in the techno-economic framework
ERA5	Fifth generation atmospheric reanalysis of the European Centre for Medium Range Weather Forecasts
FCR	Fixed charge rate
FG	Fly generation
GG	Ground generation
GNSS	Global navigation satellite system
GPS	Global Positioning System
HAWT	Horizontal axis wind turbine
HMPE	High modulus polyethylene
ICC	Initial capital cost
IEA	International Energy Agency
IMU	Inertial measurement unit
IRR	Internal rate of return
KCU	Kite control unit
LCoE	Levelised cost of energy
LEI	Leading edge inflatable
LLS	Launch and landing system
MDAO	Multidisciplinary design, analysis and optimisation
NPV	Net present value
OMC	Operation and maintenance cost
OPEX	Operational expenditure
QSM	Quasi steady model
RORI	Reel out to reel in
TEF	Techno economic framework
TPU	Thermoplastic polyurethane
WACC	Weighted average cost of capital
XDSM	Extended design structure matrix

Symbols

Symbol	Definition	Unit
<i>Latin symbols</i>		
A	Projected wing area	$[\text{m}^2]$
A_{ref}	Reference projected wing area	$[\text{m}^2]$
a	Operational availability factor	$[-]$
a_c	collapse resistance factor	$[-]$
b	Power law scaling exponent	$[-]$
b_1, b_2	Reference and scaled wing span	$[\text{m}]$
b_C	KCU cost scaling exponent	$[-]$
b_{eff}	Effective wing mass scaling exponent	$[-]$
b_i	Scaling exponent of component i	$[-]$
b_{mat}	Cost weighted material scaling exponent	$[-]$
c	Weibull scale parameter	$[\text{m s}^{-1}]$
C_D	Aerodynamic drag coefficient	$[-]$
$C_{D,i}, C_{D,o}$	Reel in and reel out drag coefficients	$[-]$
$C_{D,t}$	Tether drag coefficient	$[-]$
C_L	Aerodynamic lift coefficient	$[-]$
$C_{L,i}, C_{L,o}$	Reel in and reel out lift coefficients	$[-]$
C_0	Structural design anchor cost	$[\text{€}]$
C_i	Cost of component i	$[\text{€}]$
$C_{\text{CAPEX},i}$	Capital cost of component i	$[\text{€ yr}^{-1}]$
C_{crew}	Annual ground crew cost	$[\text{€}]$
C_{ICC}	Initial capital cost	$[\text{€}]$
C_{KCU}	KCU cost	$[\text{€}]$
$C_{\text{KCU,ref}}$	Reference KCU cost	$[\text{€}]$
C_{kite}	Total kite cost	$[\text{€}]$
C_{lab}	Structural labour cost per unit wing area	$[\text{€ m}^{-2}]$
C_{LLS}	Launch and landing system cost	$[\text{€}]$
$C_{\text{mat}}(S_0)$	Material cost at the reference wing area	$[\text{€}]$
C_{OpEx}	Annual operational expenditure	$[\text{€ yr}^{-1}]$
$C_{\text{OpEx},j}$	Annual operational expenditure of component j	$[\text{€ yr}^{-1}]$
C_{ref}	Reference manufacturer price	$[\text{€}]$
C_{sens}	Sensor package cost	$[\text{€}]$
$C_{\text{sens,ref}}$	Reference sensor package cost	$[\text{€}]$
C_{site}	Annual site overhead cost	$[\text{€ yr}^{-1}]$
C_{struct}	Wing structure cost	$[\text{€}]$
d	Bridle line diameter	$[\text{m}]$
d_1, d_2	Tube diameters at the ends of a tube segment	$[\text{m}]$
d_t	Tether diameter	$[\text{m}]$
D	Aerodynamic drag force	$[\text{N}]$
E_i	Mechanical energy consumed during reel in	$[\text{J}]$
E_{lifetime}	Discounted lifetime energy yield	$[\text{MW h}]$
E_o	Mechanical energy produced during reel out	$[\text{J}]$
E_{out}	Electrical energy produced during reel out	$[\text{J}]$
E_{stor}	Required usable storage energy	$[\text{J}]$
E_{tr}	Net mechanical energy of the transition phases	$[\text{J}]$
F	Representative or resultant aerodynamic force	$[\text{N}]$
F_{branch}	Load carried by one bridle branch	$[\text{N}]$
F_{steer}	Steering force	$[\text{N}]$
F_t	Tether force	$[\text{N}]$
$F_{t,\text{cyc}}$	Maximum tether force within a pumping cycle	$[\text{N}]$
$F_{t,\text{max}}$	Maximum design tether force	$[\text{N}]$
$F_{t,\text{min}}$	Minimum allowable tether force	$[\text{N}]$

Symbol	Definition	Unit
f_A	Load carrying fibre fraction of the tether cross section	[—]
f_{bend}	Tether replacement frequency due to bending fatigue	[yr ⁻¹]
f_{creep}	Tether replacement frequency due to creep	[yr ⁻¹]
f_{crest}	Ratio of peak mechanical power to rated electrical power	[—]
f_{fix}	Fixed fraction of KCU cost	[—]
f_k	Occurrence frequency of wind profile cluster k	[—]
f_{lab}	Labour fraction of the structural design anchor cost	[—]
f_{oper}	Tether replacement frequency due to operational wear	[yr ⁻¹]
f_{out}	Reel out fraction of the pumping cycle	[—]
$f_{\text{repl},j}$	Annual replacement frequency of component j	[yr ⁻¹]
$f_{\text{ret,end}}$	Tether length fraction at the end of retraction	[—]
$f_{\text{tr,end}}$	Tether length fraction at the end of traction	[—]
f_{wind}	Fraction of year within the operating wind speed range	[—]
h	Tube segment length	[m]
h_{op}	Operating labour hours per operating day	[h d ⁻¹]
H_{downtime}	Annual downtime	[h yr ⁻¹]
H_{flight}	Annual flight hours	[h yr ⁻¹]
H_{out}	Annual reel out hours	[h yr ⁻¹]
$H_{\text{potential}}$	Annual potential flight hours	[h yr ⁻¹]
k	Geometric scale factor	[—]
$k_{i,j}$	Wind profile cluster index	[—]
k_w	Weibull shape parameter	[—]
k_p	Projected to flat wing area ratio	[—]
l	Characteristic length or bridle line length	[m]
ℓ_c	Characteristic tributary width	[m]
$\ell_{t,\text{max}}$	Maximum tether length	[m]
L	Aerodynamic lift force	[N]
L_{canopy}	Canopy lifetime in reel out hours	[h]
L_{GS}	Ground station component service life	[yr]
L_{oper}	Operational tether lifetime	[h]
$L_{t,\text{oper}}$	Prescribed operational tether lifetime	[h]
m	Mass	[kg]
$m_{\text{additional}}$	Additional unresolved kite mass	[kg]
m_{air}	Total airborne mass	[kg]
m_{bridle}	Bridle system mass	[kg]
m_{canopy}	Canopy mass	[kg]
m_i	Mass of component or evaluated design i	[kg]
m_{KCU}	KCU mass	[kg]
m_{kite}	Wing and bridle mass	[kg]
m_{LE}	Leading edge mass	[kg]
m_{LEI}	Mass predicted by the baseline LEI scaling relation	[kg]
$m_{\text{pulley},j}$	Mass of pulley j	[kg]
m_{reinf}	Reinforcement mass	[kg]
m_{sens}	Sensor package mass	[kg]
m_{struts}	Total strut mass	[kg]
$m_{\text{wing},0}$	Reference wing mass	[kg]
M_A	Maximum bending moment	[N m]
N	Project lifetime	[yr]
N_y	Project lifetime expressed as a number of years	[yr]
N_{bridle}	Number of bridle support points	[—]
N_{cl}	Number of wind profile clusters	[—]
N_{cyc}	Annual number of pumping cycles	[yr ⁻¹]
N_{LE}	Number of leading edge segments	[—]
N_{line}	Number of bridle line segments	[—]

Symbol	Definition	Unit
N_{op}	Annual number of operating days	[d yr ⁻¹]
N_{panel}	Number of canopy panels	[-]
N_{pulley}	Number of pulleys	[-]
$N_{\text{strut,int}}$	Number of strut intervals	[-]
N_{struts}	Number of struts	[-]
p	Internal tube overpressure	[Pa]
p_n	Aerodynamic pressure normal to a canopy panel	[Pa]
p_{cyc}	Canopy damage penalty per pumping cycle	[cycle ⁻¹]
p_{LLS}	Launch and landing system cost per unit wing area	[€ m ⁻²]
p_{site}	Annual site overhead per unit rated power	[€ kW ⁻¹ yr ⁻¹]
$\mathbf{p}_i$	Position vector of panel vertex i	[m]
P_{act}	KCU actuation power	[W]
P_{bridle}	Bridle support pitch	[m]
P_e	Mean electrical power	[W]
$P_{e,k}$	Mean electrical power for wind profile cluster k	[W]
$P_{e,\text{avg}}$	Cycle averaged electrical power	[W]
$\bar{P}_e$	Cycle averaged electrical delivery power	[W]
P_{gen}	Prescribed generator mechanical power limit	[W]
$P_{\text{m,avg}}$	Mean mechanical cycle power	[W]
$P_{\text{m,i}}$	Mechanical power during reel in	[W]
$P_{\text{m,o}}$	Mechanical power during reel out	[W]
P_{rated}	Rated electrical power	[W]
q	Debt to equity ratio	[-]
q_α	Distributed aerodynamic line load	[N m ⁻¹]
r	Tube radius or discount rate	[m/ -]
r_d	Cost of debt	[-]
r_e	Cost of equity	[-]
r_{maint}	Maintenance hours per flight hour	[h h ⁻¹]
S	Flat wing area	[m ²]
S^*	Representative optimal flat wing area	[m ²]
$S_{\text{LE},i}$	Surface area of leading edge segment i	[m ²]
S_{panel}	Canopy panel area	[m ²]
$S_{\text{strut},j}$	Surface area of strut j	[m ²]
S_{tube}	Surface area of a tapered tube segment	[m ²]
t	Time	[s]
t_{cycle}	Pumping cycle duration	[s]
t_{out}	Reel out duration	[s]
T_C	Corporate tax rate	[-]
U_{ref}	Wind speed at the reference height	[m s ⁻¹]
$\bar{U}_{\text{ref}}$	Mean wind speed at the reference height	[m s ⁻¹]
v	Apparent wind speed	[m s ⁻¹]
V_{line}	Volume of a bridle line segment	[m ³]
$v_{\text{r,max}}$	Maximum drum reeling speed	[m s ⁻¹]
v_{ret}	Retraction reeling speed	[m s ⁻¹]
v_{tape}	KCU steering tape velocity	[m s ⁻¹]
v_{tr}	Traction reeling speed	[m s ⁻¹]
$v_{t,i}, v_{t,o}$	Tether reeling speeds during reel in and reel out	[m s ⁻¹]
v_w	Reference wind speed	[m s ⁻¹]
$v_{w,\text{ci}}, v_{w,\text{co}}$	Cut in and cut out wind speeds	[m s ⁻¹]
$v_{w,\text{in}}, v_{w,\text{out}}$	Cut in and cut out wind speeds	[m s ⁻¹]
$v_{w,\text{rated}}$	Rated wind speed	[m s ⁻¹]
$v_{w,\text{ref}}$	Wind speed at the reference height	[m s ⁻¹]
w	Unsupported canopy panel width	[m]
w_{fix}	Fixed KCU mass fraction	[-]

Symbol	Definition	Unit
w_{hous}	KCU housing and protection mass fraction	[–]
w_{lab}	Labour rate	[€ h ^{−1}]
w_{lin}	Linearly scaling KCU mass fraction	[–]
w_{mech}	KCU mechanical mass fraction	[–]
w_{pow}	KCU power supply mass fraction	[–]
w_{sup}	Superlinearly scaling KCU mass fraction	[–]
$W_{\text{l,max}}$	Maximum wing loading based on projected area	[N m ^{−2}]
y	Year index	[–]
z	Kite altitude	[m]
z_{ref}	Reference height	[m]
Greek symbols		
α_{auto}	Automation fraction	[–]
α_{w}	Power law wind shear exponent	[–]
$\beta_{\text{RORI,end}}$	Elevation angle at end of the reel out to reel in transition	[°]
$\beta_{\text{tr},i}$	Traction elevation angle	[°]
η_{DT}	Drivetrain efficiency	[–]
η_{gen}	Generator efficiency	[–]
η_{mot}	Motor efficiency	[–]
η_{stor}	Electrical storage efficiency	[–]
κ	Crest factor	[–]
ρ	Density	[kg m ^{−3}]
ρ_d	Dacron areal density	[g m ^{−2}]
ρ_p	Ripstop polyester areal density	[g m ^{−2}]
ρ_t	Tether material density	[kg m ^{−3}]
ρ_{TPU}	TPU bladder areal density	[g m ^{−2}]
σ	Canopy membrane stress	[Pa]
σ_C	Circumferential tube wall stress	[Pa]
σ_L	Longitudinal tube wall stress	[Pa]
σ_{panel}	Canopy panel membrane stress	[Pa]
σ_t	Operating tether fibre stress	[Pa]
$\sigma_{t,\text{allow}}$	Allowable tether fibre stress used for sizing	[Pa]
$\sigma_{t,\text{cyc}}$	Maximum operating tether fibre stress within a pumping cycle	[Pa]

Introduction

The global shift to a sustainable, low-carbon energy system has created strong incentives for innovation across the full spectrum of renewable electricity technologies. Wind energy has been one of the most successful outcomes of this development: over the past three decades, horizontal-axis wind turbines (HAWTs) have advanced from small experimental devices to multi-megawatt installations that supply some of the most cost-competitive electricity at favourable onshore and offshore sites [2]. Yet despite this progress, the support structures and installation logistics for tower-mounted turbines still account for significant portions of system cost [3]. In addition, the finite availability of high-quality wind resources at accessible locations motivates the exploration of fundamentally different approaches to wind energy conversion.

Airborne wind energy (AWE) represents one such alternative. Rather than mounting an energy-harvesting device on a tower, an Airborne wind energy system (AWES) uses a tethered flying device to convert aerodynamic loading into useful mechanical work. The tether serves as the primary structural load path between the airborne device and the ground station, replacing part of the tower's structural function in a conventional wind turbine. Figure 1.1 illustrates this conceptual analogy. By shifting much of the structural requirement from an elevated tower to a tensioned tether, airborne systems may reduce the structural material in the elevated part of the system, easing constraints related to manufacturing, transport, installation, and maintenance [4]. In addition, AWES can access higher altitudes than tower-mounted turbines, where wind speeds are typically stronger and more persistent [5, 6].

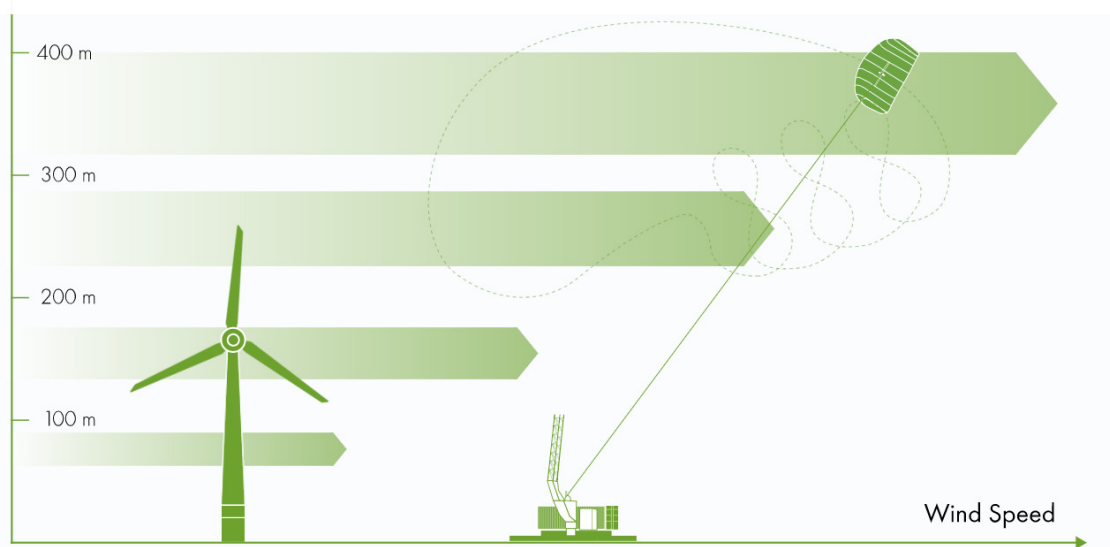


Figure 1.1: Conceptual comparison of a horizontal-axis wind turbine and a ground-generation airborne wind energy system, showing the analogous rotor/kite, hub/drum, generator and support components. The kite is depicted in a representative orientation during crosswind figure-eight flight. Adapted from SkySails GmbH [7].

Lloyd [8] established the theoretical basis for the high power potential of tethered wings, showing that a wing flown rapidly in crosswind motion experiences an apparent wind speed far in excess of the ambient wind speed. Because aerodynamic force scales with the square of apparent wind speed, even a relatively small, lightweight wing can generate large tether tensions and, consequently, high power output during crosswind flight. This crosswind principle underlies the operating concept of most AWES demonstrated to date [9].

A primary distinction within the AWES field is the location where electricity is generated. In ground-generation (GG) systems, the tether is connected to a drum on the ground that drives a generator. Electricity is produced during the traction phase, when the wing flies crosswind patterns, and the tether unwinds under high tension. Once the tether reaches its maximum length, the wing is depowered and retracted at a lower tether force to complete the pumping cycle, after which the traction phase restarts [10]. In fly-generation (FG) systems, electricity is instead produced on board the airborne unit and transmitted to the ground through an electrical cable integrated within the tether [11]. GG systems are particularly relevant for soft-wing AWES because the generator and associated power-conversion hardware remain on the ground, where mass and maintenance access are less constrained [12].

Despite this theoretical potential, airborne wind energy remains an emerging technology. Key challenges still include reliable autonomous operation, launch and landing, tether and wing durability, and designing airborne structures that can withstand repeated aerodynamic loading [13, 14]. These challenges are particularly pronounced for soft-wing systems, in which the wing is a flexible textile structure rather than a rigid wing. The literature commonly identifies three soft-wing architectures for AWES: LEI kites, ram-air kites, and single-skin concepts [9]. All three benefit from low mass per unit area, compact packability, and crash tolerance relative to rigid wings [15], but their deformation, load transfer, and durability make their scaling behaviour more complex, with direct implications for system cost [16].

This thesis analyses a ground-generation system with a LEI wing based on the TU Delft V3 reference design. The quantitative models and assessments apply specifically to this system architecture. Ram-air and single-skin wings belong to the broader soft-wing technology landscape but remain outside the scope of the coupled assessment.

The economic performance of a soft-wing AWES depends on more than its aerodynamic performance or initial capital cost. Changes in wing area, airborne mass, tether dimensions, and operating conditions can affect the generated energy as well as the loading, service life, and replacement cost of system components. The wing and tether may also require multiple replacements during the project lifetime. Assessing these systems therefore requires considering the interactions among their physical design, operation, energy production, and lifetime costs.

This thesis extends the AWE performance assessment framework developed by Bredael within the MERIDIONAL project [17, 18]. MERIDIONAL investigates the application of airborne wind energy at selected European reference sites and for various use cases. Bredael's framework assesses AWES performance under the wind conditions of these reference cases. The present work advances this assessment by coupling techno-economic evaluation with system sizing, airborne mass, component service life, replacement, and cost. The Bangor Erris reference site from the MERIDIONAL case studies provides the basis for applying the coupled framework.

This thesis develops and applies a coupled techno-economic framework for conceptual evaluation of soft-wing ground-generation AWES. The framework links system design, operational performance, energy production, component service life, and lifetime cost across system scales. The analysis investigates coupled effects, trade-offs, and scaling trends, rather than predicting a commercially optimal system.

The relevant literature on soft-wing AWES, system scaling, and techno-economic assessment is reviewed in chapter 2. Based on this review, the research gap, objective, research questions, and scope are defined. The coupled framework is presented in chapter 3, followed by the airborne mass model in chapter 4 and the economic model in chapter 5. The fixed system assessment and coupled scaling study are presented in chapter 6 and chapter 7, respectively. Finally, chapter 8 provides the conclusions and recommendations.

Literature Review

This chapter reviews the literature required for the conceptual techno-economic assessment of soft-wing airborne wind energy systems. Such an assessment requires both a consistent economic framework and an understanding of the technical characteristics that govern system performance, component lifetime, and scaling. The chapter first introduces the formulation of the levelised cost of energy and reviews existing techno-economic modelling approaches in section 2.1. Subsequently, the characteristics and operation of soft-wing AWES are discussed in section 2.2. Existing techno-economic studies for these systems are then examined to identify the modelling approaches and assumptions currently applied in section 2.3. Mass estimation and scaling methodologies are reviewed thereafter in section 2.4, with particular attention to the drivers relevant to soft-wing architectures. The chapter concludes by synthesising the limitations of the available literature and formulating the research gap and research questions addressed in this thesis.

2.1. LCoE and techno-economic frameworks

Techno-economic analysis combines technical performance modelling with economic assessment in a single framework so that energy technologies can be evaluated not only in terms of technical feasibility or energy yield, but also in terms of their economic effects over the project lifetime [19, 20]. Rather than serving only to report a final cost, it is typically used as a comparative tool to evaluate design options, identify the dominant technical and economic drivers, and support decisions during system development [19]. In energy systems, such analyses commonly integrate performance, subsystem sizing, operational assumptions, lifetime, and cost categories into a set of indicators. Among these, the LCoE is one of the most widely used metrics, as it enables consistent comparison between technology concepts and design variants on a per-unit-energy basis [21].

In wind energy, LCoE is frequently used as both an assessment metric and an optimisation objective because design changes often affect annual energy production and cost simultaneously [3, 22, 23]. This makes it more informative than considering isolated variables such as capital expenditure (CAPEX), operational expenditure (OPEX), or annual energy production (AEP) separately, since these describe only part of the design trade-off. LCoE combines both cost and energy yield in a single indicator [3].

The literature also distinguishes LCoE from project finance indicators such as net present value (NPV) or internal rate of return (IRR). Whereas those metrics depend more directly on electricity price assumptions, financing structures, and market situations, LCoE is primarily used for technology comparison and design assessment [24]. Reported LCoE values nevertheless remain sensitive to modelling scope, discounting assumptions, and subsystem boundaries, and should therefore be interpreted as comparative reference values instead of direct market prices.

2.1.1. Standard LCoE formulation and cost drivers

A discounted cash flow formulation is commonly used to define LCoE as the discounted lifetime cost divided by the discounted lifetime energy delivered [25]. A representative formulation used in recent AWES studies is

$$\text{LCoE} = \frac{\sum_{y=0}^{N_y} \frac{\text{CapEx}_y + \text{OpEx}_y}{(1+r)^y}}{\sum_{y=0}^{N_y} \frac{\text{AEP}_y}{(1+r)^y}}, \quad (2.1)$$

where CapEx_y and OpEx_y denote capital and operational expenditures in year y , AEP_y is the annual energy production, r is the discount rate, and N_y is the project lifetime in years. Replacement costs can be included as expenditures in the years in which the corresponding components are replaced. Planned and unplanned downtime are generally represented through their effect on the annual energy production.

The LCoE decreases when more energy is produced for a given lifetime cost and increases when the initial investment, operating expenditure, or replacement expenditure increases. A shorter project lifetime and a higher discount rate can also increase the LCoE, particularly for capital-intensive energy systems [3, 23]. Wind energy studies additionally show that differences in LCoE close to the economic optimum can be small, even when the corresponding design variables differ substantially. The LCoE should therefore be interpreted together with the underlying cost composition and the technical mechanisms governing energy production.

For an AWES, the influence of a design change on the LCoE cannot generally be inferred from its effect on either cost or energy production alone. Increasing the wing area can increase the aerodynamic force and energy production, but can also increase the airborne mass, tether force, tether dimensions, drivetrain requirements, and airborne-system cost. A larger tether diameter may additionally increase aerodynamic drag and reduce part of the expected performance gain. Conversely, a higher allowable tether stress can reduce the tether diameter, mass, and drag, while increasing the operating stress and possibly lowering the tether service life. The economically preferred design therefore depends on the combined effect of the design variables on performance, subsystem sizing, service life, replacement, and cost.

Those interactions are particularly important for soft-wing systems. Sector studies identify durability and reliability as key conditions for competitive cost development, and explicitly highlight soft-wing materials and operational reliability as remaining development needs [14]. Earlier pumping kite studies also show that wing and tether replacement assumptions can strongly affect the resulting LCoE [26, 27]. Replacement events increase lifetime expenditure and can reduce energy production when the associated downtime is significant. Component service life, replacement frequency, and availability are therefore important modelling inputs in the techno-economic assessment of soft-wing AWES.

2.1.2. Techno-economic modelling structures

Wind energy studies commonly use coupled techno-economic model structures, in which design variables, performance models, subsystem sizing, cost models, and LCoE are linked directly. In offshore wind farm studies, a change in turbine scale affects not only annual energy production, but also installation, electrical infrastructure, support structure requirements, and operations and maintenance [3]. At the turbine level, Ashuri et al. [22] showed that aerodynamic, structural, and cost changes interact in the same way, so further upscaling does not automatically reduce LCoE. Sieros et al. [23] further noted that similarity-based scaling and historical trend-based scaling should be treated separately, because the two approaches contain different information. A major result from this literature is that techno-economic assessment should not be treated as a simple sequential exercise in which performance is calculated first and cost is added afterwards. The same design variable often affects both the numerator and the denominator of the LCoE expression. Larger systems may increase energy production but can also increase structural mass, subsystem costs, installation effort, and replacement needs. Because these effects act in different directions, coupled frameworks are needed to identify economically favourable designs.

In the literature, this coupling is often implemented through multidisciplinary analysis or multidisciplinary design optimisation frameworks. In those frameworks, LCoE is not only reported as a final output, but is used directly as a design objective [3, 22]. This distinction is important because, in some studies, LCoE is calculated only after the engineering design has been finalised, so it primarily serves as a comparison metric. In other studies, the framework is built around LCoE minimisation, so aerodynamic, structural, and economic decisions are solved in a coupled way. The resulting optimum can then differ substantially from the result of a decoupled analysis [3].

The same methodological logic appears in fixed-wing AWES studies, but with different dominant couplings. Trevisi et al. [11] presented a unified engineering framework that combines analytic performance relations, subsystem sizing, and cost functions for both ground- and fly-generation concepts. In that framework, wing area and aerodynamic efficiency affect traction power. Flying mass affects gravity-related losses and operational limitations, and tether sizing affects both drag and subsystem cost. Joshi et al. [12] developed a system design framework based on the multidisciplinary design and optimisation (MDAO) methodology, integrating wind resource, power production, energy production, and cost models into a single workflow. They formalised this system through an extended design structure ma-

trix, which makes the coupling between performance, energy production, and economic blocks explicit and supports both operational optimisation and system-level design exploration. These AWES studies present their outcomes mainly as trend identification under stated aerodynamic, mass, and lifetime assumptions, rather than as precise absolute predictions.

Because coupled techno-economic frameworks contain interacting technical and economic assumptions, their interpretation usually relies on sensitivity analysis in addition to the baseline model result. In wind energy, this is commonly done through one-at-a-time parameter variation, ranked deterministic sensitivity visualisations such as tornado charts, Monte-Carlo based uncertainty propagation, and scenario-based comparisons [28–31]. In the soft-wing AWES literature, however, sensitivity analysis remains more limited and is used mainly to expose dominant drivers under uncertain assumptions on performance, lifetime, and replacement, rather than to quantify full probabilistic uncertainty.

2.1.3. Lifetime and replacement modelling

Existing soft-wing AWES studies generally treat lifetime as an assumed economic input rather than as the output of a validated load-driven model. Reported assumptions span a wide range: Gambier et al. [26, 27] adopt kite and tether lifetimes of the order of a few hundred to a few thousand operating hours, Grete [32] treats both as optimisation inputs with ranges spanning an order of magnitude, and more recent reference models such as Joshi and Trevisi [25] introduce replacement rates via a loading-factor concept rather than fixed operating-hour assumptions. Across these studies, lifetime assumptions are typically informed by manufacturer estimates, kitesurfing-industry analogues, or engineering judgement, rather than by empirical evidence from long-duration AWES operation. The resulting LCoE values are correspondingly sensitive to the chosen lifetime range, which is consistently identified as one of the dominant modelling uncertainties.

In more mature wind turbine practice, structural lifetime is instead coupled directly to simulated load spectra. Aeroelastic simulations produce time series of loads at critical locations, which are reduced to cycle counts via rainflow counting and evaluated against material $S - N$ curves to estimate accumulated fatigue damage [33]. This approach is embedded in wind turbine techno-economic optimisation, for example in Ashuri et al. [22], where blade and tower fatigue damage are used as explicit design constraints in an LCoE minimisation framework.

An analogous load-driven closure cannot yet be applied with confidence to soft-wing AWES, because the limiting gap lies on the materials and damage side rather than on the load side. Verheul et al. [34] show that inflatable beams and canopy panels are subject to biaxial membrane loading and that panel joining is itself a primary design issue, but their work addresses material selection and static testing of fabrics and joints rather than fatigue characterisation. No publicly available fatigue curves yet describe kite-grade textiles or seams under representative cyclic biaxial loading, folding, UV exposure, and environmental ageing. The tether is further advanced: Bosman et al. [35] provide publicly cited bending fatigue and creep relations for High-Modulus polyethylene (HMPE) tethers that can already be incorporated into cost modelling.

2.1.4. Cost model structures

Recent AWES cost models use component-based breakdowns rather than the single-factor cost relations presented by Heilman and Houle [15]. Joshi and Trevisi [25] decomposed the system into kite, tether, ground station, balance of system, and balance of plant, and assigned capital expenditures, operational expenditures, and replacements at the subsystem level. The model was developed as a reference framework for comparative studies and sensitivity analyses, with cost inputs compiled through an anonymised industry-academic workshop hosted by Airborne Wind Europe and calibrated to system ratings of 100 kW to 2 MW at early series production volumes of the order of fifty units.

A component-based structure also enables the use of cost relations from other industries when AWES-specific cost data are limited. This is particularly relevant for electrical machines, power electronics, and other standardised hardware, for which broader industrial datasets are available. In contrast, the largest AWES-specific uncertainties remain concentrated in the kite, tether, replacement logic, and operational and site-dependent assumptions [14]. The main physical subsystems can be seen in Figure 2.1.

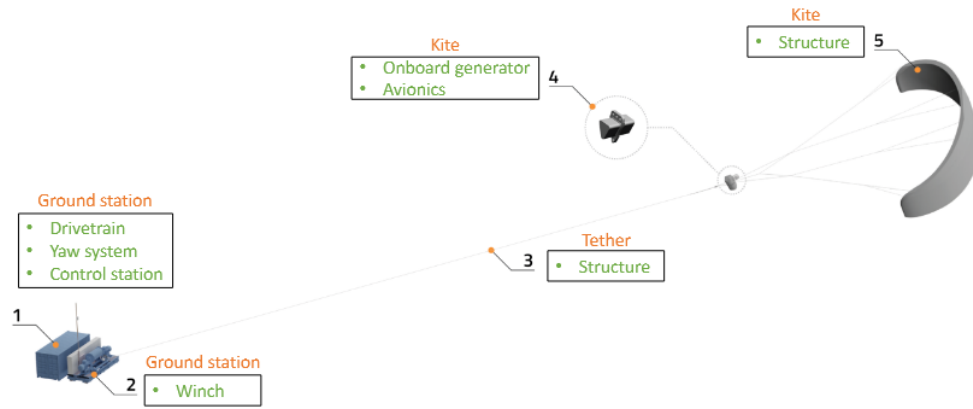


Figure 2.1: Overview of the main physical subsystems of a ground-generation soft-wing AWES. Adapted from [25]

In techno-economic modelling, a fundamental conceptual distinction exists between size-scaling, which defines cost variations against physical dimensions or capacity ratings within a single technology generation, and learning-curve effects, which capture cost reductions resulting from cumulative production volume and manufacturing maturity [23]. Conventional wind turbine studies commonly combine both mechanisms to project future costs, recognising that geometric upscaling alone may increase specific costs unless offset by simultaneous technological learning [2, 22]. However, because airborne wind energy remains in a pre-commercial phase without a historical trajectory of cumulative production, technology-specific techno-economic frameworks have to adapt their treatment accordingly. Rather than incorporating dynamic, volume-based learning curves, existing studies rely predominantly on static size-scaling relations. Unit-level cost projections apply fixed scaling exponents to physical parameters such as wing area and tether diameter [15]. At the same time, more recent multidisciplinary frameworks isolate geometric mass-scaling penalties from hypothetical manufacturing improvements [12]. This simplified treatment provides deterministic baselines for prototype comparison, while leaving long-term cost reductions from manufacturing maturity outside the scope of current AWES techno-economic models.

In the fixed-wing scaling study of Joshi et al. [12], replacement-related costs, tether-related effects, and mass-driven performance penalties are identified as dominant contributors to LCoE trends across scale. The same study uses the economic model within a coupled optimisation workflow, thereby treating cost modelling as part of the system design process rather than as a separate post-processing step. Its framework and the modelling of tether and ground-station subsystems provide a useful methodological basis for later soft-wing studies. In contrast, kite mass scaling, kite-specific cost relations, and performance modelling remain architecture-dependent and must be adapted for soft-wing systems.

2.1.5. Cost annualisation and the cost of capital

A levelised cost combines a one-off capital investment with recurring annual costs, so the capital cost must be converted to an equivalent annual charge. The simplest conversion is the capital recovery factor (CRF), the annuity that repays the investment over the project life at the discount rate. Many wind studies use a richer fixed charge rate (FCR) instead. An FCR is built on the same annuity, but it also folds in corporate tax, depreciation, insurance, and construction financing [36, 37]. An FCR-based LCoE is therefore systematically higher than a CRF-based one. The choice between them sets how much fiscal and financing detail a model resolves.

The discount rate in either annuity is the weighted average cost of capital (WACC), the blended return that debt and equity providers require. For capital-intensive technologies, the cost of capital is a leading determinant of LCoE, because most of the cost is annualised upfront investment. Vartiainen et al. find that, for utility-scale photovoltaics, the WACC is the dominant LCoE parameter after the resource [38]. Wilson et al. show that the LCoE of renewables is more sensitive to financing cost than that of fossil alternatives [39]. The discount rate is therefore not a neutral bookkeeping constant. It sets the cost level, and where technologies differ in capital intensity, it can also shift their ranking.

Financing conditions vary strongly with technology maturity and market risk. For mature onshore wind, market data place the post-tax cost of capital well below the classic model value, with a wide regional spread [2, 40]. Less mature and higher-risk technologies are financed with less debt and higher required returns [41]. Airborne wind energy sits in the demonstration and early-commercialisation phase, where technology and market risk keep commercial lenders cautious [42]. A cost of capital above the mature-wind benchmark is therefore appropriate for an early-commercial AWES, even though it would be conservative for a mature fleet.

A levelised cost must also be internally consistent in its treatment of inflation. Nominal cash flows are discounted with a nominal WACC, and constant-currency cash flows with a real WACC, the two being related through the inflation rate [37, 38]. Mixing a nominal discount rate with constant-currency costs overstates the annualised capital charge, so the convention must be stated explicitly.

For an early-stage technology assessed at the conceptual level, a simplified CRF annualisation with a conservative project-finance WACC is therefore a defensible basis. It avoids the many project-specific fiscal assumptions that a full FCR or discounted cash-flow pro-forma would require, while preserving the dominant cost-of-capital effect.

2.2. Soft-wing AWES

Soft-wing airborne wind energy systems are mainly developed in ground generation configurations, where a tethered wing produces power by pulling on a ground-based drum generator during cross-wind flight [10, 16]. Compared with fixed-wing concepts, soft-wings are attractive because of their low mass per area, compact packability, crash tolerance, and the possibility of drawing on established manufacturing knowledge from the kite industry [9]. At the same time, their performance and structural behaviour are more strongly affected by deformation, line tension, and textile durability, which makes both performance prediction and lifetime assessment more uncertain than for rigid flying structures [43, 44]. The designs of these kites are similar to the designs used in kitesurfing and paragliding. As a representative example of a soft-wing AWES architecture, the TU Delft V3 kite, shown in Figure 2.2, illustrates the typical layout of an LEI kite. The system consists of three main components: the wing, the bridle line system, and the Kite Control Unit (KCU). The wing converts the wind's kinetic energy into aerodynamic loads, which are transmitted through a complex bridle network to the main tether. The KCU, suspended below the wing, acts as the "pilot" of the system. It contains winches that adjust the relative lengths of steering lines and depower tapes, permitting precise control of the wing's pitch, yaw, and roll.

2.2.1. Soft-wing kite architectures and characteristic parameters

The literature commonly distinguishes three soft-wing architectures for airborne wind energy: LEI kites, ram air kites, and single skin concepts [9, 46, 47]. The characteristic airfoil shapes of these three configurations are shown in Figure 2.3.

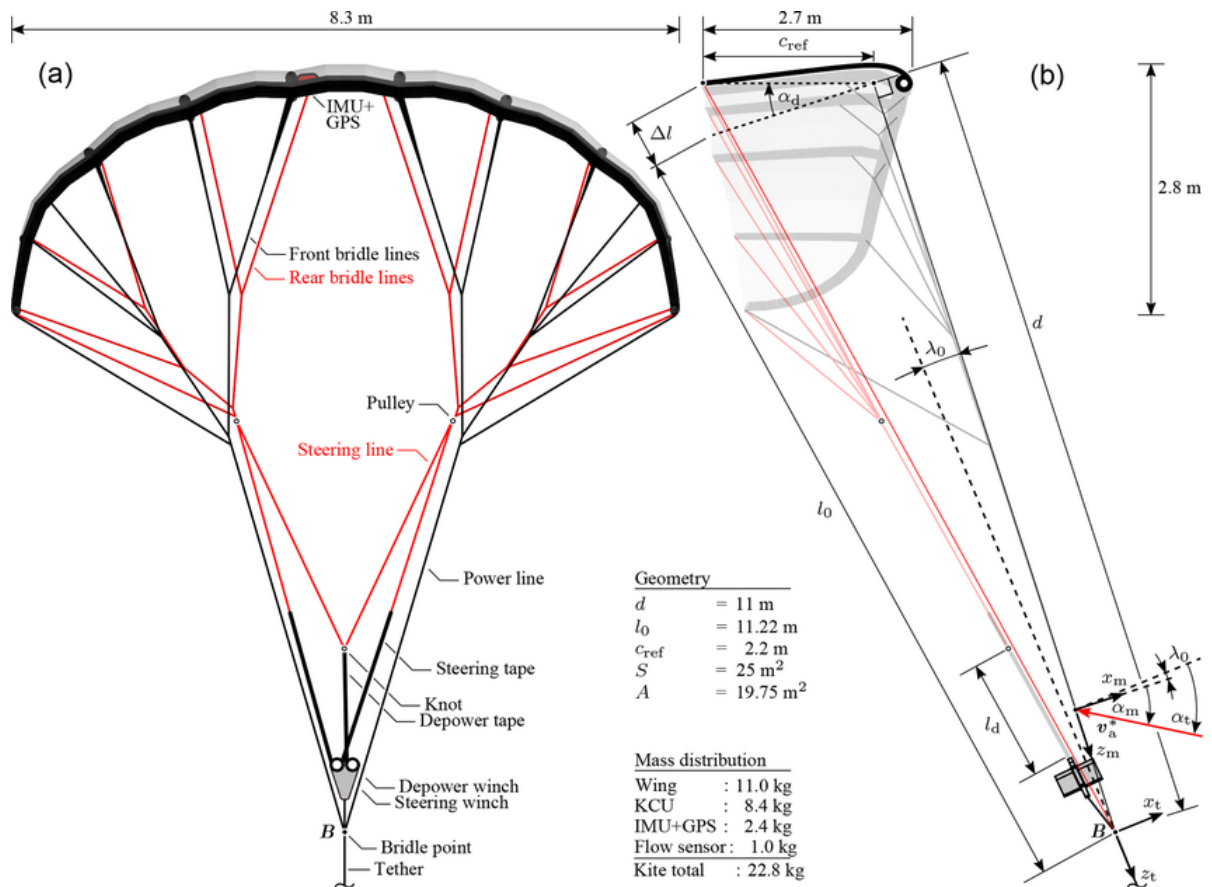


Figure 2.2: Architecture overview of an LEI kite. Displaying the front (left) and side (right) view of the TU Delft V3 kite [45].

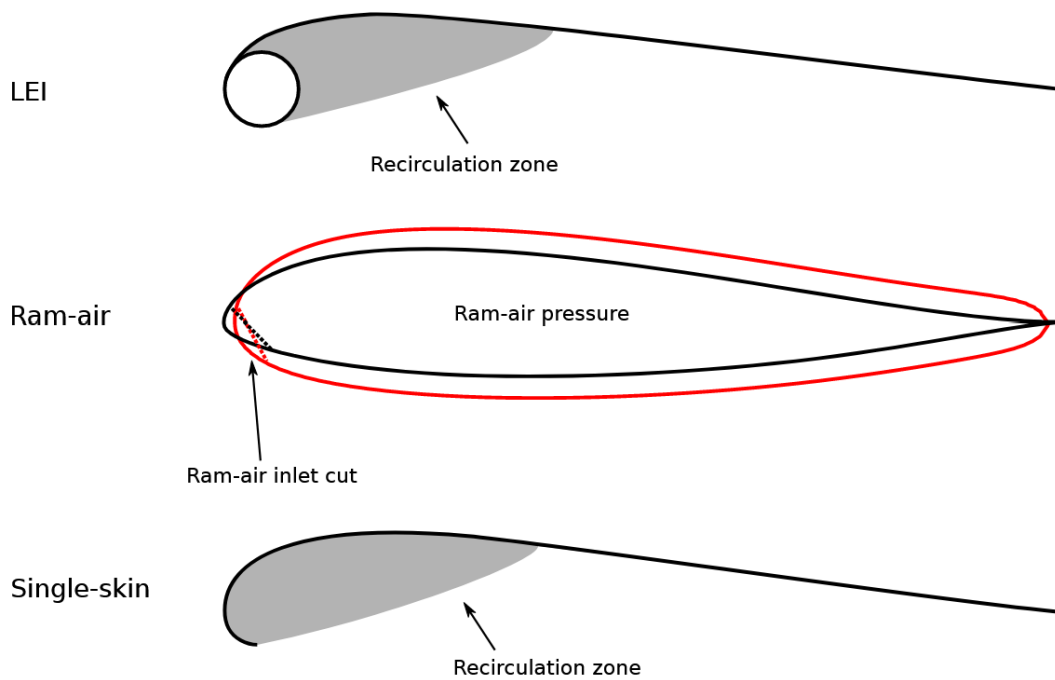


Figure 2.3: Airfoil types of the three soft-wing kite architectures. The black line represents the design shape of the airfoil, the red line shows the inflated shape of the ram-air airfoil between the ribs, and the grey area shows the recirculation zones [48].

LEI kites use an inflated tubular frame to maintain an intrinsic aerodynamic shape even when depowered. This rigidity simplifies automated launching and recovery, as the wing does not collapse in low-tension states. The structural load path runs from the distributed aerodynamic pressure on the canopy into the inflated struts and leading edge, and then through the bridle system to the tether [9]. Leading-edge diameter and strut layout are reported as important design variables because they simultaneously affect structural stability, depower range, and mass. Increasing the number of struts improves structural resistance and flight stability, but also adds mass and can introduce operational penalties such as higher minimum wind speeds for take-off. Among the three architectures, LEI kites have the highest mass-to-area ratio, because the tubes must resist high internal pressure and remain leakproof. Therefore, they are constructed from two membrane layers: a load-bearing woven fabric and an air-tight bladder [48]. In contrast, ram-air and single-skin kites lack internal rigid elements and rely on wind pressure or bridle tension to maintain their profile. Therefore, more bridle lines are required for both architectures to prevent the wing from collapsing and retain its flight shape. Ram-air kites utilise a double-skin, cell-based structure where internal ribs act as the primary load-carrying elements. Air enters through leading-edge inlets to pressurise the cells. Ram-air wings typically offer better aerodynamic efficiency than LEI kites and scale more effectively to large areas because they do not rely on heavy inflated beams. Because these wings collapse without sufficient airflow, they generally require dedicated launching infrastructure such as masts or auxiliary deployment systems, and the absence of spanwise tension limits the achievable aspect ratio [46].

Single-skin concepts are reported less frequently in the airborne wind energy literature and remain at an earlier stage of development than LEI and ram-air kites. However, these designs are already commercially available in the kite surfing and paragliding industry, mostly as hybrid single-skin concepts [49, 50]. In these systems, lift is generated by a tensioned membrane surface without either an inflated tubular frame or a closed double-skin structure, so the concept relies more strongly on bridle geometry, membrane tension, and local reinforcements to maintain a usable aerodynamic shape. The study performed by Coenen [47], mainly presents single-skin kites as a promising low-mass architecture, but also notes that the available aerodynamic and structural evidence for large-scale operation is still limited.



Figure 2.4: Industrial examples of soft-wing AWES concepts. Left: Kitepower LEI kite, image adapted from [51]. SkySails Power ram-air system, image adapted from [52]. Beyond the Sea LEI kite, image adapted from [53]. Airseas ram-air kite, image adapted from [54]

Across all three architectures, the load path from distributed aerodynamic pressure to a limited number of line attachment points makes bridles, attachment reinforcements, and allowable membrane or seam stresses central for both sizing and mass estimation [10, 55]. This also motivates the set of characteristic parameters that recur later in performance, cost, and scaling studies, including wing area S , aspect ratio AR , aerodynamic coefficients such as C_L and C_D , maximum tether force $F_{t,max}$, allowable stress levels for textile and tether materials, and operational limits such as reel out speed and admissible flight angles [45, 56]. A distinction between flat and projected area is especially important for soft-wings, because aerodynamic force scales with projected area, whereas textile mass and manufacturing effort scale more directly with flat area [45]. Because deformation, trim, and bridle loading affect the effective shape of the wing, several of these parameters are operating point dependent rather than fixed properties [43, 44]. Detailed geometric parameterisations of LEI kites, such as the recent work by Kroese [57], enable shape-level design optimisation at a finer level of detail than is captured by the characteristic parameters listed above.

2.2.2. Pumping cycle operation and performance relevant outputs

In ground-generation AWES, energy is produced through a two-phase pumping cycle: a high-tension traction phase (reel-out) and a low-tension retraction phase (reel-in), as shown in Figure 2.5. During the traction phase, the kite flies rapid crosswind manoeuvres in the "power zone" to maximise apparent wind velocity and tether force. The retraction phase begins by depowering the wing, lowering the angle of attack, to retrieve the tether with minimal energy consumption. [10]. This is achieved by the KCU actuating the depower lines to change the wing's pitch, as illustrated in the system layout in Figure 2.2. The pumping cycle is therefore commonly described by a production phase, one or more transition phases, and a retraction phase, with the operating strategy chosen to maximise cycle-averaged power while keeping loads within structural and drivetrain limits [16]. For soft-wings, reel-in performance is constrained because depowering reduces both aerodynamic force and steering authority. Minimum line tension must be maintained to keep the wing airborne and controllable, so depower remains limited in practice, which affects the amount of energy required [9].

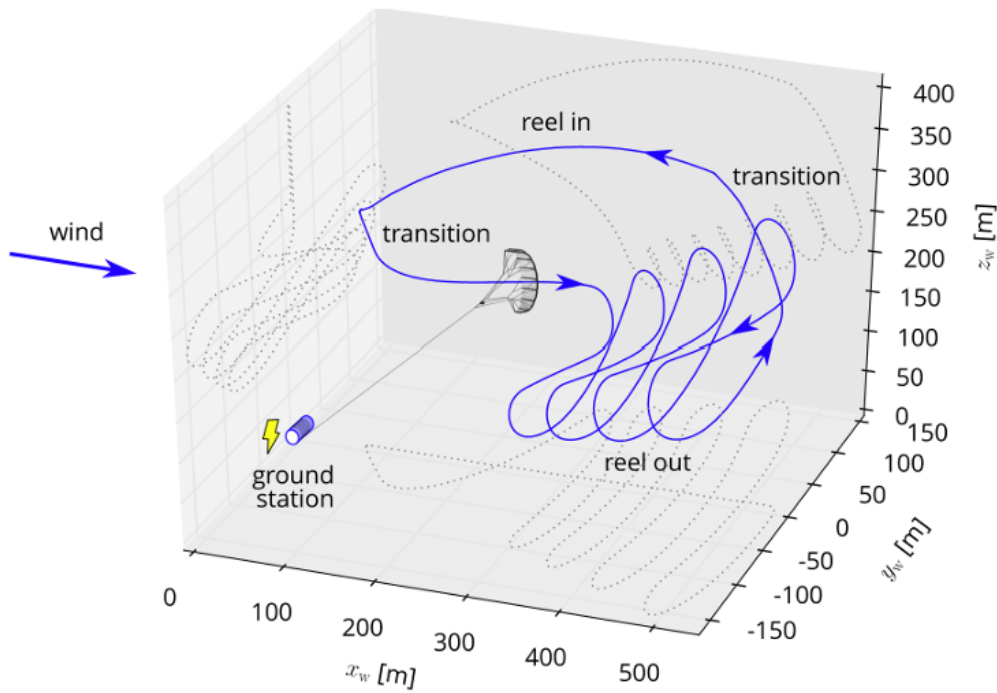


Figure 2.5: Schematic representation of pumping-cycle operation in a ground-generation AWES. During reel-out, the kite flies crosswind, generating high tether force and positive power. During reel-in, the wing is depowered, and the tether is retrieved at reduced force, resulting in lower power consumption. Image adapted from [58].

The relationship between aerodynamic force and generated power follows directly from crosswind-flight analysis, in which traction power is given by the product of tether force and reel-out speed [8, 16]. This links achievable power directly to aerodynamic efficiency and makes assumptions on C_L , C_D , and lift-to-drag ratio central when performance is propagated into component sizing and cost models [56]. In practice, steering and turning losses reduce the effective aerodynamic efficiency during pumping-cycle operation, so that cycle-averaged performance can differ from estimates based on straight crosswind flight alone [45].

The literature commonly reports power curves, cycle-averaged power, annual energy production, and tether-force statistics as the main performance outputs for system assessment [56]. Among these, peak tether force is especially important because it drives the sizing of the wing, tether, and drivetrain. The reel-out phase is therefore generally the governing case for structural sizing. By contrast, reel-in performance is more sensitive to depower capability and, at lower wind speeds, to gravity effects associated with wing mass [16]. For soft-wings, limited depower means the lift-to-drag ratio can remain relatively high during reel-in, which tends to increase the elevation angle as the tether is shortened.

A known limitation of this type of analysis is aerodynamic scaling. Reynolds-number effects are expected to become more relevant as scale and flight speed increase, and current design-oriented aerostuctural models are not yet validated robustly in this regime [44]. For the present framework, this is treated as an inherited limitation from the performance block.

Beyond the steady pumping cycle, two additional aspects shape the design and lifetime of soft-wing systems: launch and landing, and environmental exposure. Soft-wing kites are generally launched either with a winch or from a telescopic mast [48]. In a winch launch, the kite starts in a tail-sitting position downwind of the ground station. By reeling in the tether, the apparent wind speed and lift increase, helping the kite take off, especially in low-wind conditions. In a mast launch, the kite is first brought to a higher altitude before being released, and is later reeled back towards the mast for landing. Compared with a winch launch, a mast launch can offer greater autonomy by reducing ground handling and benefiting from more favourable wind conditions at altitude.

In both launch methods, launch and landing remain critical phases because the kite operates close to the ground and on a short tether. The short tether increases the angular response to disturbances, while operation in the lower atmospheric boundary layer exposes the kite to wind shear, turbulence and gusts [48, 59]. During parts of these manoeuvres, the tether tension can also be relatively low [45]. The wing is therefore more sensitive to disturbances than during normally tensioned crosswind flight. As a result, wing design cannot be based solely on maximising aerodynamic efficiency but must also account for robustness and controllability, which often conflict with purely efficiency-driven design choices. [59]

Environmental exposure further affects soft-wing design and lifetime. Membrane kites are sensitive to rain, particularly in depowered conditions, because water uptake increases the wing mass and must therefore be considered in both design and control. Ultraviolet radiation reduces membrane durability over time, and in maritime applications, saltwater and humidity introduce additional degradation pathways for the textile and bridle components. As a result, the wing typically needs to be replaced multiple times over the system lifetime [48]. Together with the cyclic loading imposed by pumping-cycle operation, these environmental factors set the context for the lifetime and replacement modelling.

2.3. Existing techno-economic studies for soft-wing AWES

The previous sections outlined how LCoE is formulated, how coupled techno-economic frameworks are structured, and how AWES cost models can be organised at the subsystem level. Against this background, the existing soft-wing literature can be compared more systematically by focusing on three aspects: the type of performance model used, the study's main contribution, and the main limitation that affects the interpretation of the reported LCoE values.

Heilmann and Houle (2013) Heilmann and Houle [15] presented an early unit-level techno-economic model for pumping kite generators, combining a simplified pumping-cycle power model with a standard LCoE formulation and parametric CAPEX scaling based mainly on kite area, reel-out power, tether force, and tether length. They explored the effect of changes in system size, wind resource and operational settings. The study showed that scaling affects both performance and cost drivers, so LCoE should be evaluated in a coupled framework rather than extrapolated from isolated quantities. Its main limitation is the simplified treatment of cost coefficients, lifetimes, and operational assumptions, which makes the model more suitable for trend analysis than for accurate prediction.

Gambier et al. (2014) Gambier et al. [26] performed an early discounted cash flow assessment for a pumping kite concept using a simplified power model and wind statistics to estimate annual energy production and LCoE. The study showed that higher-altitude operations and scaling effects could make pumping kites economically attractive. Its main limitation is the strong dependence on assumed net efficiency, availability, and aggregated operation and maintenance costs, which makes the resulting LCoE values highly uncertain.

Grete (2014) Grete [32] developed a coupled techno-economic framework that combines quasi-steady pumping-cycle physics, wind statistics, component scaling, and cost modelling, and then optimises both

operating settings and system configuration with respect to LCoE. The study showed that optimal kite size and generator rating follow from a trade-off between annual energy gains and recurring replacement costs, and that kite and tether lifetimes are major economic drivers. Its main limitations are the assumption of aerodynamic similarity during scaling and the uncertainty in cost coefficients, lifetime inputs, and operational reliability.

Argatov and Shafranov (2016) Argatov and Shafranov [60] presented a small-scale techno-economic study based on a simplified crosswind equilibrium power model combined with basic cost assumptions and fixed efficiencies. The study primarily presented order-of-magnitude trends and identified remote and stand-alone applications as the most favourable contexts for such systems. Its main limitations are the idealised aerodynamics, the neglect of kite and tether weight and availability effects, and the limited treatment of operation and maintenance costs.

Gambier et al. (2017) Gambier et al. [27] updated their earlier LCoE framework by introducing more explicit pumping-cycle assumptions and direct replacement scheduling for soft goods such as kites and tethers. The study showed more clearly that recurring replacements and ground station costs can dominate the economic outcome. Its main limitation was that long-term operation and maintenance costs and validated component lifetimes were still unavailable, leaving significant uncertainty.

Faggiani and Schmehl (2018) Faggiani and Schmehl [61] extended pumping-kite techno-economics to the wind park level by combining a quasi-steady pumping-cycle model with operational optimisation, farm aggregation, and an itemised cost model. The study showed that coordinated, phase-shifted operation can improve power quality and modestly reduce LCoE, while wing loading and tether drag remain important performance drivers. Its main limitations are the neglect of aerodynamic interactions between kites, the use of estimated cost coefficients, and the limited validation of durability and load limits.

Table 2.1 summarises the LCoE values reported in these studies.

Year	Author(s)	Power (kW)	LCoE (EUR/MWh)
2013	Heilmann and Houle	900	45
2014	Gambier et al.	200	18-22
2014	Grete	350	115
2016	Argatov and Shafranov	-	-
2017	Gambier et al.	200	45-50
2018	Faggiani and Schmehl	60	96-106

Table 2.1: Overview of LCoE values reported in public soft-wing AWES techno-economic studies.

2.4. Mass scaling methodologies

Mass scaling is needed because system-level studies require a relation between wing size and structural mass before performance, loads, and cost can be evaluated consistently. In flying systems, this follows directly from geometric scaling arguments. Tennekes [62] notes that under geometric similarity, weight scales with length cubed, whereas lifting area scales with length squared, so wing loading tends to increase with size. Noth and Siegwart [63] report a similar difficulty for lightweight solar aircraft, where structural mass becomes a limiting factor in upscaling and cannot be treated as a secondary correction.

For soft-wing kites, the same problem is more difficult because aerodynamic loading, structural deformation, and load-carrying behaviour are strongly coupled. A change in size not only changes the magnitude of the aerodynamic force but also the way in which that force is transferred through the structure. As a result, simple geometric extrapolation does not automatically preserve structural or aerodynamic similarity. This makes it difficult to relate mass growth to size through a single global scaling rule.

A further complication is that soft-wing systems are built from textile structures, pressure-supported elements, reinforcements, and bridles. All of these components respond differently to scaling [43, 55].

Their structural behaviour is therefore not governed by one single similarity condition. The literature consequently treats mass scaling in soft-wing systems as a nontrivial problem, especially when extrapolating beyond the currently demonstrated size range.

2.4.1. Mass estimation approaches reported in literature

Dababneh and Kipouros [64] provide an overview of classes of mass estimation methods used in aircraft design, distinguishing among empirical or statistical methods, quasi-analytical methods, and higher-fidelity structural methods. Although this classification is developed in the context of aircraft structures, it is relevant here because aircraft design has a much more mature literature on sizing methods for flying systems across different design stages. The structural configuration of soft-wing kites is clearly distinct. Still, the methodological distinction among low-detail empirical methods, intermediate physically informed methods, and high-fidelity structural methods provides a useful reference framework for mass-scaling studies.

Empirical methods estimate mass from statistical relations derived from existing designs. They are mainly used at very early design stages because they require only a limited number of global input parameters and can be evaluated quickly. Their main limitation is that they rely heavily on the quality and representativeness of the underlying database, thereby reducing their reliability for unconventional systems or for extrapolation beyond the available design space [64].

Quasi-analytical methods form an intermediate class. In these methods, the primary structure is sized from simplified physical relations for strength, stiffness, or stability, while secondary items are introduced through correction factors or lower-order empirical terms. Elham et al. [65] describe this class as design-sensitive and suitable for preliminary design and optimisation studies, because it retains a more physical interpretation than purely statistical methods while remaining much cheaper than full structural analysis.

At the highest fidelity level, finite element-based methods determine structural mass from a detailed geometric and material model by directly resolving the structural response. These methods can capture local stresses, stiffness effects, and structural interactions more realistically. Still, they require much more design detail and computational effort and are therefore better suited to later design stages [64, 65].

The same methodological distinction is relevant for soft-wing kites. At large scales, the available historical database is limited, reducing the reliability of purely empirical extrapolation. At the same time, high-fidelity structural methods require architecture-specific information on textile properties, joining methods, internal pressure, and reinforcement layout.

2.4.2. Reported soft-wing scaling studies in AWE literature

Only a limited number of studies report explicit mass scaling relations for soft-wing AWES. The available analytical scaling work is mainly concentrated on LEI kites. Ram-air and single-skin concepts are more commonly addressed through aerodynamic characterisation, aero-structural modelling, or design optimisation studies rather than through validated component-level mass-scaling frameworks.

Van den Heuvel [66] developed scaling rules for LEI tube kites in the context of kitesailing and energy generation. The study uses geometric similarity and introduces a one-dimensional scaling factor, such that characteristic lengths scale with this factor and surface areas scale with its square. The analysis considers the consequences of this scaling for canopy dimensions, tube dimensions, distributed loads, and material stresses. The work is based on kites in the order of tens of square metres and includes the design and testing of a 25 m² LEI kite.

Verheul et al. [34] report that the weight-to-lifting-area ratio of inflatable kite structures increases with scale, and that joining methods limit how much of the nominal fabric strength can be used in the assembled kite structure. These observations, based on kite designs in the sports and early energy-generation size ranges, indicate that LEI mass scaling is inherently component-based rather than well represented by a single areal mass coefficient.

Grete [32] applied a simplified LEI scaling approach in a techno-economic study of pumping kite power systems. The utility-scale case considers a kite with a projected area of 150 m², corresponding to a flat

area of 208 m^2 . The model assumes constant thickness and aerodynamic properties. This is consistent with the geometric similarity assumption underlying Van den Heuvel's approach. Under these assumptions, the kite mass scales with kite surface area. This makes the relation suitable for system-level cost optimisation, but the scaled design is a conceptual extrapolation rather than a validated structural scaling study at that size.

More recent work on larger LEI systems mainly reports operational prototypes, reliability considerations, or aero-structural modelling approaches. Salma et al. [67] discuss Kitepower prototypes with sizes of 25, 40, 60, and 100 m^2 in the context of reliability and safety for a 100 kW system. Cayón et al. [68] introduce a fast aero-structural model for LEI kites, coupling structural and aerodynamic modelling for design studies. Breukels [69] develops a multi-body fluid-structure interaction model for LEI kites, discretising the inflated tubular frame as rigid bodies connected by torsion springs and representing the canopy as a lattice of spring-damper elements, with aerodynamic load correlations derived from a large set of computational fluid dynamics simulations. This work characterises structural deformation and aeroelastic coupling for sports-scale LEI kites but does not reduce the results to a mass-area scaling law. These studies contribute to the understanding of larger LEI kite behaviour and structural mechanisms, but they do not introduce new closed-form mass-scaling laws comparable to those used by Van den Heuvel.

For ram-air kites, Faggiani [61] presents one of the few explicit mass scaling arguments available in the AWE-related literature. The study represents a ram-air kite as a double-skin membrane wing with upper and lower canopies separated by ribs. Two limiting design approaches are considered. In the first approach, the same relative design is retained, so the number of ribs remains constant and the material thickness increases with chord. In the second approach, the rib thickness is kept constant, and the number of cells is increased. In both cases, the specific mass increases with kite area. The study is presented as a simplified extrapolation for a wind-assisted ship propulsion application, rather than as a validated component-level design method for AWES.

Other ram-air studies focus mainly on aero-structural response and reinforcement design. Thedens and Schmehl [70] present an aero-structural model for ram-air kite simulations, while Thedens et al. [55] investigate airfoil and reinforcement optimisation. These studies provide structural and aerodynamic modelling methods for ram-air wings, but they do not reduce the results to validated analytical mass scaling laws as functions of wing area or wing loading.

On the other hand, single-skin kite literature is more limited. Coenen [47] presents single-skin kites as a possible concept for AWES and discusses their potential advantages relative to heavier soft-wing architectures. Maślanka and Korycki [71] provide a related single-skin paraglider study, reporting a reduction in mass and packing volume relative to a conventional reference wing. These studies indicate the relevance of single-skin concepts to lightweight wing design, but they do not provide a validated mass-scaling methodology for large crosswind power kites.

It is therefore important to distinguish between reported scaling relations and validated large-kite scaling methods. The available LEI relations are derived from or validated against relatively small kite geometries. When applied to larger systems, they rely on assumptions about geometric similarity, aerodynamic similarity, material thickness, and structural support that have not been validated across the large-size range relevant to utility-scale AWES. For ram-air kites, the available scaling arguments remain idealised and are not validated as component-level mass scaling laws. For single-skin kites, no comparable scaling relations have been reported.

2.4.3. Reported scaling drivers for soft-wing architectures

The studies reviewed above indicate that soft-wing mass scaling cannot generally be represented by a single universal area law. The relevant scaling drivers depend on the wing architecture because LEI, ram-air, and single-skin kites transfer aerodynamic loads through different structural mechanisms.

For LEI kites, the reported drivers are associated with multiple components, such as the canopy, leading-edge tube, struts, seams, reinforcements, internal pressure, and the bridle-supported load path. The canopy and tube dimensions follow geometric scaling, while the structural response also depends on fabric thickness, allowable stress, seam strength, and support spacing. At constant material thickness, larger LEI kites experience increasing fabric stresses as geometric scale increases. Verheul et

al. [34] further note that the weight-to-lifting-area ratio increases with scale and that joining methods limit how much of the nominal textile strength can be used in practice. The pressurised tube structure introduces additional dependencies on tube radius, inflation pressure, and local support by struts or bridle attachments. Together, these sources indicate that LEI mass scaling is inherently component-based, with the canopy, tubes, struts, seams, and reinforcements following different structural drivers.

For ram-air kites, the reported drivers include membrane deformation, rib geometry, reinforcement layout, cell spacing, and bridle loading. Dunker [46] shows from a design perspective that rib spacing is limited by topskin strength, so larger wings typically require more cells and more lines. Faggiani [61] describes this trade-off from a mass-growth perspective, identifying rib thickness and cell count as two limiting scaling choices, both of which increase specific mass with size. Thedens et al. [55] and Thedens and Schmehl [70] further show that the structural response is coupled to aerodynamic loading and deformation. The available literature, therefore, points to a locally driven scaling problem in which mass depends not only on planform area but also on cell layout, reinforcement strategy, and bridle architecture.

For single-skin kites, fewer architecture-specific scaling drivers are reported in the AWES literature. The absence of a lower skin and internal rib structure reduces the amount of membrane material relative to double-skin concepts. At the same time, the load path depends strongly on the bridle layout, local reinforcement strategy, and membrane shape stability. The available studies support the relevance of single-skin concepts for mass reduction but do not yet provide sufficient information to define a general physical mass-scaling law for large AWES kites.

2.5. Research gap and research questions

Existing research emphasises the importance of simultaneously considering system architecture, technical performance, annual energy yield, subsystem sizing, and cost in conceptual evaluations of airborne wind energy systems. The fixed-wing assessment framework proposed by Joshi et al. [12], for instance, facilitates the consistent propagation of design modifications through an integrated techno-economic system analysis. However, its underlying mass, structural, and cost correlations are tailored specifically to fixed-wing configurations and are therefore not directly applicable to soft-wing architectures. In addition, the coupled assessment does not explicitly account for component service life or replacement strategies.

This omission is particularly important for soft-wing systems. Their membrane wing and tether have substantially shorter service lives than the project lifetime and may require repeated replacement. The associated recurring costs can therefore form a major part of the lifetime cost. Changes in wing and tether design may simultaneously affect airborne mass, subsystem sizing, operational performance, annual energy production, component loading, service life, replacement frequency, and cost. Existing soft-wing techno-economic studies make assumptions about several of these quantities but do not consistently propagate their interactions through a coupled design assessment. Consequently, a framework which extends the coupling from design and performance to component service life, replacement, and lifetime cost is required to investigate the techno-economic scaling of soft-wing ground-generation AWES. Consequently, it remains unclear how soft-wing-specific mass, service-life, and replacement characteristics affect the techno-economic design and scaling trends of ground-generation AWES.

The available literature and design documentation for the TU Delft V3 reference system provide a foundation for deriving the requisite relations for a LEI wing configuration. Nevertheless, the underlying empirical and experimental evidence remains limited, particularly regarding component scaling laws and service-life extrapolation beyond the dimensions and operating durations of existing prototypes. Consequently, the formulated relations require several assumptions regarding geometric scaling behaviour, material properties, component design, replacement strategies, and supply availability. Given that component replacement can account for a significant fraction of lifetime costs, the sensitivity of the resulting techno-economic trends to these assumptions must be carefully evaluated when interpreting the framework's outcomes.

The present study addresses this gap by developing a framework for individual ground-generation systems based on the TU Delft V3 reference design. The framework integrates system sizing and component-level mass relations with the existing low-fidelity performance and AEP framework developed by Bredael [17]. The system is evaluated using the wind conditions of the Bangor Erris reference site from the MERIDIONAL case studies [18]. Component service life is represented using simplified lifetime relations and prescribed replacement assumptions. Detailed degradation modelling based on the complete load history, environmental exposure, and launch and landing cycles is outside the scope of this work. Other soft-wing architectures, interactions between multiple systems, farm-level effects, and shared infrastructure are not included in the quantitative assessment. Although the framework architecture is formulated for soft-wing ground-generation AWES more generally, its quantitative implementation and assessment are limited to systems derived from the TU Delft V3 LEI reference design.

Based on the identified research gap, the objective of this thesis is formulated as follows:

To develop and assess a coupled techno-economic framework for the conceptual evaluation of soft-wing ground-generation AWES by linking system design, operational performance, annual energy production, component service life, and lifetime cost across system scales.

This objective leads to the following main research question:

How does system scale affect the performance and LCoE of soft-wing ground-generation airborne wind energy systems?

The main research question is divided into the following four subquestions:

1. How does airborne mass scale with wing area?
2. How do component lifetime and replacements affect LCoE

3. Which performance and cost effects drive changes in LCoE across system scales?
4. How does the resulting LCoE scaling behaviour compare with fixed wing AWE assessments?

These questions are answered by applying the coupled framework to the TU Delft V3-derived soft-wing systems across a prescribed range of system scales. The analysis focuses on relative trends and the mechanisms underlying them, providing insights into trade-offs during conceptual design. It is not intended to predict commercial LCoE values or identify a definitive optimal design.

Methodology

A coupled techno-economic framework is used to assess soft-wing airborne wind energy systems across a range of system scales. The framework links system sizing, operational performance, annual energy production, and component costs, ensuring that design changes propagate consistently through the complete system. This coupling captures the physical and economic interactions that govern the LCoE and supports identifying the dominant design drivers as system scale increases. The problem formulation is introduced in section 3.1, after which section 3.2 describes the framework architecture and data flow. The subsequent sections present the system sizing, performance and energy, cost, and optimisation methods in section 3.4-3.5. The chapter concludes with the definition of the reference case in section 3.7.

3.1. Problem formulation

The conceptual design of a soft-wing GG airborne wind energy system cannot be assessed from aerodynamic performance or component cost in isolation. Changes in the wing design affect the airborne mass, tether loading, pumping cycle performance, and energy production. The resulting loads and operating conditions influence component sizing, service life, replacement requirements, and lifetime cost. The replacement of components with limited service lives further contributes to the operating expenditure. Joshi et al. [12] formalised these interactions for fixed-wing ground generation systems. The present study adapts this formulation to a soft-wing system.

Consistent propagation of changes in the physical design through the performance, energy, and economic calculations is required for the assessment. The relations used for these calculations differ in fidelity and may be revised as new data become available. The framework retains a modular structure, allowing individual relations to be adapted or replaced while preserving the overall assessment procedure and the exchange of required quantities.

The LCoE is chosen as the objective metric because it links the total lifetime costs to the electrical energy produced. Any design modification that increases energy output is assessed alongside its implications for capital investment, operating costs, and component replacement. LCoE enables comparison across different system sizes without reference to a specific future electricity price. It represents the cost of supplied energy under the given technical and economic assumptions, but does not indicate project profitability or the market value of the electricity produced.

Each prescribed generator mechanical power limit, denoted by P_{gen} , defines a separate design study. This limit distinguishes the investigated system scales and remains fixed during the grid evaluation. For each value of P_{gen} , the design problem is to identify the feasible candidate with the lowest calculated LCoE.

At the outer design level, the aim is to assess how changes in the physical system affect its lifetime LCoE. The physical design is therefore varied through the flat wing area S and the allowable tether stress $\sigma_{\text{t,allow}}$. These quantities are selected because their effects can be propagated consistently through the available sizing, performance, service life, and cost relations. The flat wing area defines the geometric scale of the kite. It changes the projected aerodynamic area, airborne component masses, design force, and component dimensions, which subsequently affect pumping cycle performance, annual energy production, and cost. The allowable tether stress determines the tether cross-sectional area required for the design force. Increasing this stress reduces the tether diameter, mass, aerodynamic drag, and material requirement, but can reduce the calculated service life and increase replacement expenditure. The two variables therefore define the installed physical configuration and expose coupled trade-offs between energy production, capital cost, and recurring cost.

The maximum wing loading remains constant to provide a consistent basis for comparing the effects of variations in wing scale and tether configuration. The aspect ratio, relative planform geometry, and aerodynamic coefficients are also held fixed. Systematic variation of these parameters would require more sophisticated aerodynamic and structural models, including design-specific aerodynamic polars and mass distributions, which are excluded from the present study. This restriction confines the analysis to design variables that can be investigated without invoking mutually incompatible aerodynamic, structural, and economic models. The modular framework allows more detailed models and additional design variables to be incorporated in future work, provided that the requisite interface quantities are available. The present study focuses on formulating and assessing the coupled framework at a level of fidelity consistent with the capabilities of the existing models.

The design and operational variables serve different purposes within the assessment. The design variables define the installed physical configuration, and candidate configurations are compared based on their resulting LCoE. Once a candidate has been defined, its physical design remains fixed during the performance assessment. The operational variables do not alter the installed system. They determine how the candidate is operated at each wind condition. The operating strategy is optimised to maximise the mean electrical power over the pumping cycle, subject to the applicable constraints. The outer design comparison therefore addresses lifetime economic performance, while the operational optimisation determines the favourable feasible operation of each candidate.

The operational variables comprise the reel-out and reel-in velocities, the minimum and maximum tether length fractions, the traction elevation angle, and the transition end elevation angle. The reeling velocities influence the power generated during traction, the energy consumed during retraction, and the duration of both phases. The tether length fractions determine the pumping stroke and the range of operating altitudes. The elevation angles influence the flight geometry, the accessible wind conditions, and the force components acting along the tether. Together, these variables determine the cycle energy balance, cycle duration, and operating feasibility. Their detailed formulation is presented with the performance model in section 3.4.

The operational optimisation is subject to the prescribed variable bounds and the physical limits of the candidate system. These include constraints on tether force, generator power, reeling speed, tether length, operating altitude, elevation angle, and site footprint. A minimum tether force is imposed to maintain canopy inflation and controllability. The economic comparison therefore includes only physical candidates for which a feasible operating strategy can be obtained.

The resulting grid evaluation enables conceptual design exploration and reveals coupled trends and trade-offs across system scales. The selected configuration achieves the lowest calculated LCoE within the investigated design space and under the adopted modelling assumptions. This configuration does not constitute a globally optimal or engineering-ready soft-wing airborne wind energy system. The results identify influential design relations and favourable regions of the investigated design space, rather than prescribing a final system design.

3.2. Framework Architecture

The techno-economic framework (TEF) evaluates each candidate design through a defined sequence of coupled model blocks. A change in flat wing area, for example, affects the wing geometry, airborne mass, maximum tether force, and tether diameter. These changes propagate through pumping-cycle performance, annual energy production, component sizing, service life, and cost calculations before the LCoE is evaluated. This sequence prevents combining performance and cost data that represent different physical systems.

Figure 3.1 provides a simplified overview of one TEF evaluation. For a prescribed generator mechanical power limit, each candidate combination of wing area and allowable tether stress is passed to the system sizing model. This model derives a consistent physical configuration, including the projected wing area, airborne masses, maximum tether force, tether diameter, and ground station dimensions. AWESPA subsequently optimises the pumping-cycle operation at each reference wind speed and calculates the corresponding performance, power curve, and annual energy production [17]. The electrical energy exchanged during the pumping cycle is used to derive the required storage capacity. EcoMo then combines the physical system specifications and performance outputs with the economic assump-

tions to calculate component costs, replacement expenditure, operating expenditure, and LCoE.

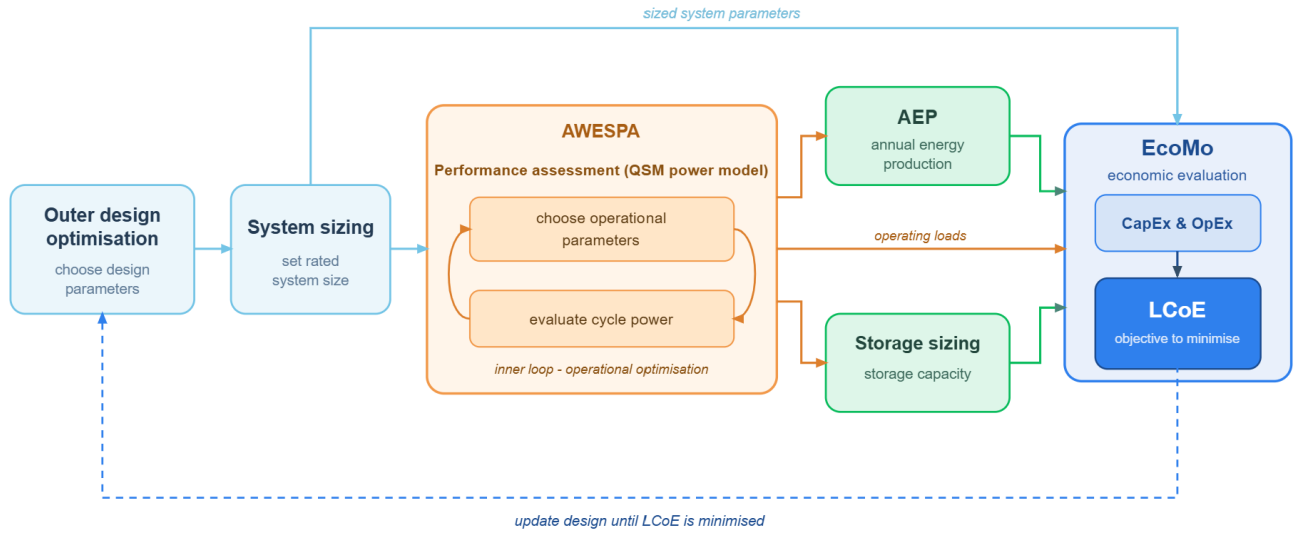


Figure 3.1: Overview of the TEF evaluation sequence, including system sizing, operational optimisation, performance assessment, storage sizing, and economic evaluation.

The resulting architecture follows an MDAO structure in which the model blocks are arranged according to their data dependencies. System design and operation are treated at separate levels. The wing area and allowable tether stress define the candidate physical system, which remains fixed during the corresponding performance assessment. Within AWESPA, the operational settings are optimised separately for each reference wind speed. After all candidate designs have been evaluated, the feasible combination with the lowest LCoE is identified. Each LCoE value therefore represents a complete coupled assessment of one evaluated grid node.

The principal model blocks have unidirectional data dependencies and are therefore evaluated in feed-forward order. No coupling variables are iterated between the disciplinary models until multidisciplinary consistency is reached. The outer driver is consequently a design-space evaluation rather than a multidisciplinary design analysis solver. The MDAO classification instead refers to the integration of multiple disciplinary models within a system-level design search.

3.2.1. Data exchange and modularity

The data exchanged between the model blocks is detailed by the extended design structure matrix (XDSM) in Figure 3.2. The simplified overview in Figure 3.1 and the XDSM represent the same framework at different levels of detail. The simplified overview communicates the principal calculation sequence, while the XDSM identifies the prescribed inputs, coupling variables, inner operational optimisation, and outer design space evaluation.

The XDSM notation introduced by Lambe and Martins [72] combines the execution order with the information exchanged between computational blocks. The diagram is read by following the black process line through the analysis and optimisation blocks along the diagonal. Grey lines show the flow of data between the models, while grey blocks contain the input and coupling variables transferred between them. The input blocks along the upper edge contain the quantities prescribed before the evaluation. By following the grey lines from these inputs through the diagonal blocks, the reader can trace how the system design is sized, evaluated, and passed to the economic assessment. The return of the LCoE to the outer design space evaluation closes the complete assessment sequence.

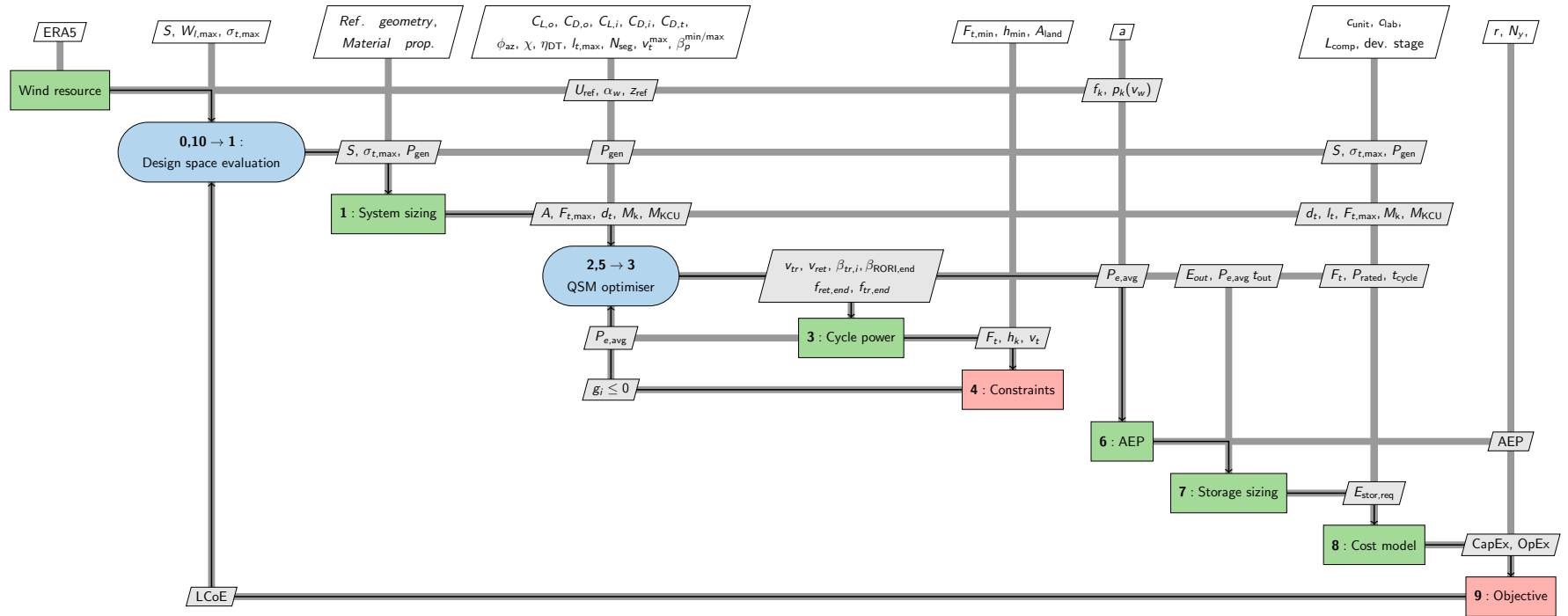


Figure 3.2: Extended design structure matrix of the soft-wing ground-generation MDAO framework. The diagram shows the model inputs, disciplinary analyses, optimisation loops and data exchanged between the framework components.

The modular organisation extends the separation of disciplinary analyses used within AWESPA to the complete techno-economic assessment [17]. The framework distinguishes between orchestrating the evaluation workflow and the models employed for system sizing, performance analysis, annual energy yield, storage dimensioning, and economic analysis. The TEF pipeline controls the order in which these models are evaluated and transfers the required information between them. The equations, numerical settings, and internal data representations remain within their respective model blocks. The models are therefore physically coupled through their exchanged quantities, while their implementations remain separate.

The use of `awesIO` provides a consistent representation of the system across the framework. It prevents each pair of models from requiring a separate data exchange definition and reduces the risk of passing inconsistent system information between analyses. Model-specific interfaces convert the common `awesIO` data into the representation required by each model and return the quantities needed by the subsequent blocks.

This separation allows individual models to be modified or replaced without restructuring the complete framework. A replacement model must accept the information available from the upstream models and return the quantities required by the downstream models. It is not required to use the same parameters or internal data structures as the original model, provided that its interface performs the necessary conversion to and from the common data representation.

The economic assessment illustrates this approach. The `EcoMo` interface converts the shared system and performance data into the model-specific inputs required for the economic calculation. An alternative economic model could be connected at the same position in the framework if its interface performs the required input preparation and returns the economic quantities used by the TEF. The required adaptation would therefore remain confined to the replacement model and its interface.

3.3. System sizing model

The system sizing model translates the design variables into a consistent system definition for the performance and economic models. The TU Delft V3 kite provides the reference geometry for the system sizing model. Flat wing area S is the geometric design variable. Geometric similarity is assumed, so the relative planform geometry, the aspect ratio, and the ratio between projected and flat wing area remain at their reference values. The projected wing area is therefore

$$A = k_p S, \quad (3.1)$$

where $k_p = A_{\text{ref}}/S_{\text{ref}}$ is the projected-to-flat area ratio of the reference kite. The distinction between projected and flat wing area is retained because the flat wing area is varied in the design study, while the maximum wing loading is defined on the projected area.

The total airborne mass is assembled from the wing and bridle, kite control unit, and sensor package contributions,

$$m_{\text{air}} = m_{\text{wing+bridle}} + m_{\text{KCU}} + m_{\text{sens}}, \quad (3.2)$$

where each contribution is expressed in kg. The wing and bridle mass and the kite control unit mass follow from the component-specific scaling relations of section 4.3. The sensor package is treated as a fixed mass. The total airborne mass is supplied to the performance model, and the separate contributions are retained for the component cost and service-life calculations.

Airborne mass enters the force balance throughout the pumping cycle and therefore affects the feasible operating conditions and the power curve. Its influence is largest during retraction, where the tether force is low and gravitational loading is a larger fraction of the force balance [10, 13]. The component-specific mass relations therefore affect both the performance and the economic assessment.

3.3.1. Load and tether sizing

Each point in the outer design space specifies the maximum allowable tether stress used for sizing. This is the intended fibre stress at the maximum design load. It is distinct from the operating stress, which follows from the tether forces of the simulated pumping cycles.

The maximum wing loading relates the projected wing area to the maximum design tether force. The reference wing loading and the resulting force limit of a scaled design are

$$W_{l,\max} = \frac{F_{t,\max,\text{ref}}}{A_{\text{ref}}}, \quad F_{t,\max} = W_{l,\max} A, \quad (3.3)$$

where $F_{t,\max}$ is the maximum design tether force and $W_{l,\max}$ is the maximum wing loading. Retaining the reference wing loading preserves the maximum design force per unit projected area, so the maximum tether force increases proportionally with projected wing area.

The maximum wing loading is not an outer design variable. A higher maximum wing loading would permit a greater aerodynamic force per unit wing area, but would require a stronger airborne structure and a thicker tether. These changes would affect airborne mass, tether drag, and cost. Fixing the maximum wing loading isolates the effects of wing area and allowable tether stress investigated in the present design study.

The tether is represented by a circular cross section in which a fraction f_A carries the load. The required diameter is

$$d_t = \sqrt{\frac{4F_{t,\max}}{\pi f_A \sigma_{t,\text{allow}}}}, \quad (3.4)$$

where d_t is the tether diameter, f_A is the fraction of the tether cross section occupied by load carrying fibres, and $\sigma_{t,\text{allow}}$ is the allowable fibre stress. A lower allowable stress gives a thicker tether for the same maximum force, which raises the tether mass and drag but lowers the fibre stress at a given operating load.

For a simulated tether force $F_t(t, v_w)$, the operating fibre stress is

$$\sigma_t(t, v_w) = \frac{F_t(t, v_w)}{f_A \pi d_t^2 / 4}, \quad (3.5)$$

where t denotes the position within the pumping cycle and v_w is the reference wind speed. Substitution of Equation 3.4 in to Equation 3.5 gives

$$\sigma_t(t, v_w) = \sigma_{t,\text{allow}} \frac{F_t(t, v_w)}{F_{t,\max}}. \quad (3.6)$$

The operating stress equals the allowable sizing stress only when the tether force reaches the maximum design force. At lower forces, it decreases in proportion to the force ratio. The allowable stress therefore determines the operating stress distribution through the resulting tether diameter.

For the tether life calculation, EcoMo uses the maximum tether force reached during the pumping cycle at each reference wind speed, denoted $F_{t,\text{cyc}}(v_w)$. The maximum operating stress within the pumping cycle at each wind speed is

$$\sigma_{t,\text{cyc}}(v_w) = \sigma_{t,\text{allow}} \frac{F_{t,\text{cyc}}(v_w)}{F_{t,\max}}. \quad (3.7)$$

EcoMo combines these stresses with the wind occurrence probabilities, cycle frequency, material properties, and drum geometry to evaluate degradation due to bending and creep. The limiting mechanism

determines the predicted tether service life and replacement frequency. A higher allowable stress reduces the tether diameter, mass, and drag but increases the operating stress and may shorten the service life. Tether life is therefore evaluated using the simulated operating loads rather than the allowable sizing stress alone.

3.3.2. Ground station specification and storage sizing

Three power quantities are distinguished when specifying and reporting the ground station. The generator mechanical power limit is the prescribed generator constraint in the performance model and the swept parameter of the generator study. The peak mechanical power is the highest realised instantaneous mechanical power in the simulated pumping cycles. The rated power is the maximum cycle-averaged net electrical power obtained from the power curve, and it is used to report the system scale and calculate the capacity factor.

Each ground-station component is sized on the power or load that governs it. The generator power limit is prescribed for each design study and written to the generator entry of the system configuration. The drivetrain and foundation are sized on the peak mechanical power, because this is the largest load they carry. The drum is sized on the maximum tether force and the tether diameter, which set its torque and wall dimensions. The electrical storage is sized on the cycle-energy exchange derived below. Prescribing the generator power limit therefore does not fix the remaining ground-station hardware, which still varies with the candidate design.

Electrical storage buffers the electrical energy produced during reel-out that exceeds the cycle-average delivery, and releases it during reel-in. The required usable storage energy at each wind speed is this buffered surplus,

$$E_{\text{stor}}(v_w) = E_{\text{out}}(v_w) - \bar{P}_e(v_w) t_{\text{out}}(v_w). \quad (3.8)$$

Here, E_{out} and t_{out} denote the electrical energy and duration of the reel-out phase, while $\bar{P}_e$ represents the cycle-averaged electrical power. Evaluation across the wind range applies a rolling mean to minimise sensitivity to isolated numerical variations between adjacent operating solutions. The installed storage capacity results from the largest smoothed value.

3.4. Performance and energy model

The performance modelling consists of three main blocks. First, the site wind resource is represented by a vertical wind profile model and a wind speed probability distribution. Second, a cycle power model is used to compute the electrical power curve from the site wind resource and the system design and operational parameters. Third, this power curve is integrated against the site wind-speed frequency distribution to obtain the AEP. The analysis is carried out within the AWESPA toolchain developed by Bredael [17], which implements the quasi-steady pumping-cycle methodology of Schelbergen et al. [73].

3.4.1. Wind resource representation

The wind resource provides the environmental input to the performance model. In an airborne wind energy system, the vertical variation in wind speed is especially relevant because the kite does not operate at a fixed height. During a pumping cycle, the kite moves over a range of altitudes, so the cycle power depends not only on the reference wind speed but also on the assumed vertical wind profile.

AWESPA can use different wind resource representations with different levels of fidelity. A common simplified approach is to describe the vertical wind speed variation with a fitted power-law profile,

$$v_w(z) = U_{\text{ref}} \left(\frac{z}{z_{\text{ref}}} \right)^{\alpha_w}, \quad (3.9)$$

where U_{ref} is the wind speed at the reference height z_{ref} , and α_w is the power law shear exponent. The exponent can be prescribed from literature or fitted to wind profile data. If the exponent is fitted to wind profile data, the resulting profile represents the average vertical shear at the site using a single continuous profile shape.

The main advantage of the fitted power law representation is its computational efficiency. Since only one wind profile is used, the QSM needs to compute only one power curve. This makes the power-law profile suitable for repeated evaluations in the framework, where the performance model is called many times for different system designs. Its main limitation is that it assumes a constant profile shape across all reference wind speeds and directions. Variations in atmospheric stability, low-level jets, and changes in wind direction with altitude are therefore not explicitly represented.

A logarithmic wind profile can also be used to represent the vertical wind speed variation. Compared with the power law, this profile has a stronger physical basis in atmospheric boundary-layer theory because it relates wind-speed variation to surface roughness and boundary-layer stability. However, in the current modelling context, it remains a single-profile approximation. It does not retain the wind profile's temporal variability over time, nor does it explicitly represent wind direction shear or different profile shapes under different atmospheric conditions. In addition, the assumptions underlying the logarithmic profile are most appropriate near the surface, whereas airborne wind energy systems can operate over a broader altitude range. For this reason, the logarithmic profile is useful as an available reference representation, but it is not selected for the main performance calculations in this study.

A more detailed representation is obtained by clustering wind profile shapes. In this approach, multi-year reanalysis data are reduced to a limited number of representative vertical wind profiles using principal component analysis and K means clustering, following Schelbergen et al. [73]. Each cluster k contains a normalised wind profile, an occurrence frequency f_k , and a conditional wind speed distribution $p_k(v_{w,\text{ref}})$. This representation can capture different typical profile shapes at a site, including variations in shear and wind direction with altitude.

The selected representation therefore reflects a trade-off between computational effort and environmental detail. The fitted power law is used during the iterative techno-economic assessment, where one candidate design may require many power-model evaluations. Bredael [17] used the clustered resource to assess whether the simplified profile changes the predicted annual energy production, and the observed offsets are assumed to be negligible small in the final results.

3.4.2. Cycle power model

The cycle power model converts the system design and wind resource input into an electrical power curve. This is done with the inertia-free quasi-steady model (QSM) implemented in AWESPA. For each reference wind speed $v_{w,\text{ref}}$, the model simulates a complete pumping cycle and determines the corresponding mean electrical power output. Repeating this calculation for a range of reference wind speeds yields the power curve used in the AEP calculation.

The selected wind resource representation is used to determine the local free-stream wind speed at the kite altitude during the pumping cycle. In the main analysis, this is done using the fitted power-law profile from Equation 3.9. When the clustered wind resource is used, the same QSM formulation is applied, but the local wind speed follows the corresponding normalised cluster profile. The cycle model itself is therefore unchanged, and only the wind profile supplied to the model differs.

The pumping cycle consists of a traction phase, a retraction phase, and two transition phases. During the traction phase, the kite follows a crosswind trajectory to generate tension in the tether. The figure eight path is not explicitly resolved. Instead, the motion is represented by a fixed representative flight state, characterised by prescribed azimuth and course angles. For each segment of the reel-out stroke, the force balance is solved by accounting for aerodynamic force and drag, tether drag, the kite weight, and the KCU weight. The resulting tether force F_t and reel out speed $v_{t,o}$ determine the instantaneous mechanical traction power,

$$P_{m,o} = F_t v_{t,o}. \quad (3.10)$$

During the retraction phase, the kite is depowered and reeled back towards its minimum tether length. The kite and KCU weight are again included in the force equilibrium. This contribution is particularly relevant for soft-wing systems because the depowered kite has a lower lift-to-drag ratio, thereby reducing the aerodynamic force margin relative to gravity. The corresponding mechanical power consumption is

$$P_{m,i} = F_{t,i} v_{t,i}. \quad (3.11)$$

The mean mechanical cycle power is calculated from the net mechanical energy over one pumping cycle,

$$P_{m,avg} = \frac{E_o - E_i + E_{tr}}{t_{cycle}}, \quad (3.12)$$

where E_o is the mechanical energy produced during the traction phase, E_i is the mechanical energy consumed during the retraction phase, E_{tr} accounts for the contribution of the transition phases, and t_{cycle} is the total cycle duration. This formulation is used here to represent the complete cycle energy balance implemented in AWESPA.

For each reference wind speed, an inner optimisation loop can be used to maximise the mean cycle power. The optimiser varies six operational parameters, listed in Table 3.1. These variables define the traction and retraction reeling speeds, the tether length range of the pumping cycle, and the elevation angles used during traction and transition. The optimisation is subject to bounds on reeling speed, tether length fraction, elevation angle, and minimum operating height. A minimum tether force constraint is also imposed to maintain LEI canopy inflation. The azimuth angle, course angle, and aerodynamic coefficients are kept fixed throughout the optimisation.

Table 3.1: Operational variables and bounds used in the QSM optimisation.

Parameter	Symbol	Bounds
Traction reeling speed	v_{tr}	$0.01 \leq v_{tr} \leq 15.0 \text{ m s}^{-1}$
Retraction reeling speed	v_{ret}	$-15.0 \leq v_{ret} \leq -0.01 \text{ m s}^{-1}$
Tether length fraction at end of retraction	$f_{ret,end}$	$0.20 \leq f_{ret,end} \leq 0.80$
Tether length fraction at end of traction	$f_{tr,end}$	$0.50 \leq f_{tr,end} \leq 0.95$
Traction elevation angle	$\beta_{tr,i}$	$20^\circ \leq \beta_{tr,i} \leq 80^\circ$
Reel out to reel in transition end elevation angle	$\beta_{RORI,end}$	$30^\circ \leq \beta_{RORI,end} \leq 80^\circ$

The mean mechanical cycle power is converted to electrical output power by applying a constant drivetrain efficiency,

$$P_{e,avg}(v_{w,ref}) = \eta_{DT} P_{m,avg}(v_{w,ref}), \quad (3.13)$$

where η_{DT} is the assumed drivetrain efficiency. This gives one electrical power curve for the fitted power-law representation, or one electrical power curve per wind-profile cluster when the clustered representation is used.

The QSM neglects inertial forces and assumes that the kite reaches quasi-steady equilibrium within each cycle segment. This assumption is appropriate for conceptual system-level studies, but it introduces uncertainty for soft-wing kites because canopy deformation and transient effects during the transitions are not explicitly resolved. The resulting power curve should therefore be interpreted as a conceptual level performance estimate.

3.4.3. Annual energy production

The annual energy production is obtained by combining the electrical power curve with the wind speed probability distribution at the reference height. The calculation method depends on the selected wind resource representation.

For a single profile representation, such as a fitted power-law profile, only one electrical power curve is computed. The AEP is then obtained by integrating this power curve over the wind speed probability density,

$$AEP = 8760 \int_{v_{w,ci}}^{v_{w,co}} P_e(v_{w,ref}) p(v_{w,ref}) dv_{w,ref}, \quad (3.14)$$

where $P_e(v_{w,ref})$ is the mean electrical power at the reference wind speed, $p(v_{w,ref})$ is the wind speed probability density at the reference height, and $v_{w,ci}$ and $v_{w,co}$ are the cut in and cut out wind speeds.

For the clustered wind resource representation, the AEP is calculated by integrating the power curves

of each wind profile cluster and weighting them by each cluster's occurrence frequency.

$$\text{AEP} = 8760 \sum_{k=1}^{N_{cl}} f_k \int_{v_{w,ci}}^{v_{w,co}} P_{e,k}(v_{w,ref}) p_k(v_{w,ref}) dv_{w,ref}, \quad (3.15)$$

where N_{cl} is the number of wind profile clusters, f_k is the fractional occurrence frequency of cluster k , $P_{e,k}(v_{w,ref})$ is the electrical power curve computed for that cluster, and $p_k(v_{w,ref})$ is the conditional wind speed probability density for that cluster.

The single profile expression in Equation 3.14 can be interpreted as a special case of the clustered expression with one representative profile. The clustered formulation is more general but requires multiple power-curve evaluations. In this study, the single-profile formulation is used during the main techno-economic iterations, while the clustered formulation is used to assess AEP sensitivity to wind-profile representation.

The annual wind speed distribution is assumed to remain constant over the project lifetime. Interannual variability and long-term climate trends are therefore not modelled. The lifetime energy yield is calculated as

$$E_{\text{lifetime}} = \sum_{y=0}^{N_y} \frac{\text{AEP}}{(1+r)^y}, \quad (3.16)$$

where N_y is the project lifetime, y denotes the year index, and r is the discount rate.

3.5. Cost model

EcoMo maps the physical system and its operating behaviour onto the system cost, following the component-based hierarchy of Joshi's reference economic model [25]. The framework evaluates a single onshore soft-wing ground-generation system. The costs are grouped into four subsystems, namely the kite, the tether, the ground station, and the balance of system. Each subsystem retains the physical origin of its contribution to the LCoE. The kite subsystem contains the soft-wing structure and the kite control unit. The balance of system covers site preparation, foundation, installation, maintenance, and decommissioning. The balance of plant is excluded, because it concerns farm-level infrastructure that applies only to multi-system deployments.

The tether and ground-station relations are retained from the reference model, because their physical role does not depend on whether the airborne device is fixed-wing or soft-wing. The kite relations are adapted to the soft-wing through its component masses. Component lifetimes, maintenance labour, and supervision are set separately from the physical scale, because they follow from operational maturity rather than from wing area or generator power.

EcoMo receives the physical quantities computed by the preceding models. These are the component masses, tether properties and loads, generator and storage requirements, annual energy, and operating hours. EcoMo combines them with unit costs, component lifetimes, labour and availability assumptions, and financial parameters. It returns the initial capital cost, the annual operating and replacement expenditure, the subsystem contributions, and the LCoE.

The tether cost depends on the design through two paths. Its diameter sets the material cost, mass, and drag. Its simulated loads set the degradation, lifetime, and replacement cost. Using a single fixed lifetime would remove this dependence from the optimisation.

The detailed cost equations, lifetime relations, and financial parameters are given in chapter 5. The cost inputs are subject to substantial uncertainty regarding manufacturing maturity, labour intensity, supply chains, and site conditions. The resulting LCoE is therefore interpreted relative rather than as a definitive commercial cost.

3.6. Design optimisation strategy

The outer design search is performed separately for each prescribed mechanical power limit of the generator. Rated electrical power would provide a more conventional basis for comparing system scales. Within the present framework, however, the rated electrical power is unknown prior to the

performance assessment. It is derived from the optimised power curve, which represents the maximum cycle-averaged net electrical power, and therefore depends on both the candidate physical design and its optimised pumping-cycle operation. Prescribing rated electrical power would require an additional iterative procedure for every candidate, in which the generator mechanical power limit is adjusted until the resulting rated power reaches the prescribed value.

The generator mechanical power limit is instead a direct input to AWESPA and is imposed as an instantaneous constraint during the operational optimisation. It can therefore be specified before the candidate designs are evaluated and held constant throughout each evaluation of the design space. This provides a feasible and reproducible means of investigating increasing power handling levels with the present framework. The prescribed mechanical limit should not be interpreted as equivalent to the rated electrical power, since candidates evaluated under the same limit may produce different rated powers. The rated electrical power is consequently derived and reported as an output for each selected design.

Within each prescribed generator mechanical power limit, the flat wing area and allowable tether stress are varied to identify the candidate with the lowest calculated LCoE. Repeating this procedure allows the resulting physical designs and their derived performance characteristics to be compared across the investigated power limit cases.

The design space is evaluated using a bounded full factorial grid. Both design variables are sampled at linearly spaced values, after which every combination is evaluated using the complete TEF. This approach is computationally feasible because the design space contains only two variables. It also does not require gradients of the objective function or an assumption of a smooth LCoE surface. Such assumptions may not be appropriate because every candidate evaluation contains a nested operational optimisation and may approach boundaries between feasible and infeasible operation. The individual grid points are mutually independent and can therefore be evaluated separately.

Evaluating the complete grid provides information beyond the location of the lowest calculated LCoE. The resulting surface shows the sensitivity of LCoE to both design variables, their combined effects, and the extent to which the lowest LCoE region is clearly defined. A candidate is included in the design selection only when the complete framework returns a valid LCoE. Designs that violate physical or operational constraints are classified as infeasible. An evaluation that terminates because no numerical solution is obtained does not necessarily demonstrate physical infeasibility and is therefore recorded separately where its cause can be identified. Infeasible designs and unsuccessful evaluations are excluded from the selection of the lowest LCoE candidate.

The grid search evaluates discrete combinations of the design variables and does not determine the exact minimum of the continuous design space. The selected design therefore represents the lowest calculated LCoE among the sampled candidates. Its neighbouring feasible points are taken into account when interpreting the result. A broad region with similar LCoE values is treated as a range of economically comparable designs rather than as evidence of a uniquely defined optimum.

The position of the selected candidate relative to the design bounds is also examined. A lowest LCoE candidate located on a bound indicates that the evaluated range may not enclose the complete lowest LCoE region. The corresponding range is extended and re-evaluated as long as this remains consistent with the validity of the underlying models. If further extension cannot be justified, the selected candidate is reported as a boundary solution. Separate wing area bounds are used for each generator mechanical power limit to retain an appropriate resolution for the corresponding design range. The final bounds, sampling intervals, and number of evaluated points are reported as part of the study configuration.

3.7. Case study definition

The case study uses the RWE airborne wind energy test centre near Bangor Erris on the north-western coast of Ireland. The site lies on the exposed Erris peninsula, with flat, open terrain and a strong Atlantic wind resource. It has hosted flight testing of Kitepower systems and serves as the reference location for the small-scale isolated inland use case of the MERIDIONAL project [18].

Bangor Erris is chosen because its wind resource has already been characterised by [17] using AWESPA, so wind-resource inputs for the power-curve model are directly available. The key parameters of the fit-

ted power-law wind profile are listed in Table 3.2. In addition, the site is operational, and measurement data from both the instrumented AWES and an upstream lidar are available, enabling future validation of the computed power curve against flight-test data

The main analysis uses the fitted power-law representation of the Bangor Erris wind resource, as given by Equation 3.9. The annual wind-speed probabilities are the empirical ERA5 distribution at the reference height, resolved into 50 wind-speed bins over 2010 to 2019. This representation requires one power curve per candidate design. The clustered representation requires a separate power curve for each representative profile and is used for the wind-profile sensitivity analysis.

Table 3.2: Wind-resource inputs for Bangor Erris, derived from ERA5 data for 2010 to 2019 using AWESPA [17].

Parameter	Symbol	Value
Mean wind speed at reference height	$\bar{U}_{\text{ref}}$	9.94 m s^{-1}
Reference height	z_{ref}	200 m
Power-law shear exponent	α_w	0.189
ERA5 data period	—	2010–2019

The fixed system, operational, and conversion assumptions are collected in Table 3.3. The aerodynamic coefficients, the aspect ratio, and the relative planform geometry are held at the TU Delft V3 values [45] because varying them consistently would require an aerodynamic and structural redesign beyond the framework's fidelity. The results therefore represent system scaling at fixed relative geometry and aerodynamic performance.

Table 3.3: Fixed system, operational, and conversion assumptions for the reference case.

Quantity	Symbol	Value
Flat wing area	S	25 m^2
Projected wing area	A	19.75 m^2
Wing and bridle mass	m_{kite}	10.6 kg
Kite control unit mass	m_{KCU}	8.4 kg
Sensor package mass	m_{sens}	3.4 kg
Total airborne mass	m_{air}	22.4 kg
Reel-out lift coefficient	$C_{L,o}$	0.63
Reel-out drag coefficient	$C_{D,o}$	0.14
Reel-in lift coefficient	$C_{L,i}$	0.40
Reel-in drag coefficient	$C_{D,i}$	0.12
Tether diameter	d_t	6 mm
Tether density	ρ_t	970 kg m^{-3}
Tether drag coefficient	$C_{D,t}$	1.1
Aspect ratio	AR	5.3
Projected-to-flat area ratio	k_p	0.79
Maximum tether force	$F_{t,\text{max}}$	8.2 kN
Maximum wing loading	$W_{l,\text{max}}$	0.42 kN m^{-2}
Maximum tether length	$\ell_{t,\text{max}}$	500 m
Minimum tether force	$F_{t,\text{min}}$	750 N
Generator efficiency	η_{gen}	0.85
Motor efficiency	η_{mot}	0.85
Storage efficiency	η_{stor}	0.95
Maximum drum reeling speed	$v_{r,\text{max}}$	10 m s^{-1}
Generator power limit	P_{gen}	40 kW

Mass Scaling Model

The mass of the airborne system can be divided into two parts, namely the kite mass and the kcu mass. This decomposition defines the structure of the chapter. Before developing the scaling model, the modelling approach is motivated in section 4.1. The first part focuses on the wing mass. A geometry-based mass model of the TU Delft V3 kite is first constructed to decompose the wing mass into its main structural components and to identify the dominant mass drivers in section 4.2. Based on this decomposition, scaling relations are then developed for the individual components using simplified design equations derived from the underlying physical loads and structural requirements in section 4.3. After that, the scaling of the kite control unit is discussed separately in section 4.4.

4.1. Modelling approach

Three classes of mass estimation methods are available for soft-wing kites, as reviewed in section 2.4. Each is evaluated here against the requirements of the present framework.

A purely empirical approach fits mass trends to existing kite data. For LEI kites, publicly available measurements are largely restricted to sports kites and early AWE demonstrators in the $10\text{-}25\text{m}^2$ range [15, 32]. Extrapolating these trends to the wing areas relevant to utility-scale AWES would depend strongly on the selected dataset. It would offer no physical basis for projecting structural changes with scale. This approach is therefore used only to calibrate reference values, not to define the scaling behaviour.

A design-sensitive model, such as the data-driven relation of Joshi et al. [56], relates kite mass to wing area, aspect ratio, and maximum wing loading. These parameters are physically meaningful for soft-wings. Applying such a model to LEI kites, however, requires more detailed structural layout data of components, textile properties, bridle topology, and reinforcement geometry that are not publicly available at the scales of interest. Using this approach would therefore require assumptions that cannot be grounded in the available data.

A coupled aero-structural model provides the most physically complete description, resolving deformation, load redistribution, and material requirements simultaneously [68]. For soft-wing kites beyond the demonstrator scale, neither the validated aero-structural models nor the required design data exist in the public domain. This fidelity level is therefore beyond the scope of the present work.

The adopted approach decomposes the kite into its principal structural components and assigns each a scaling relation derived from the governing physical loads. This is consistent with the quasi-analytical class of methods identified in section 2.4 as the appropriate intermediate fidelity for conceptual techno-economic assessment. The available V3 prototype data [45] anchor the model at the reference scale. The derived physical relations determine the scaling behaviour above this scale rather than extrapolating an empirical fit.

4.2. Mass decomposition

To gain insight into the distribution of wing mass across its main structural components, a geometry-based component model of the TU Delft V3 kite is constructed. The aim of this model is not to provide a highly accurate mass prediction, but rather to estimate the relative contribution of the different components and to identify the dominant mass drivers of the V3 kite. The model combines geometric information, material properties, simplified representations of additional reinforcement elements, and available verification data.

For a soft-wing LEI kite, the total wing mass can be decomposed into the following six component groups:

$$m_{\text{kite}} = m_{\text{canopy}} + m_{\text{LE}} + m_{\text{struts}} + m_{\text{bridle}} + m_{\text{reinf}} + m_{\text{additional}}. \quad (4.1)$$

The canopy is the single-layer membrane that carries the aerodynamic tension. The leading-edge tube and the struts are the inflatable beam elements that provide bending and torsional stiffness, with the bladder mass included in both terms. The bridle is the line system that transfers the aerodynamic loads into the tether. The reinforcement term collects local patches, seam-related reinforcements, and profile-maintaining inserts. The additional term collects discrete components such as valves and minor hardware [32, 34, 66].

The kite geometry is obtained from the CAD-based design files through SurfplanAdapter [74]. This allows the main geometric quantities of the kite to be loaded directly. An image of the resulting geometric representation is shown in Figure 4.1. Since the fabric and tube materials contribute significantly to the total mass, the component masses can be estimated by determining the amount of material associated with the current geometry, such as the canopy surface area, the tube surface area, and the bridle line lengths.

Several reinforcement and additional mass contributions are not explicitly contained in the exported geometry files. These terms are therefore estimated separately based on layout images of the V3 kite and discussions with experts.

The V3 kite is reported to have a total mass of 10.6 kg, including the bridle system, in its original design configuration [45]. This reference configuration excludes several reinforcements that were added later in the development process to strengthen local weak points. An overview of the materials used for the V3 kite [75], together with the corresponding density values adopted here, is given in Table 4.1.

Table 4.1: Material properties of the TU Delft V3 kite.

Material	Areal density [g m^{-2}]	Component used for	Symbol
Dacron	170	Tubes outer layer	ρ_d
Ripstop polyester	50	Canopy	ρ_p
100 micron TPU	100	Bladder	ρ_{TPU}

These material densities are used to estimate the mass of the fabric-based components through

$$m = \rho S \quad (4.2)$$

where ρ is the areal density of the material and S is the associated surface area.

As shown in Figure 4.1, the canopy consists of panels bounded by two leading-edge nodes and two trailing-edge nodes. Since the local chord length varies along the span, these panels are generally quadrilateral rather than rectangular. In the present model, the area of each canopy panel is obtained by splitting the quadrilateral into two triangles and summing their corresponding areas. This ensures that the local panel area is represented correctly, even when the four corner nodes are not coplanar. Between two neighbouring struts, the canopy is represented by three such panel segments. Near the tips, the number of segments is increased to more accurately capture the local rounding of the leading edge.

For a quadrilateral with vertices $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, and $\mathbf{p}_4$, ordered consecutively around the panel, the panel area is given by

$$S_{\text{panel}} = S_1 + S_2 \quad (4.3)$$

with

$$S_1 = \frac{1}{2} \|(\mathbf{p}_2 - \mathbf{p}_1) \times (\mathbf{p}_4 - \mathbf{p}_1)\| \quad (4.4)$$

and

$$S_2 = \frac{1}{2} \|(\mathbf{p}_4 - \mathbf{p}_2) \times (\mathbf{p}_3 - \mathbf{p}_2)\|. \quad (4.5)$$

The canopy mass is then obtained by summing the mass of all canopy panels:

$$m_{\text{canopy}} = \rho_p \sum_{i=1}^{N_{\text{panel}}} S_{\text{panel},i} \quad (4.6)$$

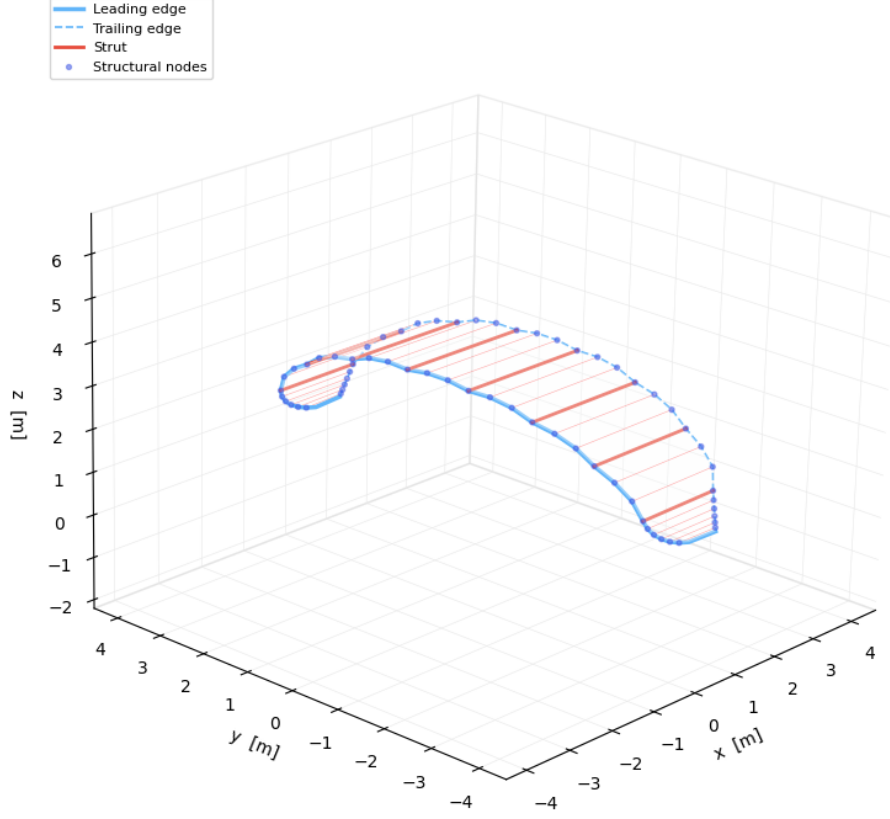


Figure 4.1: Geometrical representation of the TU Delft V3 kite geometry.

To determine the mass of the inflatable tubes, two components are considered: the outer Dacron shell and the internal bladder material. The tube surface area is calculated using

$$S_{\text{tube}} = \pi h \frac{d_1 + d_2}{2} \quad (4.7)$$

where h is the segment length, and d_1 and d_2 are the diameters at the first and second node, respectively. Both the leading edge and the struts are therefore modelled as straight tapered tubes. For the leading edge, the segment length is defined as the distance between two consecutive nodes along the leading edge. For the struts, the segment length is taken as the distance between the leading edge node and the trailing edge node at the corresponding spanwise location, which is equal to the local chord length.

For the struts, the slight curvature of the centreline is neglected, since its contribution to the total surface area is assumed to be small. The tube masses are then determined as

$$m_{\text{LE}} = (\rho_d + \rho_{\text{TPU}}) \sum_{i=1}^{N_{\text{LE}}} S_{\text{LE},i} \quad (4.8)$$

and

$$m_{\text{struts}} = (\rho_d + \rho_{\text{TPU}}) \sum_{j=1}^{N_{\text{struts}}} S_{\text{strut},j} \quad (4.9)$$

where $S_{LE,i}$ is the surface area of leading edge segment i , and $S_{strut,j}$ is the surface area of strut j .

The mass of the bridle system is determined from the volume of each individual line segment and the density of Dyneema, taken as $\rho = 970 \text{ kg m}^{-3}$. A complete overview of the bridle layout is adopted from the work presented by Roeleveld [76]. The line volume and mass are calculated using Equation 4.10 and Equation 4.11:

$$V_{\text{line}} = \frac{\pi d^2 l}{4} \quad (4.10)$$

where d is the line diameter and l is the line length. The total bridle system mass is then given by

$$m_{\text{bridle}} = \rho \sum_{i=1}^{N_{\text{line}}} V_{\text{line},i} + \sum_{j=1}^{N_{\text{pulley}}} m_{\text{pulley},j} \quad (4.11)$$

where $V_{\text{line},i}$ is the volume of line segment i , and $m_{\text{pulley},j}$ is the mass of pulley j . In this way, both the bare line material and the pulley masses are included in the bridle system term.

Although many of these components can be derived from the SurfplanAdapter geometry files, several additional reinforcement elements contribute to the total mass and are not explicitly included in the exported geometry. These components were identified from layout images of the TU Delft V3 kite and discussions with experts. An overview of the included reinforcement elements is given below:

1. **Node patches.** These are applied at every bridle line attachment point to reinforce the local load-carrying region. They consist of multiple elements, including additional fabric layers and the internal attachment pigtail.
2. **Reinforcement triangles.** As shown in Figure 4.2, each canopy panel located between two struts contains a Dacron reinforcement triangle at the trailing edge. These reinforcements strengthen the weakest part of the canopy. Similar, but smaller, reinforcement triangles are also placed at the trailing edge near each strut.
3. **Trailing edge strip.** A reinforcement strip is applied along the trailing edge to locally increase the stiffness and strength of this region, which is particularly susceptible to deformation and wear. This strip is made of the same material as the canopy, as indicated by its white appearance, and provides additional support while maintaining material consistency in the trailing-edge construction.
4. **Profile strips.** Stiff laminate strips are applied at the chordwise canopy seams directly behind the leading edge tube, as shown in Figure 4.2. These reinforcements help maintain a smooth transition between the leading edge tube and the canopy, reduce local wrinkling, and preserve the intended airfoil profile in this aerodynamically sensitive region. At the strut junctions, the reinforcements are larger and also contribute to local load transfer and support for bridle attachment. In contrast, the smaller strips between the struts mainly act as profile-maintaining inserts.
5. **Tube seams.** Both the leading edge and the struts contain seams along their full length. These seams add mass because extra material is required for overlap, stitching, and load transfer. The leading edge seam is made of Dacron, while the strut seam is made of canopy material.



Figure 4.2: TU Delft V3 kite on the ground, showing the reinforcement applied on the canopy and tubes, adapted from [77].

4.2.1. Total mass overview

With all components defined, the total mass is estimated from the geometry-based component model described above. Table 4.2 presents the resulting mass of each component together with its contribution to the total kite mass. The relative distribution of the mass over the main components is additionally visualised in Figure 4.3.

The specific masses reported in Table 4.2 are obtained by dividing the component mass by the reference flat kite area of 25 m^2 . These values provide a useful normalised measure of the relative importance of the different components and allow a more direct comparison of mass contributions. In this way, the specific mass not only shows which component contributes most in absolute terms, but also how much mass is associated with a unit reference area. These values are therefore primarily intended as comparative quantities for interpretation and later scaling, rather than as precise local design quantities.

Table 4.2: Overview of the estimated component masses of the TU Delft V3 kite. The specific mass is obtained by dividing the component mass by the reference flat area of 25 m^2 .

Component	Mass [kg]	Share [%]	Specific mass [kg m^{-2}]
Canopy	1.71	16.1	0.068
LE tube	2.15	20.3	0.086
Struts	1.97	18.6	0.079
Bridle system	1.55	14.6	0.062
Reinforcements	2.03	19.1	0.081
Additional	1.19	11.2	0.048
<i>Total</i>	10.60	100.0	0.424

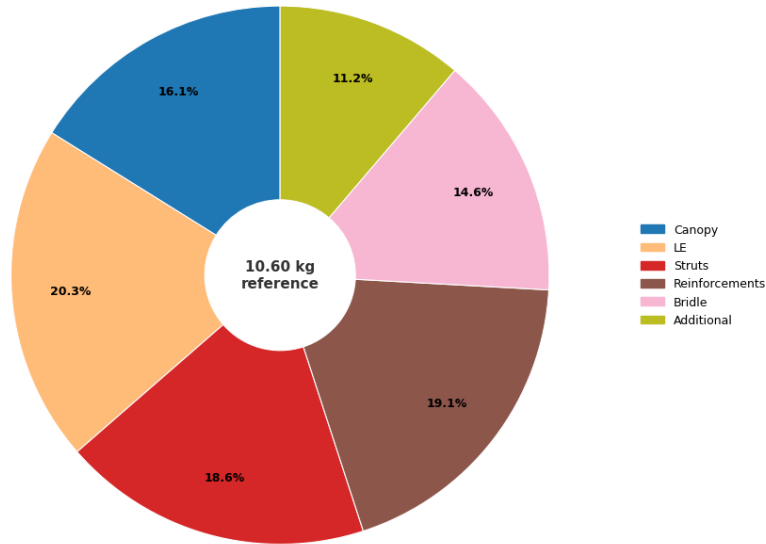


Figure 4.3: Estimated mass distribution of the TU Delft V3 kite over the main component groups, based on the geometry-based low fidelity mass model. The values indicate the relative contribution of each component to the total reference mass of 10.6 kg.

This bottom-up approach should be interpreted as a low-fidelity estimation method intended to provide insight into the relative importance of the different components, rather than as a highly accurate prediction tool. Although most geometric quantities can be derived directly from the Surfplan files, several reinforcement and additional components are not explicitly included in the exported geometry and therefore still require simplified assumptions. The additional mass term represents components that do contribute to the total system mass, but are not modelled separately in the geometric build-up, such as valves, pulley hardware, line attachment hardware, local fittings, and other small manufactured details. To account in a first-order manner for further mass contributions that are not included explicitly, such as coatings, stitching, overlaps, and similar production-related details, an additional correction of 20 g m^{-2} is added to all fabric areal densities.

It can be observed that the leading-edge tube, reinforcements, and struts constitute the dominant mass contributions. While the additional mass term and the bridle system still account for a substantial fraction of the total mass.

4.3. LEI mass scaling

The purpose of this section is to derive physically motivated scaling relations for the mass of a soft-wing LEI kite. The model is a low-fidelity, component-based framework that relates wing mass to wing area for use in techno-economic analysis. It is not a structural sizing method for a specific large-scale kite design. The aim is to identify how the dominant structural components are expected to scale under simplified but physically meaningful assumptions.

A complete structural scaling analysis of an LEI kite would require the aerodynamic pressure distribution, aeroelastic deformation, internal tube moments, membrane stresses, bridle force distribution, seam loads, and reinforcement loads to be recomputed for each scaled geometry [48]. Since such a validated large-scale basis is not yet available in the public literature, the present approach uses representative loads and geometric similarity. The resulting relations define a plausible mass envelope rather than exact design equations.

The mass scaling model is built from component-level structural arguments. First, the geometric scaling assumptions and the resulting scaling of aerodynamic forces are introduced. Next, the structural response of the canopy, tubes, and bridle support system is considered. These relations are then translated into component-specific mass scaling laws, from which lower and upper bound trends for the total LEI wing mass are constructed.

4.3.1. Scaling assumptions and force scaling

A single one-dimensional scale factor k is introduced to describe geometric similarity with respect to the reference kite. Under this assumption, all characteristic lengths of the reference design scale linearly with k , including the wingspan, chord length, tube radius, and tube segment length. Surface areas therefore scale with k^2 . This geometric scaling concept is illustrated in Figure 4.4, which shows a representative inflatable tube segment and its attached canopy section before and after scaling.

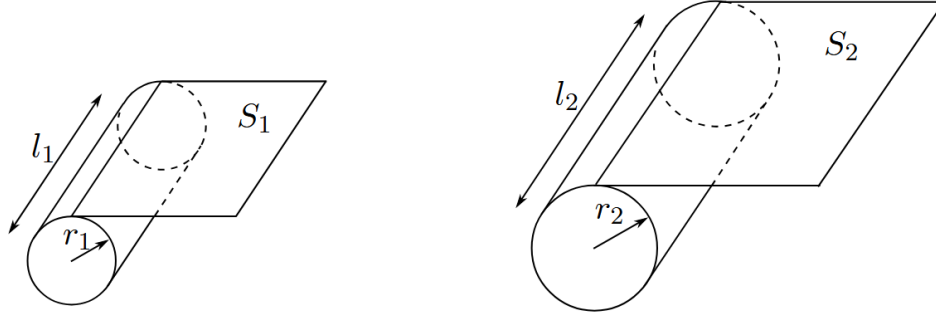


Figure 4.4: Geometric scaling of an inflatable tube segment with an attached canopy section. The reference configuration (1) is shown on the left and the scaled configuration (2) on the right. All characteristic lengths scale with k , while the canopy surface area scales with k^2 [66].

The corresponding geometric scaling relations are given by

$$l_2 = l_1 k, \quad r_2 = r_1 k, \quad S_2 = S_1 k^2, \quad (4.12)$$

where subscript 1 refers to the reference kite, subscript 2 to the scaled kite, l is a characteristic length, r is a tube radius, and S is the flat wing area. The flat area is used here rather than the projected area, since the constituent materials and manufacturing effort scale with the actual material area of the wing.

The derivations in this section are based on the following modelling assumptions:

1. Geometric similarity according to Equation (4.12);
2. A representative maximum design wing loading that is taken as constant across scale;
3. Constant aerodynamic coefficients C_L and C_D ;
4. Similar stress distributions, such that only the stress magnitudes change with scale;
5. Static load scaling, so dynamic, gust, and fatigue effects are not resolved explicitly;
6. Unchanged material classes unless stated otherwise.

These assumptions define the scope of the present low-fidelity model. They are revisited later in sub-section 4.3.8, where the corresponding limitations are discussed in more detail.

Under these assumptions, both lift and drag scale with the flat wing area. The lift force is given by

$$L = \frac{1}{2} \rho V^2 S C_L, \quad (4.13)$$

and the drag force by

$$D = \frac{1}{2} \rho V^2 S C_D. \quad (4.14)$$

If the representative design wing loading, C_L , and C_D are assumed constant across scale, then both lift and drag are directly proportional to the wing area. Substituting Equation (4.12) into Equations (4.13) and (4.14) gives

$$L_2 = L_1 k^2, \quad D_2 = D_1 k^2. \quad (4.15)$$

The total aerodynamic resultant force, therefore, also scales with k^2 :

$$F_2 = F_1 k^2. \quad (4.16)$$

Two different distributed load measures are used in the following derivations. The first is the distributed aerodynamic line load q_α in N m^{-1} , which represents the aerodynamic force per unit length acting on a tube section or panel boundary. Since the total force scales with k^2 and the relevant length scales with k , the line load scales as

$$q_{\alpha,2} = q_{\alpha,1} k. \quad (4.17)$$

The second is the aerodynamic pressure difference p_n in Pa, acting normal to a canopy panel. Under the present assumptions, this pressure remains approximately scale independent. The two quantities are related through a characteristic tributary width $\ell_c \propto k$, such that

$$q_\alpha \sim p_n \ell_c. \quad (4.18)$$

This distinction is important for the structural arguments developed later in this section. The line load q_α governs tube bending and bridle support spacing, while the normal pressure p_n governs canopy membrane tension. The main conclusion of the present subsection is therefore that, under geometric similarity and constant representative wing loading, the total aerodynamic force scales with area, the distributed line load scales with the characteristic length, and the canopy pressure remains approximately constant. These relations form the basis for the component-specific mass scaling laws derived in the following subsections.

4.3.2. Canopy membrane tension

The canopy is modelled as a single-layer membrane under tension and subjected to aerodynamic loading. For the present scaling analysis, the dominant quantity of interest is the membrane stress in the canopy fabric, since this determines whether the same fabric class can be retained as the kite is scaled up. The canopy is therefore treated as a tension-carrying sheet of thickness t , subjected to a representative transverse aerodynamic load.

Following the panel relation used by Van den Heuvel [66], the membrane stress can be written as

$$\sigma = \frac{F}{t l}, \quad (4.19)$$

where F is the representative aerodynamic force acting on the panel, t is the canopy thickness, and l is a characteristic panel length. Under geometric similarity, the characteristic length scales with the geometric scale factor k , while the representative aerodynamic force scales with k^2 , according to Equation (4.16). Substituting these relations into Equation (4.19) gives

$$\sigma_2 = \frac{F_1 k^2}{t l_1 k} = \sigma_1 k. \quad (4.20)$$

Equation (4.20) shows that if the same canopy fabric thickness is retained, the membrane stress increases linearly with the geometric scale factor k . The canopy, therefore, becomes progressively more highly loaded as the kite is scaled up, even when geometric similarity and constant representative wing loading are assumed.

If, instead, the stress level is held constant by increasing the fabric thickness proportionally with k . The resulting areal density therefore scales with k , so that the canopy mass scales as

$$m_{\text{canopy},2} = m_{\text{canopy},1} k^3 \quad \Longleftrightarrow \quad m_{\text{canopy}} \propto S^{3/2}. \quad (4.21)$$

In practice, commercially available fabrics exist only in discrete grades, so a real design will respond through a combination of fabric selection, localised reinforcement, and support topology changes rather than through continuous thickness scaling.[32, 34].

4.3.3. Tube membrane stress and pressure scaling

The leading edge tube and the struts are modelled as thin-walled, pressurised cylindrical shells. For these inflatable structural elements, the governing quantities are the membrane stresses generated by the internal overpressure, since these determine whether the same tube material and seam construction can be retained as the kite is scaled up. The tube scaling problem, therefore, differs from the canopy scaling problem: for the canopy, the critical quantity is membrane tension under aerodynamic loading, whereas for the tubes, the relevant starting point is the balance between internal pressure, tube radius, and wall stress.

Under the thin-walled pressure vessel assumption, the circumferential and longitudinal membrane stresses can be determined by the membrane approach described by Veldman. The equations are given by

$$\sigma_C = \frac{pr}{t}, \quad \sigma_L = \frac{pr}{2t}, \quad (4.22)$$

where σ_C is the circumferential, or hoop, stress, σ_L is the longitudinal stress, p is the internal overpressure, r is the tube radius, and t is the wall thickness. The thin-walled tube idealisation used in the present derivation is illustrated in Figure 4.5.

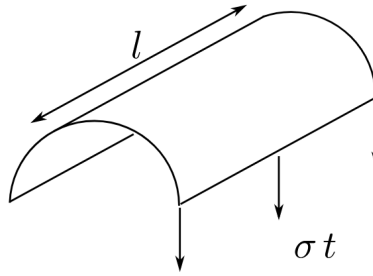


Figure 4.5: Schematic representation of the thin-walled tube idealisation used for the inflatable structural elements. The tube is modelled as a cylindrical shell of characteristic length l and wall thickness t .

In the lower-bound scaling regime, the same tube material and seam construction are retained. Following the design logic of Van den Heuvel [66], the product pr is then taken as the governing structural quantity, since keeping pr constant at constant wall thickness implies that the membrane stresses in Equation (4.22) also remain constant. Under geometric similarity, the tube radius scales with the one-dimensional scale factor k , so that

$$r_2 = r_1 k. \quad (4.23)$$

If the product pr is required to remain constant, it follows that the allowable internal pressure must decrease as the tube radius increases:

$$pr = \text{const} \quad \Longrightarrow \quad p_2 = \frac{p_1}{k}. \quad (4.24)$$

Equation (4.24) shows that, under geometric similarity and unchanged tube material, the inflation pressure cannot remain constant as the kite is scaled up. Instead, the required pressure level decreases inversely with the geometric scale factor to maintain a similar membrane stress level in the tube wall.

Equation (4.24) should be interpreted as an optimistic structural idealisation. Maintaining constant pr preserves the membrane stress in the tube wall. Still, it does not guarantee that the inflatable structure remains sufficiently stiff or collapse-resistant at a larger scale, since the internal pre-stress also decreases. Larger kites may therefore require higher-performing materials, modified seam constructions, or additional structural support. These adaptations motivate the bridle- and strut-support criteria developed in the following subsections.

4.3.4. Leading-edge collapse and bridle-point count

A reduction in inflation pressure decreases the bending resistance of the inflatable leading edge. The spacing of the bridle attachment points, therefore, becomes an important design parameter, since these points provide discrete support to the leading-edge tube and limit the unsupported span over which bending can develop.

The leading edge between two neighbouring bridle attachment points is modelled as a supported inflatable beam segment of length P_{bridle} , subjected to a uniformly distributed aerodynamic line load q_α . The corresponding support mechanism and idealised local force balance are illustrated in Figure 4.6.

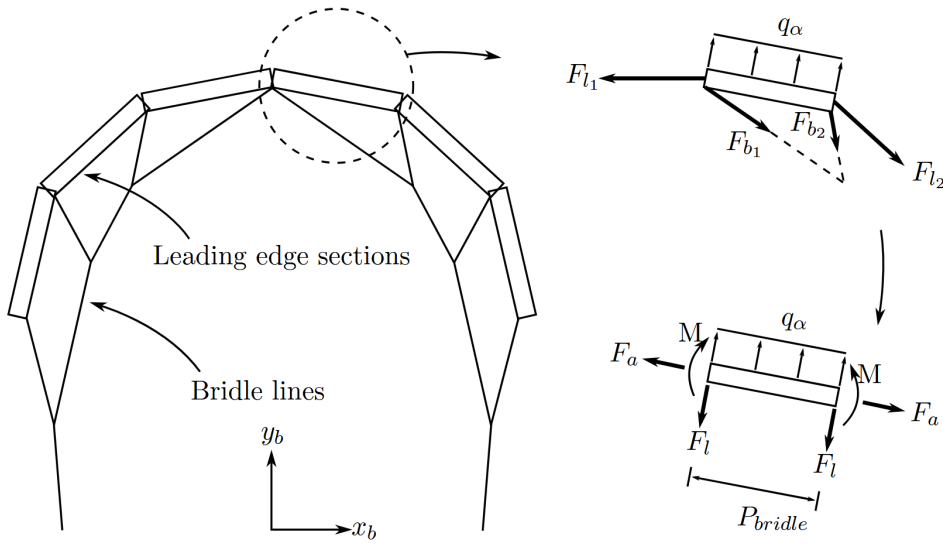


Figure 4.6: Schematic representation of the leading-edge bridle support system and the corresponding local force balance. The left-hand side shows the leading-edge sections and attached bridle lines; the right-hand side illustrates the distributed aerodynamic loading and the equivalent support forces and bending moments acting on a segment of length P_{bridle} . Adapted from [66].

For a supported segment under uniform line loading, the maximum bending moment at mid-span is

$$M_A = \frac{1}{8} q_\alpha P_{\text{bridle}}^2. \quad (4.25)$$

Following Van den Heuvel [66], wrinkling onset is taken to occur when the longitudinal bending stress becomes comparable to the longitudinal inflation stress. Combining Equation 4.25 with the thin-walled circular-section approximation gives the characteristic bridle-pitch relation

$$P_{\text{bridle}} = \sqrt{\frac{8\pi pr^3}{q_\alpha}}, \quad (4.26)$$

where p is the internal overpressure, r is the local tube radius, and q_α is the distributed aerodynamic line load.

Under the geometric scaling assumptions of subsection 4.3.1, $r_2 = r_1 k$, $p_2 = p_1/k$, and $q_{\alpha,2} = q_{\alpha,1} k$. Substitution into Equation 4.26 then gives

$$P_{\text{bridle},2} = P_{\text{bridle},1} \sqrt{k}. \quad (4.27)$$

Since the kite span scales linearly with k , while the allowable bridle pitch increases only with $\sqrt{k}$, the required number of bridle support points increases with scale:

$$\frac{N_{\text{bridle},2}}{N_{\text{bridle},1}} = \frac{b_2/P_{\text{bridle},2}}{b_1/P_{\text{bridle},1}} = \sqrt{k}. \quad (4.28)$$

Introducing a collapse-resistance factor a_c , representing a desired increase in allowable aerodynamic load before collapse relative to the reference design, reduces the allowable pitch by a factor $1/\sqrt{a_c}$, giving

$$P_{\text{bridle},2} = P_{\text{bridle},1} \sqrt{\frac{k}{a_c}}, \quad \frac{N_{\text{bridle},2}}{N_{\text{bridle},1}} = \sqrt{ka_c}. \quad (4.29)$$

The bridle system is both a geometric and a load-carrying component and therefore cannot be scaled from total line length alone. The characteristic length of each bridle branch scales with k , while the total aerodynamic load scales with flat area as $F_2/F_1 = k^2$. When this load is distributed over $\sqrt{ka_c}$ times as many branches, the average load per branch scales approximately as

$$\frac{F_{\text{branch},2}}{F_{\text{branch},1}} \approx \frac{k^2}{\sqrt{ka_c}}. \quad (4.30)$$

Assuming that line-specific mass scales with the required line strength, the total bridle mass scales with the number of branches, the characteristic line length, and the load per branch:

$$\frac{m_{\text{bridle},2}}{m_{\text{bridle},1}} \approx \sqrt{ka_c} \cdot k \cdot \frac{k^2}{\sqrt{ka_c}} = k^3. \quad (4.31)$$

The bridle system therefore scales with k^3 , making it an increasingly significant contributor to the total wing mass at larger scales.

4.3.5. Strut spacing

As with leading-edge bridle spacing, the longitudinal spacing of the struts defines a structural support criterion that, in general, cannot be preserved under strict geometric similarity. Between two neighbouring struts, the canopy behaves as an unsupported membrane panel subjected to aerodynamic normal pressure. As the kite is scaled up, the spanwise width of this panel increases, which leads to higher membrane stresses, increased billow, and a reduced ability to maintain the intended aerofoil geometry.

In the present model, the canopy segment between two adjacent struts is idealised as a membrane panel with unsupported spanwise width w , subjected to a representative aerodynamic normal pressure p_n . Based on the Young–Laplace-type panel relation used by Verheul [34], the membrane stress is approximated by

$$\sigma_{\text{panel}} = \frac{p_n w}{2t}, \quad (4.32)$$

where p_n is the aerodynamic pressure normal to the canopy, w is the unsupported spanwise width, and t is the canopy thickness. Equation (4.32) replaces the distributed panel curvature by a single representative unsupported width.

Under the scaling assumptions of subsection 4.3.1, p_n remains scale-independent, while w scales with k . If the number of struts is kept constant,

$$w_2 = w_1 k, \quad (4.33)$$

and the membrane stress therefore increases linearly with k when canopy thickness is held constant.

If, instead, the unsupported canopy width is kept approximately constant in order to preserve a comparable stress level and degree of billowing, the number of strut intervals must increase in proportion to the span:

$$N_{\text{strut,int},2} = N_{\text{strut,int},1} k. \quad (4.34)$$

Equation (4.34) should be interpreted as a canopy-support criterion rather than as a validated design rule. In practice, a real design may respond through a combination of additional struts, changes in fabric grade, added profile reinforcements, or acceptance of increased local deformation. The present relation, therefore, represents a conservative support strategy rather than a unique design solution.

The mass implication follows directly. Each strut has a surface area proportional to radius and chord, with both scaling linearly with k . The mass of an individual strut therefore scales with k^2 . If the number of struts also scales with k , the total strut mass scales as

$$m_{\text{struts},2} = m_{\text{struts},1} k^3. \quad (4.35)$$

This introduces a clear structural trade-off: additional struts reduce canopy-panel loading, but shift the associated scaling penalty into the strut mass. This trade-off is one of the main contributors to the uncertainty in the LEI wing mass envelope developed later in this chapter.

4.3.6. Component mass scaling relations

The component-level scaling arguments developed in subsection 4.3.2–subsection 4.3.5 are collected here. Each component is scaled from its reference mass according to

$$m_{i,2} = m_{i,1} \left(\frac{S_2}{S_1} \right)^{b_i}, \quad (4.36)$$

where b_i is the adopted scaling exponent for component i . The adopted exponents and their physical basis are summarised in Table 4.3. For the reinforcement group, the exponent spans a range because this group contains both area-related subcomponents, such as seams and strips, and load-transfer-related subcomponents, such as strut-root reinforcements and attachment patches.

Table 4.3: Adopted scaling exponents for the component-based LEI mass model.

Component group	b_i	Physical basis
Canopy	1	Constant fabric class; mass scales proportionally with surface area.
Leading edge	1	Tube and bladder mass scale with surface area under the constant pr assumption.
Struts	$3/2$	Strut count increases with span, while each strut also scales geometrically.
Bridle system	$3/2$	Additional branches, greater line length, and proportionally stronger lines are required with increasing scale.
Reinforcements	$1-3/2$	Area-related reinforcements scale with $b = 1$, whereas load-transfer reinforcements scale with $b = 3/2$.
Additional	1	Lumped unresolved contributions, scaled with flat wing area.

4.3.7. Baseline curve and mass uncertainty envelope

Collecting the component-level relations from subsection 4.3.6, the mass scaling of the LEI wing is represented by a baseline scaling curve together with two bounding relations. The baseline curve is used as the nominal mass relation in the techno-economic framework, while the lower and upper bounds are retained as sensitivity cases to quantify the influence of uncertainty in the mass-scaling assumption on the resulting LCoE trends.

The lower bound corresponds to the most optimistic scaling case. It assumes that the reference material classes and support topology remain adequate, such that the dominant structural components scale primarily through geometric similarity. Under this assumption, the specific mass remains constant and the wing mass scales linearly with flat wing area,

$$m \propto S \quad \Longleftrightarrow \quad m \propto k^2. \quad (4.37)$$

This relation is consistent with the geometric-similarity scaling arguments of Van den Heuvel [66]. For the inflated tube components, it is associated with the constant pr assumption introduced earlier: as the tube radius increases with scale, the inflation pressure is reduced such that the membrane stress in the tube wall remains unchanged. This preserves membrane stress, but does not guarantee sufficient practical stiffness or collapse resistance at larger scale. The lower bound should therefore be interpreted as an optimistic area-linear extrapolation of the reference design, rather than as a structurally conservative design rule.

The upper bound represents the opposite limiting case. It assumes that the increasing structural demand with scale must be accommodated by an increase in effective material thickness, effective areal density, additional structural support, or a combination of these effects. If the required tensile or membrane-carrying capacity increases with the geometric scale factor k , while the structural area increases with k^2 , the corresponding mass scales with k^3 . Since $S \propto k^2$, this gives

$$m \propto S^{3/2} \quad \Longleftrightarrow \quad m \propto k^3. \quad (4.38)$$

This $S^{3/2}$ scaling represents a conservative stress-similarity-type response, in which the structure becomes progressively heavier as the kite is scaled up. Similar behaviour is reported in the LEI scaling work of Verheul, where direct structural extrapolation leads to rapidly increasing wing mass at larger scale [34]. In the present work, this relation is therefore used as an upper sensitivity bound rather than as the nominal mass prediction.

Together, these two relations define the LEI mass uncertainty envelope. For a given flat wing area, the wing mass can be written in the generic power-law form

$$m(S) = m_{\text{wing},0} \left(\frac{S}{S_0} \right)^b \quad (4.39)$$

where $m_{\text{wing},0}$ is the reference wing mass at S_0 , and the exponent b lies between 1 and $3/2$. The lower bound is recovered for $b = 1$, while the upper bound is recovered for $b = 3/2$.

A baseline curve is then introduced within this envelope using the geometry-based mass decomposition of the TU Delft V3 kite. The reference wing mass is decomposed into canopy, leading-edge tube, struts, bridle system, reinforcements, and additional hardware. Each component group is assigned a simplified scaling relation based on the physical arguments developed in the previous subsections. Components dominated primarily by material area are scaled closer to the lower bound, whereas components associated with structural support, load transfer, and local reinforcement are allowed to increase more strongly with scale.

The purpose of this component-based calculation is not to define the exact structural layout of a future large LEI kite. In a real design, the number of struts, bridle attachment points, reinforcement layout, tube dimensions, and material selection would be determined jointly as part of a detailed aero-structural design process. Such a process is beyond the scope of this thesis. Instead, the component-based calculation is used to obtain a physically informed intermediate trend that remains anchored to the V3 reference design while accounting for the expected non-linear increase in structural mass.

The component-based totals are evaluated at a sequence of doubled flat wing areas, starting from the reference V3 kite with $S_0 = 25 \text{ m}^2$. This sequence is used because each step corresponds to a constant increase in area ratio, while the corresponding geometric scale factor follows directly from $k = \sqrt{S/S_0}$. It therefore provides a transparent way to evaluate how the mass trend evolves as the kite is progressively scaled. The largest point, $S = 400 \text{ m}^2$, corresponds to an area increase by a factor of 16 and defines the upper end of the LEI range considered in the present framework.

Table 4.4: Component-based LEI mass totals used to fit the baseline scaling curve.

Flat area S [m ²]	Scale factor k	Component-based mass [kg]	Cumulative b_{eff}
25	1.00	10.60	—
50	1.41	23.97	1.18
100	2.00	55.79	1.20
200	2.83	133.75	1.22
400	4.00	330.21	1.24

The final column gives the cumulative exponent between the V3 reference point and each scaled evaluation point,

$$b_{\text{eff}} = \frac{\ln(m_i/m_0)}{\ln(S_i/S_0)} \quad (4.40)$$

The increase in b_{eff} with scale reflects the growing contribution of components that scale more strongly than area, such as struts, bridle lines, and structural reinforcements. Since the effective exponent is not constant over the full range, a single representative exponent is obtained by fitting a power law to the component-based totals in logarithmic space. The exponent is found by minimising the sum of squared logarithmic residuals,

$$b = \frac{\sum_i \ln\left(\frac{S_i}{S_0}\right) \ln\left(\frac{m_i}{m_0}\right)}{\sum_i \left[\ln\left(\frac{S_i}{S_0}\right)\right]^2} = 1.22 \quad (4.41)$$

The baseline LEI mass scaling relation used in the techno-economic framework is therefore

$$m_{\text{LEI}}(S) = 10.60 \left(\frac{S}{25}\right)^{1.22} \quad (4.42)$$

The lower and upper bounds are calibrated to the same TU Delft V3 reference kite at $S_0 = 25 \text{ m}^2$. The resulting envelope is intentionally broad, particularly at larger wing areas. This is a direct consequence of extrapolating from a relatively small reference kite to areas up to 400 m^2 , corresponding to a factor 16 increase in flat area. If reliable mass and component data for larger LEI kites were available, the same methodology could be recalibrated around a larger reference design, thereby reducing the extrapolation distance and narrowing the uncertainty envelope.

The adopted baseline relation and the corresponding sensitivity bounds are shown in Figure 4.7. The baseline curve represents the component-based scaling relation used in the framework, while the lower and upper curves define the sensitivity bounds. The widening envelope illustrates the increasing uncertainty associated with extrapolating from the 25 m^2 V3 reference kite to larger wing areas.

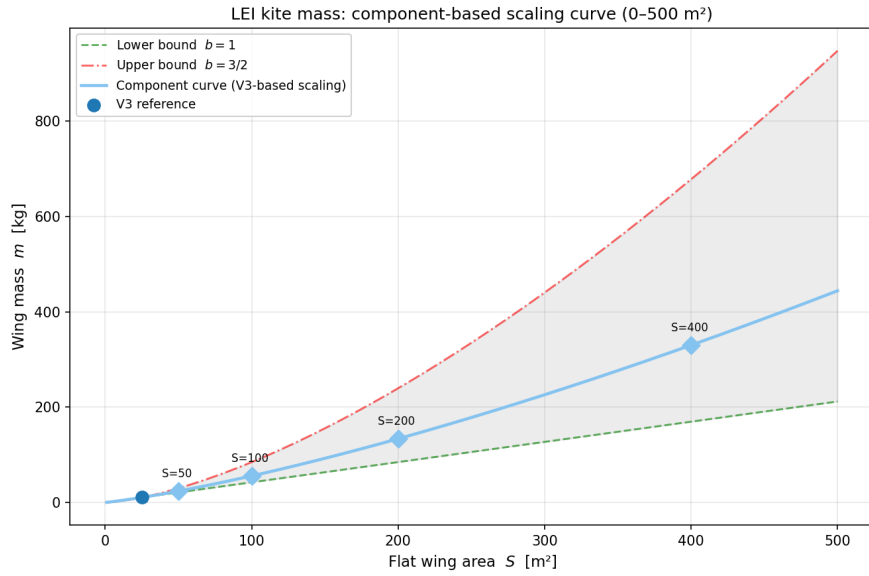


Figure 4.7: Adopted LEI mass scaling relation and uncertainty bounds. The baseline curve represents the component-based scaling relation with $b = 1.22$, used in the techno-economic framework. The dashed curves define the lower bound with $b = 1$ and the upper bound with $b = 3/2$. The widening envelope illustrates the increasing uncertainty associated with extrapolating from the 25 m^2 V3 reference kite to larger wing areas.

In the remainder of the framework, Equation 4.42 is used as the nominal LEI mass scaling relation. The lower and upper bounds are propagated separately as sensitivity cases. This allows the influence of mass-scaling uncertainty on LCoE to be assessed without treating either bounding relation as the most likely outcome for a large-scale design.

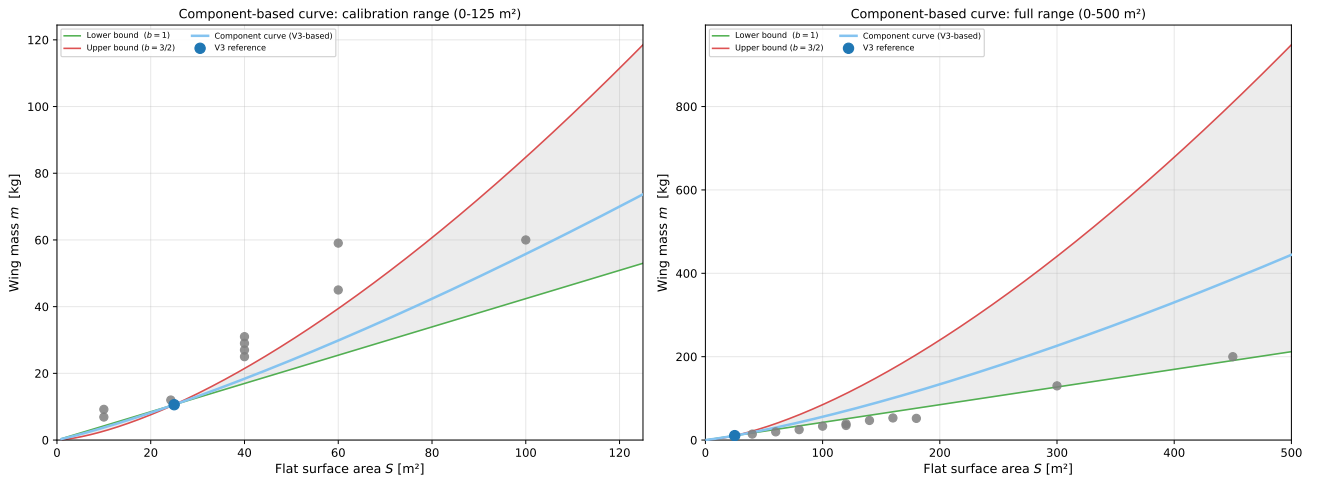


Figure 4.8: Mass-scaling curves are shown with anonymised reference data points, enabling a quantitative assessment of the fit of the underlying scaling relationships. The grey markers represent a heterogeneous set of soft-wing kite architectures: LEI kites, ram-air kites, and single-skin kites. Plotting these kite types on the same mass–size diagram shows how well empirical data from different design families match the derived scaling laws, indicating the robustness and generality of the fitted curves across a wide range of structural concepts and construction techniques.

4.3.8. Assumptions and limitations

The present model is intended as a low-fidelity framework for estimating mass-scaling trends, rather than as a structural sizing tool for a specific large-scale kite design. Its scope is therefore limited by the assumptions introduced in subsection 4.3.1.

Only static structural load scaling is considered. Dynamic effects, including gust loading, cyclic fatigue,

transient deformation, local seam degradation, and aeroelastic coupling, are not resolved. These effects may become more important at larger scale, but the publicly available literature does not yet provide sufficient validated data to include them consistently at the present level of fidelity.

Stress distributions are assumed to remain similar across scale, such that the dominant load paths are preserved and only the stress magnitudes change. Local stress concentrations at seams, bridle attachments, profile reinforcements, and trailing-edge terminations are not resolved explicitly. Their contribution to mass is represented only approximately through the reinforcement group and the width of the uncertainty envelope.

Material selection is treated only at first order. The model does not predict discrete redesign choices in fabric grade, seam type, or reinforcement layout. Where such changes would become necessary, their effect is represented indirectly through the widening gap between the lower and upper bounds, rather than through explicit structural redesign.

The model is calibrated to a single reference design, namely the TU Delft V3 kite at $S_0 = 25 \text{ m}^2$. The resulting mass curves should therefore be interpreted as reference-based trend lines rather than as universal LEI scaling laws. Their primary value lies in comparing scaling assumptions and quantifying sensitivity within the techno-economic framework, not in predicting the detailed structural layout of a future large-scale kite.

4.4. KCU scaling

The purpose of this section is to derive a KCU mass scaling relation consistent with the component-based approach used for the wing. The KCU is scaled separately because its components do not share a single dependence on kite size: the control electronics are set by the flight task, the onboard power supply by the power demand, the housing by the unit envelope, and the actuation drivetrain by the steering loads. Rather than compress this into a single fixed-plus-variable split, the KCU mass is built up from four functional groups, each assigned the scaling that follows from its role. Physically, a controller, motors, and a pulley system pull on the steering tapes, while onboard sensors, electronics, batteries, and an airborne wind turbine support flight control; the unit is mounted near the kite and protected by foam padding, since it lands with the airborne system each cycle.

The mass shares of these groups are taken from the soft-wing life cycle inventory of Coutinho [6] and listed in Table 4.5. The division into control, power supply, protection, and steering subsystems follows the KCU product tree of Braun [78] and the controller design of Milandri [79], while the protective foam is identified in the same inventory. The shares are therefore not all independently validated, but the grouping gives a transparent, component-level basis for the scaling rather than an assumed fixed fraction.

Table 4.5: Component mass breakdown of a 60 m^2 -class reference KCU ($m_{\text{KCU},60} = 26.2 \text{ kg}$), adapted from the soft-wing life cycle inventory of Coutinho [6] and grouped into the scaling classes used in the mass model. The reported shares sum to 99%; the 1% rounding residual is left unassigned.

Group	Scaling	Constituent	Mass share	Group weight
Electronics	S^0	Electronic boards	1%	$w_{\text{fix}} = 0.01$
Power supply	S^1	Airborne wind turbine	8%	
Housing/protection	S^1	Batteries	5%	$w_{\text{pow}} = 0.13$
		Cover foam	33%	$w_{\text{hous}} = 0.40$
		Casing and protection	7%	
Mechanical	$S^{1.5}$	Tape guides	23%	$w_{\text{mech}} = 0.45$
		Pulleys and drivetrain	15%	
		Mounting structure	7%	

The control electronics measure, compute, and communicate the flight state; a single set suffices regardless of wing area, so this group is treated as size-independent, consistent with reference economic models that treat avionics as fixed and with the observation by Grete [32] that the electronics do not

scale with the mechanical parts. The onboard power supply, the batteries, and the airborne wind turbine are scaled with the average power the KCU must supply rather than with the wing area directly; assuming this demand grows in proportion to kite size, the group is scaled linearly, S^1 . The casing, protection, and cover foam follow the growing envelope of the unit rather than the drivetrain loads and are likewise scaled as S^1 .

The mechanical group is the only one whose scaling requires a derivation, since it is sized by the steering loads and the power needed to actuate the kite. Under similar operating conditions, the steering force follows the tether force, which scales with wing area as established for the wing loads in subsection 4.3.1,

$$F_{\text{steer}} \propto F_t \propto S, \quad (4.43)$$

while the steering displacement, and hence the tape velocity, grows with the characteristic kite length, which scales as $S^{0.5}$ under the geometric similarity of subsection 4.3.1. The actuation power therefore scales as

$$P_{\text{act}} \sim F_{\text{steer}} v_{\text{tape}} \propto S \times S^{0.5} = S^{1.5}, \quad (4.44)$$

and the mechanical group is scaled accordingly. The exponent captures the combined growth of steering force, actuation speed, thermal loading, and drivetrain integration, rather than a single drum-torque relation.

Combining the two linearly scaling groups, the model reduces to a fixed, a linear, and a super-linear term,

$$m_{\text{KCU}}(S) = m_{\text{KCU,ref}} \left[w_{\text{fix}} + w_{\text{lin}} \left(\frac{S}{S_{\text{ref}}} \right) + w_{\text{sup}} \left(\frac{S}{S_{\text{ref}}} \right)^{1.5} \right], \quad (4.45)$$

where $S_{\text{ref}} = 60 \text{ m}^2$, $m_{\text{KCU,ref}} = 26.2 \text{ kg}$, $w_{\text{fix}} = 0.01$, $w_{\text{lin}} = w_{\text{pow}} + w_{\text{hous}} = 0.53$, and $w_{\text{sup}} = w_{\text{mech}} = 0.45$ (Table 4.5). Only about 1% of the KCU mass is therefore truly fixed with size, the remainder scaling with area, most of it linearly and a smaller share super-linearly through the actuation drivetrain.

Evaluated at the TU Delft V3 kite ($S = 25 \text{ m}^2$), Equation 4.45 returns 9.2 kg against a reported 8.4 kg [45], about 10% high. As the model is calibrated only at the 60 m^2 unit, this is an independent check rather than a fit, and the agreement indicates that resolving the KCU into functionally distinct groups describes its mass more consistently than a single area or square-root exponent applied to the whole unit. The relation lies close to a linear extrapolation at small scale and rises above it as the mechanical group comes to dominate.

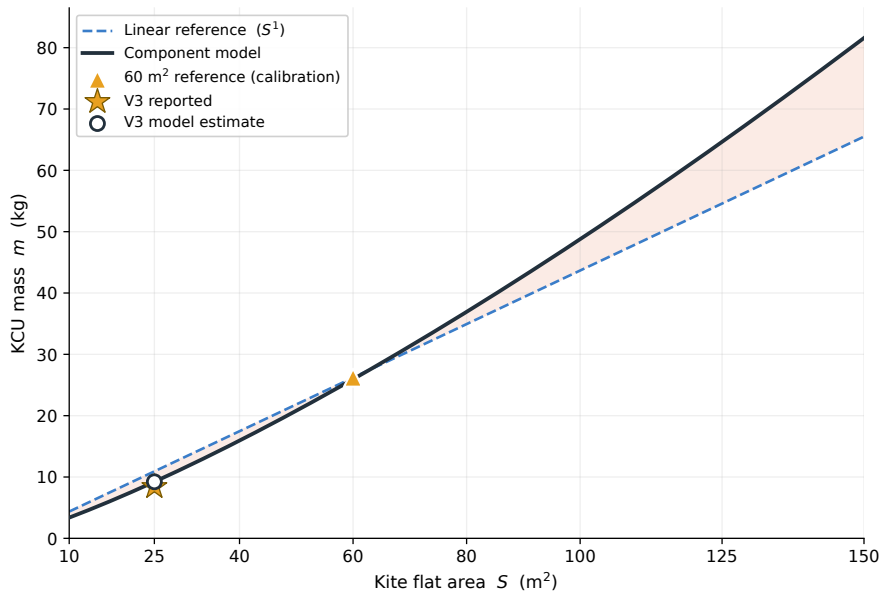


Figure 4.9: Component-based KCU mass versus flat area (solid), compared with a purely linear reference (dashed). The relation is calibrated at the 60 m^2 reference unit and checked against the independent TU Delft V3 mass; the shaded region marks the super-linear mass contributed by the mechanical group above the reference scale.

The relation should be read as an engineering scaling law rather than a validated design model. The breakdown is based on a single 60 m²-class KCU, and the V3 mass provides only one additional check, so the two reference masses constrain the calibration but not the exponents themselves. The chosen exponents describe the expected scaling of each group, but a future KCU will also depend on design choices such as the steering architecture, redundancy level, battery strategy, impact protection, cooling, and certification requirements. A wider dataset of measured KCU masses and actuator loads would be needed to validate the relation and to confirm whether the mechanical group continues to scale as $S^{1.5}$ at larger sizes.

4.5. Flight-dynamics sensor mass

The airborne flight-dynamics sensor package provides the attitude, position, and apparent-wind measurements required by the autopilot. Because only one measurement of each state is needed, a single set of sensors suffices regardless of kite size, so the sensor mass is treated as constant and independent of the wing area. A much larger kite might eventually carry additional or redundant sensors, but the package is light relative to the wing and the KCU, so its relative contribution decreases with scale, and this effect is neglected. For the V3 kite, the package has a reported mass of 3.4 kg [45], which is adopted as the fixed sensor mass. Together with the wing mass from Equation 4.42 and the KCU mass from Equation 4.45, it completes the total airborne-system mass used in the framework.

Cost, Replacement and LCoE model

This chapter presents the component cost relations, lifetime assumptions, annual operational expenditure, and financial formulation used to translate a physical system design and its simulated performance into LCoE. The formulation provides a consistent economic basis for comparing candidate designs and maturity scenarios, with particular attention to the cost drivers that differ between soft-wing and fixed-wing systems. The scope and structure of EcoMo are first defined in section 5.1. The relations transferred from the reference economic model and those requiring adaptation are subsequently identified in section 5.2. The adapted soft wing cost relations are developed in section 5.3. Component lifetimes, replacement expenditure, operation and maintenance expenditure, availability, and the maturity scenarios are described in section 5.4. Finally, the capital and annual operational expenditure are combined with the energy yield in section 5.5 to calculate LCoE.

5.1. EcoMo scope and structure

EcoMo adopts the reference economic model of Joshi and Trevisi [25] to provide a consistent basis for comparing AWE system costs. Cost inputs were aggregated through an anonymised industry-academic collaboration and calibrated for rated powers between 100 kW and 2 MW. The model allocates expenditure to the kite, tether, ground station, balance of system (BoS), and balance of plant (BoP). Capital, operational, and replacement costs are assessed at the component level within this hierarchy, as outlined in section 3.5. The cost structure is retained in EcoMo and re-implemented in Python to permit direct coupling to the system-sizing and performance models. The smaller systems analysed in this thesis have rated powers below the calibrated range, which increases the uncertainty in their absolute cost estimates. These absolute values are interpreted as indicative, with greater confidence placed on comparative trends and cost mechanisms.

EcoMo receives the physical configuration from the system-sizing model, along with the power curve and annual energy production from AWESPA, within the coupled framework presented in section 3.2. The shared system definition, expressed using the common awesIO data standard [80], ensures that component dimensions, performance, energy yield, and costs correspond to the same candidate design. A change in wing area or allowable tether stress therefore propagates consistently through the sizing, performance, and cost calculations.

EcoMo distinguishes the initial capital cost from annual operating and component-replacement expenditure. Replacement costs are determined from the component costs and their assumed service lives, allowing the recurring expenditure associated with the soft wing and tether to be represented explicitly. These expenditure streams are combined with the financial assumptions and annual energy production to calculate LCoE. The initial capital cost (ICC), annual operating cost, and component-level contributions to LCoE are retained to identify the economic mechanisms underlying the design and scaling trends.

LCoE serves as the optimisation objective because it combines lifetime expenditure and energy production without relying on assumptions about electricity prices or market revenue. This approach suits the conceptual development stage addressed here, where revenue-related assumptions would introduce additional uncertainty. Initial capital cost and component-level contributions to LCoE remain as supporting indicators for interpreting the design and scaling trends.

Within this thesis, only the soft-wing GG model is further developed, and its implementation is verified against the reference MATLAB model. The fixed-wing GG and FG functions are preserved in the codebase for future work. This shared structure is the basis for the subsystem analysis that follows. That analysis identifies which cost relations transfer directly to the soft-wing GG case, and which require adaptation.

5.2. Soft-wing model adaptation

The cost model is designed for early-stage design exploration: its role is to reveal LCoE trends and highlight the most influential design variables, rather than to provide certified, absolute LCoE figures. This purpose dictates which modelling choices must be precise. Errors that shift costs uniformly across the design space—such as inaccuracies in unit prices—scale every design option by the same factor, leaving both the optimal design and the direction of improvement unchanged. In contrast, errors in scaling behaviour or in the way subsystems are coupled do not cancel out. Any relationship that scales incorrectly with a design variable distorts the resulting trends. Consequently, the main effort is focused on capturing the physical scaling and coupling of each cost driver, while unit prices and reference values are treated as calibration parameters.

The reference model decomposes the system into five subsystems: kite, tether, ground station, balance of system, and balance of plant. This decomposition is physically meaningful regardless of wing topology, so it is retained, with the soft-wing performance model supplying the power curve, the annual energy production, and the tether-force envelope that size each subsystem. Figure 5.1 shows the resulting component tree: the five groups and the subcomponents modelled within each, with the colouring distinguishing the relations transferred from the reference model, those that are new or adapted for the soft-wing case, and the components left outside the current scope.

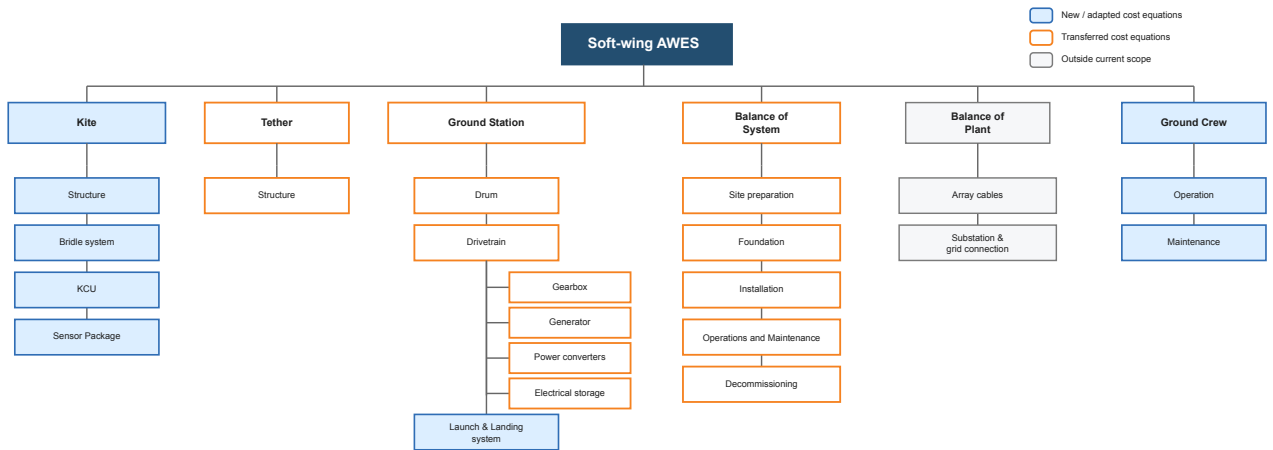


Figure 5.1: Breakdown of cost components of a soft wing AWES used in this report. The structure tree shows the main system at the top, breaking down into the five component groups. Each of these groups is then broken down into its subcomponents. The orange blocks represent the transferred cost equations, and the blue blocks represent the new cost equations. The balance-of-plant group is outside the scope of this thesis.

The transferred relations are taken from the reference economic model of Joshi and Trevisi [25]. That model is a published reference model developed with the AWE industry through an IEA Wind Task 48 collaboration, with cost inputs contributed by manufacturers, suppliers, and research groups across the sector; therefore, its relations represent the physical cost drivers well and are reused here without change. Because the model was published only recently, in 2024, the drift in unit prices since then is assumed to be small enough and enters the LCoE as an absolute offset, which does not affect the trends the framework is built to reveal. The reference model, however, was developed for a fixed-wing system that operates within a higher power and force envelope. Each relation was therefore assessed against the soft-wing ground-generation case to determine whether it transfers directly, requires adaptation, or must be researched a new. The Table 5.1 summarises the outcome.

The tether, ground-station, and balance-of-system relations transfer unchanged, each keeping the power or force basis of its reference equation. The reference model sizes the gearbox, generator, and foundation on the peak reel-out power. The winch drum is sized based on the peak tether force, and the installation, site preparation, and balance-of-system overhead are based on the rated electrical power. Lastly, the power converter is sized on both the rated and peak power. These bases are

Table 5.1: Transfer of the reference economic model of Joshi and Trevisi [25] to the soft-wing ground-generation case. Adapted and new relations are developed in the sections indicated. Transferred relations are reused from the reference model.

Subsystem	Status	Basis
Kite	Adapted / new	LEI component cost relations (section 5.3)
Tether	Transferred	Same cost per mass and fatigue replacement
Ground station	Transferred	Power- and force-based sizing, based on the performance model
Launch and land	New	Mast-assisted structure, scaling with kite area (subsection 5.3.4)
Balance of system	Transferred	Power-based sizing
Balance of plant	Out of scope	Array cables and grid connection not modelled
Ground Crew	New	Operational and maintenance cost added
Finance and LCoE	Transferred	Capital-recovery annualisation, topology-independent

copied, not altered. The soft-wing performance model supplies both the rated and the peak power, so each relation is evaluated with the appropriate soft-wing value, and only the magnitudes differ from the fixed-wing case. The peak reel-out power is therefore used directly where the reference relations call for it, rather than reconstructed from a prescribed crest factor. That crest factor is not available as a free design parameter here because the performance model does not produce a time-resolved power signal. For the same reason, the instantaneous force peaks within the figure-of-eight are not resolved. So the peak values are cycle-level, and the ground-station and tether sizing, and hence their cost, are mildly optimistic.

Within the ground station, a few configuration choices are adopted from the reference model, for reasons that carry over to the soft-wing case. The electrical drivetrain is used rather than a hydraulic or mechanical one. Following Joshi et al. [12], who identify the electrical drivetrain as the most commercially ready and most suitable for market entry among the three options, it is the relevant configuration to assess. Its components are the winch drum, gearbox, generator, power converters, and electrical storage. The winch drum is modelled as a steel structure. Steel is chosen over aluminium for its higher strength and stiffness at a low cost per unit mass. This suits the heavily and repeatedly loaded drum and sets its mass and hence its cost.

For electrical storage, an ultracapacitor is used instead of a battery. The pumping cycle exchanges energy over short intervals, with frequent high-power charge and discharge and no need for long-term storage. An ultracapacitor matches this duty, with fast charge and discharge, high burst-power capability, and a very high cycle life. A battery would be stressed by the rapid cycling and sized by power rather than energy. The same choice is made in the reference model, so the storage relation transfers with the ultracapacitor's energy-priced cost and cycle-driven replacement.

The tether uses the same Dyneema fibre and the same cost-per-mass and fatigue relations as the reference, since these depend on the tension history and the material, not on the wing topology. Its lifetime treatment, in which the soft-wing case differs, is developed in section 5.4.

The launch-and-land system is the one genuinely new ground-side relation. It is a mast-assisted structure whose cost scales with kite area, developed in subsection 5.3.4. It is distinct from the foundation: the foundation is the structural support, sized for the peak power. Whereas the launch-and-land system is the deployment and recovery apparatus itself, costing as a separate line. The reference model already treats the two as separate items, with the launch-and-land apparatus noted as a cost driver of the foundation it stands on, so charging both does not double-count.

5.3. LEI cost relations

Soft-wing kite cost comprises three parts: the wing structure (including the bridle), the KCU, and the flight-dynamics sensor package. Each scales differently and is modelled separately, then summed to give the total kite cost,

$$C_{\text{kite}}(S) = C_{\text{struct}}(S) + C_{\text{KCU}}(S) + C_{\text{sens}}. \quad (5.1)$$

The reference anchors that calibrate these relations derive from confidential manufacturer data. They are collected in Appendix A. The relations below are therefore stated symbolically, and the cost figure

is shown in reference-relative form.

5.3.1. Structural cost

The structural cost separates a material and a labour term, which follow different physical drivers,

$$C_{\text{struct}}(S) = C_{\text{mat}}(S_0) \left(\frac{S}{S_0} \right)^{b_{\text{mat}}} + C_{\text{lab}} S, \quad (5.2)$$

where $S_0 = 60 \text{ m}^2$ is the reference area, b_{mat} is the material scaling exponent, and the reference material cost $C_{\text{mat}}(S_0)$ and labour coefficient C_{lab} are given in Appendix A. Labour scales with total seam length, which scales linearly with area, whereas material cost scales with the geometric scaling of the individual components. A constant cost per unit area would underestimate the cost of larger kites, whose superlinearly scaling structural parts require heavier, more expensive cloth. Labour is dominated by sewing seams, attaching reinforcements, and assembling the bridle, and is modelled with a constant seam density at a fixed sewing rate, giving a size-independent C_{lab} . Since the true seam density may fall slightly with scale, this is a conservative upper bound on the labour trend.

Each material item is assigned an exponent from geometric similarity, consistent with the mass model presented in section 4.2: area-driven items scale as S^1 , while structural items scale as $S^{1.5}$, the bridle following its mass derivation in subsection 4.3.4. The cost-weighted mean of the component exponents is

$$b_{\text{mat}} = \frac{\sum_i C_i b_i}{\sum_i C_i} = 1.14, \quad (5.3)$$

raised above unity mainly by the strut skin, strut bladder, and bridle, which carry the structural exponent. Figure 5.2a shows the component shares and exponents.

The model is anchored to a manufacturer cost for a 60 m^2 kite. This prototype is over-reinforced, with a reinforcement-to-area ratio roughly four times the V3 value, so a design anchor reduced by the corresponding reinforcement saving is adopted as the primary scenario and the unadjusted value is retained as an as-built reference price. Both are propagated through the exponent band in Figure 5.2b, shown relative to the design anchor: the as-built band brackets the 2013 V3 sale price, the design anchor lies lower, and the bottom-up component estimate falls below both as a lower bound that excludes overhead.

The structural cost is modelled as a function of area only, at a fixed design wing loading. With the maximum wing loading $W_{l,\text{max}}$ as a free variable, it would scale with $W_l S$ rather than S . The material strength, seam, and load-path data required to support that dependence are unavailable here, so the relation is an area-based scaling law based on a fixed reference design philosophy.

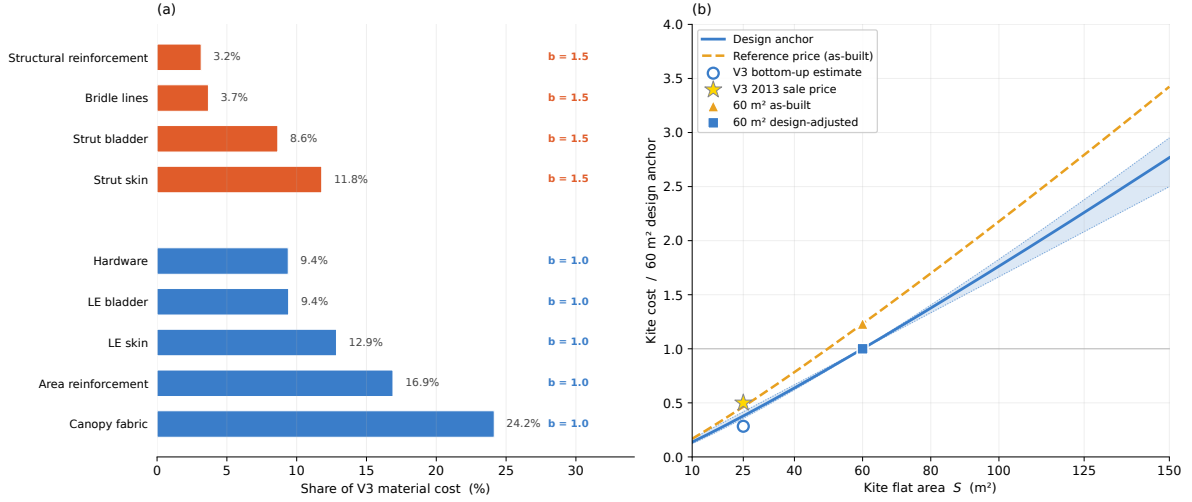


Figure 5.2: Basis for the LEI kite cost scaling model, in normalised form. (a) Material cost shares at the V3 reference (25 m²), coloured by scaling exponent: blue, $b = 1.0$; orange, $b = 1.5$. (b) Kite cost relative to the 60 m² design anchor versus flat area, for the design anchor (solid, $b = 1.00$ - 1.22 band) and the as-built reference price (dashed), with the V3 bottom-up estimate, 2013 sale price, and both 60 m² anchors marked.

5.3.2. KCU cost

The KCU cost maintains the fixed-variable structure of Grete [32] but sets the variable exponent based on cost drivers rather than the mass model. The component grouping of section 4.4 cannot be transferred to cost, since no cost breakdown exists for the reference unit and mass shares are not cost shares. The breakdown of Coutinho [6] shows that protection and drivetrain dominate the KCU mass while electronics are a small mass fraction, but low mass does not imply low cost. The cost is therefore split into a fixed and a variable contribution,

$$C_{\text{KCU}}(S) = f_{\text{fix}} C_{\text{KCU,ref}} + (1 - f_{\text{fix}}) C_{\text{KCU,ref}} \left(\frac{S}{S_{\text{ref}}} \right)^{b_C}, \quad (5.4)$$

where $f_{\text{fix}} = 0.30$, $S_{\text{ref}} = 60 \text{ m}^2$, and the reference cost $C_{\text{KCU,ref}}$ is given in Appendix A. The fixed fraction represents the avionics, electronics, sensors, and integration, taken as size-independent. These serve the kite regardless of its area, so their cost does not scale with size. The chosen value is consistent with Grete's split and with generic servo-system cost data, in which the controller and electrical integration carry a large share of the cost.¹

The baseline uses $b_C = 1$: the steering force follows the tether force and hence approximately the wing area (section 4.4), and at a constant effective drum radius the actuator torque scales linearly with area. This is deliberately below the $S^{1.5}$ mechanical mass scaling, since heavier actuation hardware at larger scale need not cost proportionally more once economies of scale and component choice are accounted for. A sensitivity case with $b_C = 1.5$, matching the mechanical mass scaling of section 4.4, tests the effect of stronger scaling. With only one cost reference, the fixed fraction and exponent cannot be fitted independently, so the relation is an engineering estimate. A wider dataset of KCU prices and actuator ratings would be needed to validate it. A KCU lifetime of five years is adopted for the replacement model, consistent with the interval reported by Coutinho [6].

5.3.3. Sensor cost

The flight-dynamics sensor package, comprising the inertial measurement unit (IMU), the global navigation satellite system (GNSS) receiver, and the flow sensors, provides the states required by the autopilot regardless of kite size, since only one measurement of attitude, position, and apparent wind is needed. Its cost is therefore fixed, independent of S , and excluded from the KCU reference cost. This gives the following formulation for the sensor cost

$$C_{\text{sens}} = C_{\text{sens,ref}} \quad (5.5)$$

¹<https://thundersaidenergy.com/downloads/servo-motors-cost-calculator/>

With $C_{sens,ref} = 1000$, it should be confirmed against supplier data as it becomes available. The assumed lifetime of the sensors is similar to the KCU.

5.3.4. Launch and Land system

The launch and land system (LLS) considered in this study is based on a mast-assisted concept for a soft wing kite. It consists primarily of a lowerable mast and supporting steel structure, along with a coupling and reefing mechanism for deploying and recovering the kite. During launch, the folded kite is raised clear of the ground, unfolded and inflated by the wind, then released as the tether is reeled out. Landing follows the reverse sequence: the kite is depowered, reeled towards the mast, reefed and lowered for storage.

consists primarily of a solid supporting structure. Its cost is therefore assumed to scale with the amount of structural material required. Under the assumption of constant wing loading, the characteristic loads acting on the LLS system increase proportionally with kite area. Due to the unavailability of detailed structural and cost data, the cost is consequently modelled as a linear function of kite area:

$$C_{LLS} = p_{LLS} S \quad (5.6)$$

where $p_{LLS} = EUR \ 165 \text{ per m}^2$. This relation should be interpreted as a first-order estimate. The relation does not account for possible changes in the LLS architecture, geometric scaling effects, standardised components, manufacturing complexity or economies of scale.

5.4. Lifetime, replacement and availability

Lifetime is a dominant economic driver for soft-wing systems for three reasons. First, the kite and tether are consumables, and their repeated replacement is a large share of the lifetime cost. Second, current soft-wing kites reach only a few hundred flying hours, far below the several-thousand-hour targets assumed for mature systems. Third, scaling amplifies the risk. Flutter and wrinkling are more pronounced in the large unsupported membrane panels of a bigger kite, and they accelerate material degradation. Lifetime is therefore modelled explicitly, rather than fixed to a generic operational assumption.

Every replaceable component follows the same rule. Its annual operational cost is the replacement frequency times its capital cost,

$$C_{OpEx,j} = f_{repl,j} C_{CAPEX,j}, \quad (5.7)$$

where $f_{repl,j}$ is the number of replacements per year. This is a smeared continuous-wear model. Rather than discounting discrete replacement events, the average annual replacement cost is charged every year. No salvage or recycling value is credited because soft-wing recycling is not yet established at the technology's current maturity.

The components fall into two logics of lifetime. The kite and tether are consumables whose wear tracks how much the system flies, so their lifetimes are usage-based, expressed as operating-hour budgets rather than calendar years. The ground-station hardware consists of standard electromechanical components whose degradation is not meaningfully described in terms of flight hours. Instead, their lifetime is defined in calendar terms. These differing lifetimes are discussed separately below. The lifetime relationships are illustrated in Figure 5.3, which also shows their influence on the LCoE, since the resulting replacement intervals are reflected in the annual replacement costs.

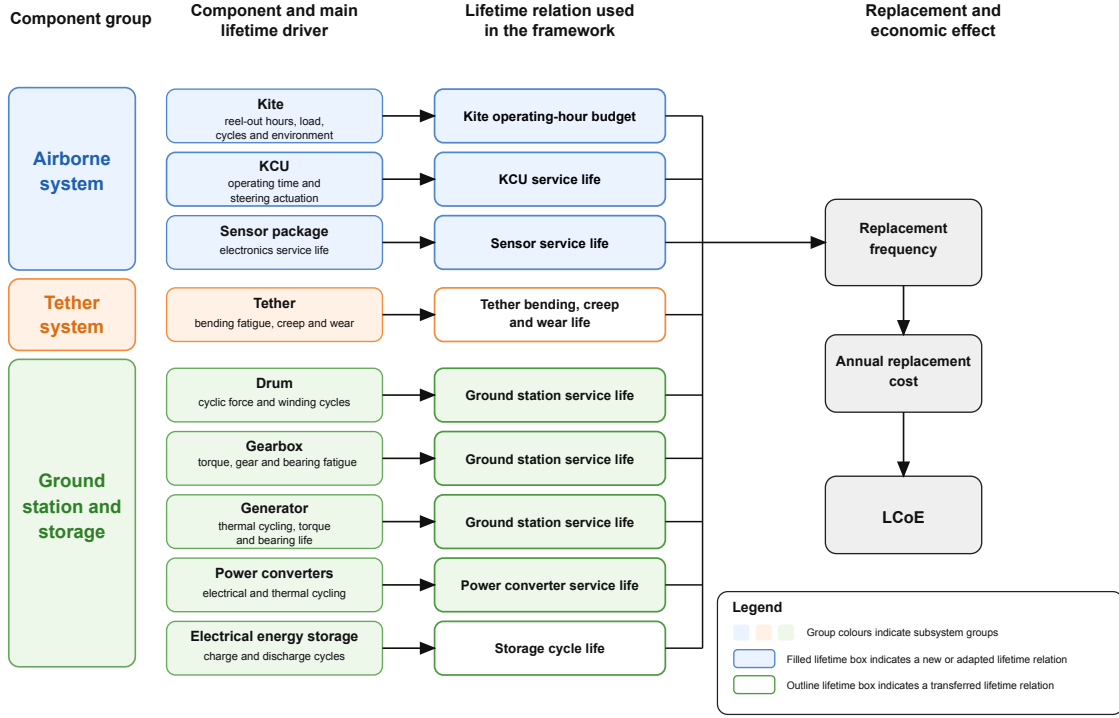


Figure 5.3: Lifetime and replacement structure used in the techno-economic framework. Components are grouped into the airborne system, tether system, ground station and storage system. Filled lifetime boxes indicate relations that are newly introduced or adapted in this thesis, while outline-only boxes indicate relations transferred from the Reference Economic Model. Each lifetime relation determines a replacement frequency and recurring replacement cost. Replacement downtime affects availability only where it is represented explicitly in the model.

The usage-based lifetime models use the annual flight and reel-out hours after operational availability has been applied. The fraction of the year within the operating wind speed range is

$$\int_{v_{in}}^{v_{out}} pdf(v_w) dv_w, \quad (5.8)$$

where $p(v_w)$ is the wind speed probability density. The corresponding annual flight and reel-out hours are

$$H_{flight} = 8760 a \int_{v_{in}}^{v_{out}} pdf(v_w) dv_w, \quad H_{out} = 8760 a \int_{v_{in}}^{v_{out}} pdf(v_w) f_{out}(v_w) dv_w, \quad (5.9)$$

where a is the availability, $pdf(v_w)$ the wind-speed distribution between cut-in and cut-out, and $f_{out}(v_w)$ the reel-out fraction of the cycle.

The availability-adjusted reel-out hours determine canopy replacement, while the availability-adjusted flight hours determine operational tether wear and maintenance labour. Where annual cycle counts enter a replacement relation, these are obtained from the same availability-adjusted operating distribution. Availability is consequently applied once to each usage quantity. Calendar-based component lifetimes and the standing operator requirement are not scaled with these hours.

The kite canopy wears only during reel-out, when it carries the traction load. Its lifetime L_{canopy} is therefore expressed as a budget of reel-out hours, and the replacement frequency is

$$f_{repl}^{kite} = \frac{H_{out}}{L_{canopy}} + p_{cyc} N_{cyc}, \quad (5.10)$$

where the optional second term adds cycle-count fatigue through a per-cycle penalty p_{cyc} and the annual number of pumping cycles N_{cyc} . This replaces the reference calendar lifetime, and it couples replacement to how the kite is actually flown. The kite capital cost is taken from section 5.3.

The tether can fail by bending fatigue, by creep, or by operational wear. Each mode gives a replacement frequency, and the governing mode is the one with the shortest life, that is, the highest frequency,

$$f_{\text{repl}}^{\text{tether}} = \max(f_{\text{bend}}, f_{\text{creep}}, f_{\text{oper}}), \quad f_{\text{oper}} = \frac{H_{\text{flight}}}{L_{\text{oper}}}. \quad (5.11)$$

The bending and creep models are derived from the reference model in section 5.2 and are the two modes active by default. Both are stress-driven, and at the low fibre stress of a soft-wing tether, the creep life is very long, so bending governs instead: the lower half of the tether crosses the winch drum twice each pumping cycle, hundreds of thousands of times per year. Its life, therefore, depends on the drum sizing, since a larger drum-to-tether diameter ratio bends the fibres less on each pass and lengthens the bending life, at the cost of a larger winch drum. As the winch is inexpensive relative to repeated tether replacement, a larger drum-to-tether diameter ratio is observed in practice, with increased ultraviolet exposure, abrasion, particle ingress, and reduced bending life of the drum. Bosman [35] identifies creep and bending as the two limiting design drivers for the Dyneema fibre tethers used in airborne wind energy, the drum-crossing lower half bending-driven and the airborne upper half creep-driven, and is referred to for a fuller treatment.

The operational-wear mode is an optional, empirical flight-hour life for the non-fatigue degradation seen in practice, ultraviolet exposure, abrasion, particle ingress and handling. It is disabled in the baseline, so the tether is fatigue-governed and is enabled only to test the shorter lifetimes currently observed in the field. A component whose governing life exceeds the project duration is never replaced, so its replacement frequency is set to zero.

The ground-station lifetimes follow a calendar convention rather than a usage budget. The winch, gearbox, generator and power converter are conventional electromechanical hardware, not consumables, so their wear is not usefully tied to flight hours. Following common practice in the wind industry, where drivetrain components are designed to a fixed calendar service life, each is assigned a service life of 20 years, independent of how much the system flies. The replacement frequency is the inverse of that life, capped at the project duration,

$$f_{\text{repl}}^{\text{GS}} = \begin{cases} 1/L_{\text{GS}}, & 0 < L_{\text{GS}} \leq N, \\ 0, & \text{otherwise,} \end{cases} \quad (5.12)$$

with N the project lifetime, so a 20 year component in a 25 year project is re-bought once. The initial unit is already annualised through the capital-recovery factor as part of the installed capital cost, which is discussed in section 5.5. The replacement term charges only the re-buy, so no component is both annualised and re-charged without an actual replacement. The launch-and-recovery system is treated the same way, with its own 20 year service life. The one ground-station item with a usage-based life is the ultracapacitor storage, whose replacement follows the number of charge-and-discharge cycles rather than a calendar life, since cycling, not age, drives its degradation.

Availability enters through the annual hours of Equation 5.9. The factor a scales the operating time, thereby reducing both the energy yield and the wear on the usage-based components. Its value and derivation are given in subsection 5.4.2, where a baseline of $a = 0.90$ is adopted.

5.4.1. Operation and maintenance

The operational expenditure is modelled as three streams: component replacement, ground crew costs, and fixed site overhead. A single fraction of capital would misrepresent a soft-wing system, whose operating cost is dominated by the wear of the flying parts and by the crew that operates and maintains the system. The separation follows the reference AWE economic model and wind industry practice [25, 37]. The streams sum to

$$C_{\text{OpEx}} = \sum_j f_{\text{repl},j} C_{\text{CAPEX},j} + C_{\text{crew}} + C_{\text{site}}, \quad (5.13)$$

where the last two terms form the fixed operation and maintenance cost of Equation 5.7.

The replacement stream is the per-component wear of section 5.4, each component charged at its full unit capital cost. Sectional repair of the tether is not supported by soft wing data, so a worn tether is renewed as a whole. This overstates the tether replacement cost, and a sectional model is planned as a refinement.

The ground crew cost has two terms with different drivers,

$$C_{\text{crew}} = w_{\text{lab}} [(1 - \alpha_{\text{auto}}) N_{\text{op}} h_{\text{op}} + r_{\text{maint}} H_{\text{flight}}], \quad (5.14)$$

where w_{lab} is the labour rate with a value of 50 €/h, N_{op} the annual operating days, h_{op} the operating hours per operating day, r_{maint} the maintenance hours per flight hour, and H_{flight} the annual flight hours of Equation 5.9. The operating component includes launch, supervision, landing, and recovery. In the current system, this part of the process is not automated [32] and is billed per operating day, since a crew must be on site for every day the system is in operation. The maintenance component encompasses inspection and servicing of airborne parts subject to wear and is billed per flight hour, as this wear accumulates with time spent flying.

The operating days follow from the wind resource. The system operates whenever the wind lies in its operating band, so

$$N_{\text{op}} = f_{\text{wind}} \cdot 365 \text{ days}, \quad f_{\text{wind}} = \frac{H_{\text{flight}}}{a \cdot 8760 \text{ h}}, \quad (5.15)$$

where f_{wind} is the share of the year the wind lies between cut-in and cut-out, the operability fraction that also sets the flight hours of Equation 5.9. For the reference site $f_{\text{wind}} \approx 0.87$, giving $N_{\text{op}} \approx 318$ days per year.

The labour rate w_{lab} is a fully loaded technician rate held fixed across designs. The intensities h_{op} and r_{maint} depend on the maturity of the operation and are set by the development stage in subsection 5.4.3. Both are size independent, so one crew operates one system regardless of wing area. Scaling intensities by area would offset crew costs against energy, which also scales with area.

The automation fraction α_{auto} scales only the operating term. It is zero for the manual baseline and approaches unity for an autonomous launch, landing, and remote supervision system. The maintenance term is independent of automation, because the hardware wears whether or not the operation is automated. The economic value of automation is examined through α_{auto} in the sensitivity analysis.

The site overhead covers land lease, insurance, and general upkeep, modelled as a rate per rated power,

$$C_{\text{site}} = p_{\text{site}} P_{\text{rated}}, \quad (5.16)$$

where p_{site} is the generic AWE operation and maintenance rate of the reference model [25]. This rate is applied as non-labour overhead only, so it does not double-count the ground crew cost of Equation 5.14.

Availability enters through the flight hours of Equation 5.9, which drive the wear of the flying parts and the maintenance term, and through the net energy in Equation 5.21. Lower availability reduces delivered energy and costs that scale with flight hours. It does not change the operating term, which is a standing cost tied to the operating calendar rather than to the hours flown. The baseline value $a = 0.90$ is derived in subsection 5.4.2.

Several operational costs are excluded. Unplanned failures and emergency recovery are not modelled as a separate term; their effects are absorbed into the availability and maintenance assumptions. Certification, permitting, and administrative costs are excluded because they are site- and company-specific. Consumables and crew transport are not modelled separately. These omissions reduce the modelled operational cost, so they constitute a lower bound. The model represents a single system and does not capture the crew and monitoring economies of a multi-system farm.

5.4.2. Availability

The annual energy production computed by AWESPA is the potential energy production from the wind resource, the system power curve, and the operational wind-speed limits. The corresponding annual potential flight time is case-specific. For the reference case, AWESPA predicts approximately 7644 h/yr

of potential flight time. Operational interruptions for launching, landing, routine inspection, and maintenance are not represented in that calculation.

In the absence of detailed operational data for commercial soft-wing systems, a single daily interruption of 2 h is assumed, giving an annual downtime of

$$H_{\text{down}} = 365 \times 2 \text{ h} = 730 \text{ h/yr.} \quad (5.17)$$

Relative to the potential flight time of the reference case, the operational availability is

$$a = 1 - \frac{H_{\text{down}}}{H_{\text{potential}}} = 1 - \frac{730}{7644} = 0.904. \quad (5.18)$$

On this basis an operational availability factor of $a = 0.90$ is adopted for all evaluated designs, and the net annual energy production is

$$\text{AEP}_{\text{net}} = a \text{ AEP}_{\text{AWESPA}}. \quad (5.19)$$

The same factor scales the annual flight and reel-out hours in Equation 5.9, so any downtime proportionally reduces the usage-based wear on the airborne components, in line with the energy not produced. Availability is therefore not reapplied in the LCoE equation. All subsequent energy assessments directly use AEP_{net} . The maturity scenarios discussed below maintain the stated availability unless it is specifically altered in a dedicated sensitivity analysis.

The two-hour daily interruption provides a physical basis for the factor and collectively accounts for launch and landing, routine inspection, and maintenance. The result should be read as a conservative first-order assumption rather than a validated availability estimate. Applying a constant factor further assumes that downtime causes a similar relative energy loss across the evaluated designs, and that it is not systematically concentrated in periods of high or low wind speed.

5.4.3. Maturity scenarios

Three maturity scenarios examine the combined economic effect of improved component reliability and reduced operational intensity. Each scenario is a coordinated set of assumptions for canopy lifetime, tether lifetime, operating intensity, and maintenance intensity, given in Table 5.2. The scenarios are not tied to calendar years.

Table 5.2: Default maturity scenario inputs used by EcoMo. Explicit configuration values take precedence over these presets.

Scenario	L_{canopy} [h]	$L_{\text{t,oper}}$ [h]	h_{op} [h/day]	r_{maint} [h/flight-h]
Early	100	250	3.27	0.25
Mid	500	1000	0.82	0.10
Mature	5000	5000	0.16	0.01

The scenarios act on the three factors that dominate operational costs and are least developed for a soft-wing system: canopy lifetime, tether lifetime, and ground crew intensity. Early represents the current state reached in practice. Mid represents an intermediate state consistent with the developments currently underway. Mature represents the values the developers strive to uphold. Staging the three factors provides a roadmap for studying the effect of maturity on the LCoE.

Current soft wing canopies reach a few hundred loaded hours, over a wide range that depends on the operating environment and on flutter and wrinkling. A baseline for this lifetime would allow canopy replacement to dominate operational costs. The mature value of 5000 h is the lower end of the several thousand-hour target pursued through improved materials and structural understanding. The tether lifetime is limited by wear and by the large safety factors used to avoid catastrophic failure. The operating and maintenance intensities are high because launch, landing, and system supervision are not yet automated.

The development stage setting selects the default scenario values from Table 5.2. Any value explicitly set in the case configuration overrides the corresponding default, so the evaluated assumptions are reported using their effective configuration values rather than the scenario name.

The preset supplies the operational tether lifetime but does not select the tether lifetime model. For the maturity roadmap, operational tether wear is enabled and the staged lifetime applies to a fixed physical design. For the design optimisation, the diameter-dependent bending model is used, and the operational lifetime is disabled, thereby retaining the dependence of the tether lifetime on stress and drum geometry.

Material prices, geometry, aerodynamic performance, labour rates, automation, availability, and financial assumptions are fixed across scenarios unless otherwise stated. Consumables and the annual site overhead retain their configured values. The roadmap isolates the effect of longer canopy and tether lives, lower supervision, and reduced maintenance intensity, so the sensitivity of the LCoE from the current range to the target is examined across the staged values.

The scenarios do not prescribe the materials, structural changes, control developments, or operational procedures required to achieve these improvements. They are controlled economic scenarios, not a forecast of how or when the technology matures.

5.5. LCoE and financial model

The subsystem costs of the previous sections are assembled into a single levelised cost of energy. This section describes that assembly and justifies the financial parameters. The general discounted cash flow definition of LCoE and the annualisation and financing background were introduced in section 5.5 and subsection 2.1.5.

The full discounted cash-flow definition divides the present value of all costs by the present-value-weighted electricity production. The discount factor in the denominator does not mean that the physical energy production is reduced. It follows from solving for the constant electricity price at which the project's net present value is zero. Future production enters the financial balance through future revenue, and that revenue is discounted in the same way as the future costs.

The initial capital cost is the sum of the subsystem capital expenditures,

$$C_{ICC} = \sum_i C_{CAPEX,i}, \quad (5.20)$$

over the kite, tether, ground station, balance of system, and balance of plant. The annual operational cost is the sum of the component replacement costs of Equation 5.7 and a fixed operation-and-maintenance term. This is defined in Equation 5.13. It includes component replacement, ground crew cost and annual site overhead.

For constant annual energy and constant annual operating cost, the discounted definition reduces to the annualised form used here. The initial capital cost is annualised with the capital recovery factor, and the result is divided by the annual available energy,

$$LCoE = \frac{CRF C_{ICC} + C_{OpEx}}{AEP_{net}}, \quad (5.21)$$

$$CRF = \frac{r(1+r)^N}{(1+r)^N - 1}, \quad (5.22)$$

where r is the discount rate, N the project lifetime. The annual energy is not explicitly discounted because the discounting is already accounted for by the capital recovery factor. This form is algebraically equivalent to the discounted definition under constant annual energy and constant annual cost.

The recovery factor annualises only the initial capital cost. Replacements do not pass through it. They enter directly as the flat annual operational cost of Equation 5.13. A component therefore contributes an annualised purchase $C_{CAPEX} CRF / AEP_{net}$ and a running replacement cost $f_{repl} C_{CAPEX} / AEP_{net}$. This constant-rate treatment is exact when the cost and energy streams are steady, which holds well for the frequently replaced soft-wing kite. It loses only the discrete timing of rare replacements, which contribute little.

EcoMo uses the capital recovery factor rather than a fuller fixed-charge-rate annualisation. The fiscal detail that an FCR resolves, namely tax, depreciation, and insurance, rests on project-specific assumptions that are not warranted at the maturity of AWE. It lies outside the model's trend-finding aim. The

CRF is therefore adopted as the simplest consistent annualisation, exact for the constant-annual case above. A full FCR or a discounted cash flow treatment is the appropriate next step for a later business case analysis.

The discount rate is the weighted average cost of capital,

$$r = \frac{q}{1+q} r_d (1 - T_C) + \frac{1}{1+q} r_e, \quad (5.23)$$

where q is the debt-to-equity ratio, r_d the cost of debt, r_e the cost of equity, and T_C the corporate tax rate. The adopted values are listed in Table 5.3. They are taken directly from the AWES reference economic model of Joshi and Trevisi [25], and give a post-tax WACC of $r \approx 0.078$.

Table 5.3: Financial parameters, adopted from the AWES reference economic model of Joshi and Trevisi [25] and read as conservative project-finance assumptions for an early-commercial AWES, not as generic present-day wind values.

Parameter	Symbol	Value	Basis
Debt-to-equity ratio	q	70/30	Classic wind project-finance structure, a conservative middle between mature European debt shares and equity-heavy first deployments
Cost of debt	r_d	8%	Reference economic model [25]
Cost of equity	r_e	12%	Reference economic model [25]
Corporate tax rate	T_C	25%	Generic reference rate, not country-specific
Post-tax WACC	r	7.8%	From Equation 5.23
Project lifetime	N	25 yr	Project-finance input [6], [25]

This WACC is a conservative reference assumption for an early-commercial AWES, not a generic present-day wind value. It is consistent with historical wind benchmarks, which sit near 8% [37, 81], but it is above the cost of capital of mature onshore wind in current markets [82]. The higher value is appropriate because AWE remains in the early-commercialisation phase, as set out in subsection 2.1.5.

The financial model transfers unchanged from the reference model of Joshi and Trevisi [25], because the annualisation depends on the cost and energy totals, not on wing topology. Its correctness is checked at the implementation level. In standalone mode, EcoMo reproduces the reference MATLAB outputs for a common configuration, which verifies the Python translation of section 5.1 before the soft-wing adaptations are applied.

Reference System Economic Assessment

This chapter applies the techno-economic framework to the TU Delft V3 kite at the Bangor Erris reference site. First, the verification, calibration and validation status of the models used is discussed in section 6.1. The assessment basis and the selection of the reference maturity scenario are established in section 6.2. The cost breakdown of the fixed V3 design, which establishes the baseline against which design changes are measured, is reported in section 6.3. The effect of operational maturity on the levelised cost of energy is then isolated in section 6.4, with the physical design and material prices held constant.

6.1. Verification and validation status

The framework combines relations with different levels of empirical and numerical support. Their evidential status is therefore distinguished explicitly to avoid treating implementation verification, parameter calibration, plausibility assessment, and experimental validation as equivalent. Table 6.1 summarises the evidence supporting each principal model and output.

Table 6.1: Evidential status of the principal models and outputs used in the framework.

Model or output	Evidential status	Basis
EcoMo implementation	Verified against reference implementation	Comparison with the original MATLAB implementation
Reference-system sizing	Implementation verified	Reproduction of reference geometry and derived quantities
Component mass relations	Calibrated	Reference-system and partner-supplied mass estimates
Cost coefficients	Inherited from reference model	Industry-informed data reported by Joshi and Trevisi [25]
Pumping-cycle formulation	Validated for reference configuration	Flight-test comparison of the 25 m ² V3 kite reported by Schelbergen [73]
Tether bending-fatigue relation	Experimentally informed	HMPE bending-fatigue tests reported by Bosman [35]
Integrated economic results	Plausibility checked, not validated	Comparison with published cost structures and LCoE estimates

This study directly verifies two elements. The EcoMo cost model replicates the original MATLAB implementation for identical inputs, confirming that the translation preserves component costs, financial calculations, and LCoE. The sizing implementation reproduces the reference-system geometry and derived quantities, confirming correct implementation of sizing relations at the reference scale. Verification does not extend to extrapolation for larger systems or to the physical assumptions underlying these relations.

The remaining evidential basis is inherited or calibrated rather than generated within this study. Component mass relationships rely on physically motivated scaling exponents tied to reference-system data and partner-provided mass estimates, meaning the mass model is tuned but not independently verified. Agreement at the aggregate level does not by itself confirm the accuracy of the component-level relations, because offsetting errors among individual components may still be present. Cost coefficients

are taken from the industry-informed reference model of Joshi and Trevisi [25], which was developed for systems with rated powers between 100 kW and 2 MW. The smaller systems analysed here have rated powers below this range, increasing the uncertainty in their absolute cost estimates.

Two model elements have experimental support from prior studies, but with limitations. The pumping-cycle formulation matches flight-test data for the reference wing size, supporting its predictive capability for the reference kite and tested operating conditions. This comparison does not validate the aerodynamic inputs, operational optimisation, resulting power curves, or extrapolation to larger wings in this study. The tether-life model uses experimental HMPE bending-fatigue data, but its translation into operational service life depends on simulated loading histories and omits several degradation mechanisms encountered in practice.

No independent economic validation is available. The resulting subsystem cost structure matches the industry-informed data underlying the reference model, as expected since those data inform the adopted cost relations. The integrated LCoE falls within the broad and largely outdated range of published estimates. These comparisons indicate consistency with available economic evidence but do not validate the results against an independently characterised system. The absolute LCoE is indicative. Greater confidence applies to the comparative design trends and identified cost mechanisms, particularly where these remain consistent under the investigated sensitivities later discussed in section 7.3.

6.2. Assessment Basis

The fixed TU Delft V3 system defined in section 3.7 is evaluated at the Bangor Erris reference site to establish the technical and economic baseline for the subsequent assessments. The physical configuration, wind-resource representation, operational limits, and conversion assumptions are retained from the case study definition. The economic relations, availability treatment, and financial assumptions follow chapter 5.

The reference assessment uses the mid-maturity scenario defined in section 6.4. This scenario represents an intermediate development stage between the present prototype and the long-term technology target. It retains the replacement and operational costs that are characteristic of a soft-wing system without evaluating the design exclusively under either current operational limitations or mature-system targets. The Early and Mature scenarios are evaluated separately in subsection 5.4.3.

Throughout the maturity assessment, the physical design, aerodynamic performance, wind resource, material prices, availability, and financial assumptions remain fixed. Only the canopy lifetime, operational tether lifetime, operating intensity, and maintenance intensity vary between the maturity scenarios. The resulting differences therefore represent the combined economic effect of the prescribed operational-development assumptions.

6.3. TU Delft V3 Assessment

6.3.1. Power performance and annual energy production

The fixed V3 design at the mid-maturity stage establishes the cost model baseline before any design variable is changed. The energy yield of this design follows from the site wind resource and the AWESPA power curve, and the resulting annual energy sets the denominator of every cost metric. The energy yield is characterised first, and the cost breakdown follows.

The empirical wind-speed distribution at the 200 m reference height is shown in Figure 6.1. The distribution has a mean wind speed of 9.94 m s^{-1} , indicated by the dashed line. Most of the probability density lies between 5 m s^{-1} and 20 m s^{-1} , which largely coincides with the productive wind-speed range of the system.

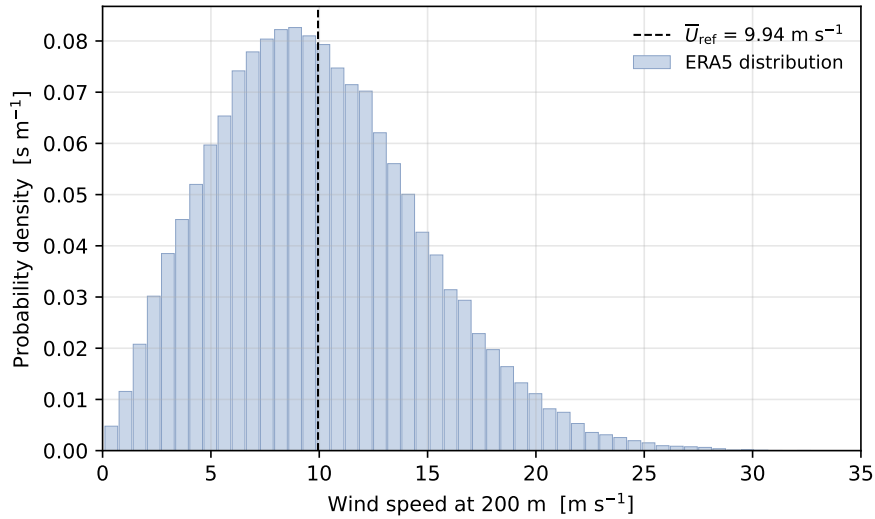


Figure 6.1: Wind-speed distribution at the 200 m reference height for Bangor Erris, based on the 2010 to 2019 ERA5 dataset [17]. The fitted Weibull distribution has a shape factor of $k = 2.24$ and a scale factor of $c = 11.25 \text{ m s}^{-1}$, and the dashed line marks the mean wind speed.

The electrical power curve of the fixed V3 system is shown in Figure 6.2. Electrical power increases from the cut-in wind speed of 5 m s^{-1} and reaches its maximum cycle-averaged value of 9.3 kW at approximately 16 m s^{-1} . The mean electrical power subsequently decreases as the operational optimisation adapts the pumping cycle to the active system constraints. The power curve extends to the cut-out wind speed of 25 m s^{-1} .

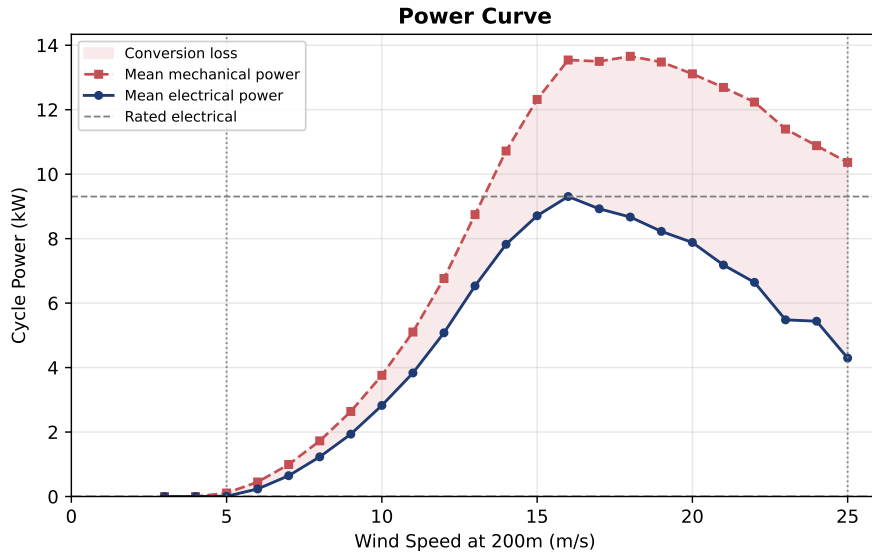


Figure 6.2: Mean electrical power curve of the fixed V3 design against the reference wind speed at 200 m, computed with AWESPA [17]. The dashed line marks the rated electrical power and the dotted lines mark the cut-in and cut-out wind speeds.

The net electrical power remains below the net mechanical power throughout the operating range, with the difference increasing towards higher wind speeds. This difference cannot be represented by applying a single efficiency to the net mechanical power because energy is converted in opposite directions during reel-out and reel-in. Mechanical power generated during reel-out passes through the generator with an efficiency of $\eta_{\text{gen}} = 0.85$. During reel-in, the required electrical energy is supplied through the storage system and motor, with efficiencies of $\eta_{\text{stor}} = 0.95$ and $\eta_{\text{mot}} = 0.85$, respectively. The conversion losses therefore depend on the gross energy exchanged during the pumping cycle

rather than only on the net mechanical energy. At the rated wind speed of 16 m s^{-1} , the net cycle-averaged mechanical power of 13.5 kW consequently corresponds to a net electrical power of 9.3 kW .

The resulting cycle conversion efficiency varies with wind speed. It increases from approximately 52 % at 6 m s^{-1} to approximately 75 % between 10 m s^{-1} and 13 m s^{-1} , where the net output is large relative to the reel-in and transition losses. At higher wind speeds, the gross energy exchanged during reel-out and reel-in continues to increase while the net mechanical power levels off. The conversion losses therefore increase relative to the net cycle output, reducing the efficiency to approximately 69 % at the rated wind speed and 41 % near the cut-out wind speed.

Integration of the net electrical power curve over the empirical wind-speed distribution results in an annual energy production of 26.8 MWh . Relative to the rated electrical power of 9.3 kW , this corresponds to a capacity factor of 0.33.

6.3.2. Lifetime cost and LCoE

The annual energy production forms the denominator of the LCoE assessment. The corresponding lifetime cost of the fixed V3 system is evaluated at the mid-maturity stage. The fixed V3 system achieves an LCoE of 2992 €/MWh . The initial capital expenditure amounts to 72.9 k€ , while the annual operating expenditure is 73.5 k€/year . Annualisation of the capital investment results in a contribution of 6.7 k€/ , approximately one order of magnitude below the annual operating expenditure.

The subsystem contributions to the capital expenditure, annual operating expenditure, and LCoE are shown in Figure 6.3 and reported in Table 6.2. The ground station accounts for 73 % of the capital expenditure, with the ultracapacitor storage forming its largest individual cost contribution. Ground-crew labour instead accounts for 65 % of the annual operating expenditure, reflecting the manual launch, landing, and supervision assumed for the mid-maturity system. The kite contributes approximately one fifth of the initial capital expenditure. Its similar contribution to the annual operating expenditure results from repeated replacement of the textile wing over the project lifetime.

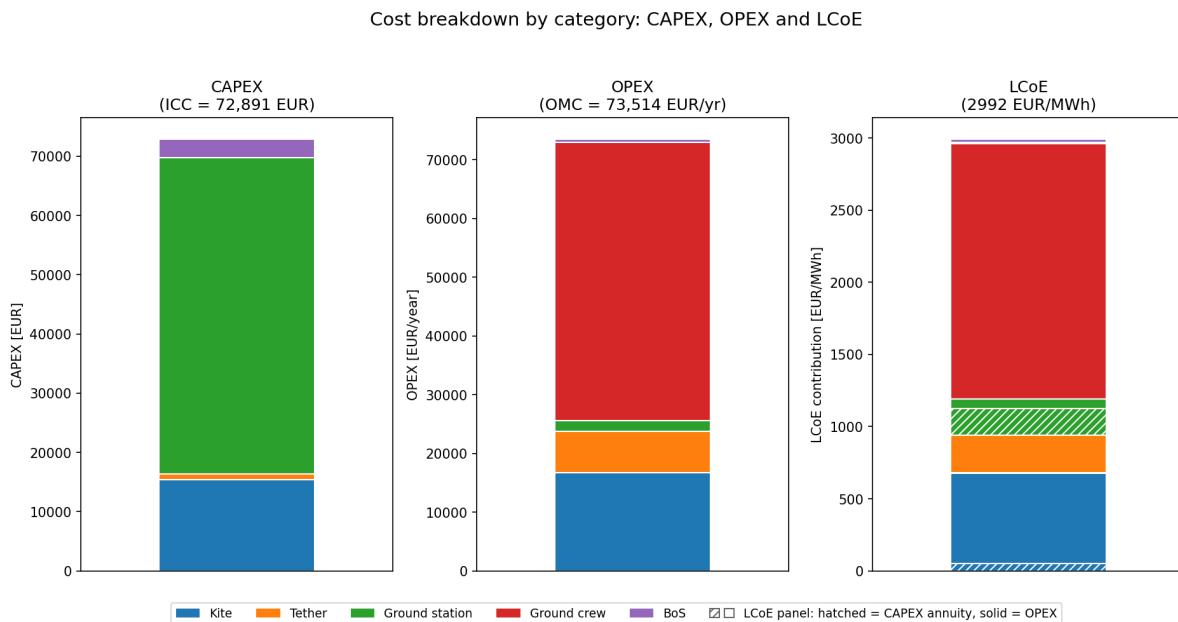


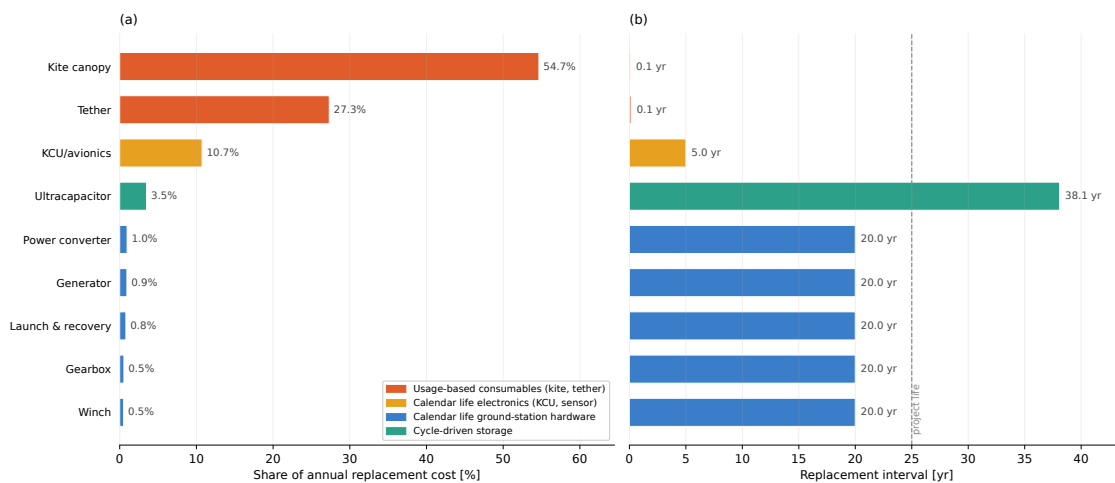
Figure 6.3: Subsystem contributions to the capital expenditure, the operating expenditure, and the levelised cost of energy for the fixed V3 baseline at the mid-maturity stage. The capital and operating panels show absolute costs, and the levelised-cost panel separates the capital annuity (hatched) from the operating contribution (solid).

Table 6.2: Subsystem shares in the capital expenditure, the operating expenditure, and the levelised cost of energy for the fixed V3 baseline at the mid-maturity stage.

Subsystem	CapEx [%]	OpEx [%]	LCoE [%]
Ground crew	0.0	64.5	59.1
Kite	21.1	22.7	22.6
Tether	1.4	9.5	8.8
Ground station	73.2	2.5	8.4
Balance of system	4.3	0.8	1.1
Total	100.0	100.0	100.0

The LCoE breakdown closely follows the operating-cost distribution because the annual operating expenditure substantially exceeds the annualised capital cost. The economics of the fixed V3 system are therefore dominated by labour and component replacement rather than by the initial capital investment. The result is compared with confidential LCoE data provided by companies, as a consistency check rather than a validation. The framework and the companies' cost model use different formulations and describe a different system. So, only the subsystem cost shares and the order of magnitude of the LCoE are compared. The comparison therefore shows the framework produces an LCoE of realistic magnitude and composition. It is not an experimental validation, because the values are partly modelled estimates rather than measured operational costs.

The distribution of replacement expenditure among the system components is shown in Figure 6.4. The kite and tether account for the largest contributions because their service lives are substantially shorter than those of the ground station components. Recurring replacement expenditure is therefore governed primarily by the consumable flying components.

**Figure 6.4:** Component contributions to the annual replacement expenditure of the fixed TU Delft V3 system at the mid-maturity stage.

The canopy lifetime is an influential assumption underlying the kite replacement contribution. The lifetime prescribed for the mid-maturity stage remains below the service life targeted for future commercial systems and therefore retains frequent canopy replacement in the reference cost baseline. The influence of extending the canopy lifetime towards the development target is examined in the following chapter.

6.4. Maturity Assessment

Mature scenarios defined in Table 5.2. The physical system, annual energy production, component prices, and financial assumptions remained unchanged between the scenarios. Only the canopy and

tether service lives, ground-crew requirements, and maintenance intensity were varied. The comparison therefore isolates the economic influence of longer component lifetimes and increasing operational autonomy. This roadmap does not consider learning effects, cost reductions for other components, or performance improvements. The resulting LCoE and its subsystem contributions are given in Table 6.3.

Under the prescribed maturity assumptions, the LCoE decreases from 9235 €/MWh in the Early scenario to 2992 €/MWh in the Mid scenario and 772 €/MWh in the Mature scenario. This corresponds to a reduction of approximately 92 % between the Early and Mature stages.

Table 6.3: Subsystem contributions to the levelised cost of energy across the three maturity stages, in €/MWh.

Subsystem	Early	Mid	Mature
Ground crew	5146	1767	225
Kite	2760	676	207
Tether	1045	264	56
Ground station	253	253	253
Balance of system	32	32	32
Total	9235	2992	772

The reduction is primarily caused by lower recurring expenditure. In the Early scenario, ground-crew labour and canopy replacement dominate the LCoE because the system requires extensive manual supervision and the canopy has a short service life. The Mid scenario assumes greater operational autonomy and longer component service lives, substantially reducing both contributions. Since the annual energy production and capital expenditure remain constant, the lower LCoE follows entirely from the reduction in operating expenditure. The staged reduction and the shifting subsystem shares are shown in Figure 6.5. The operating-cost terms fall as component lifetimes lengthen and supervision reduces, while the fixed capital terms of the ground station stay constant across the stages.

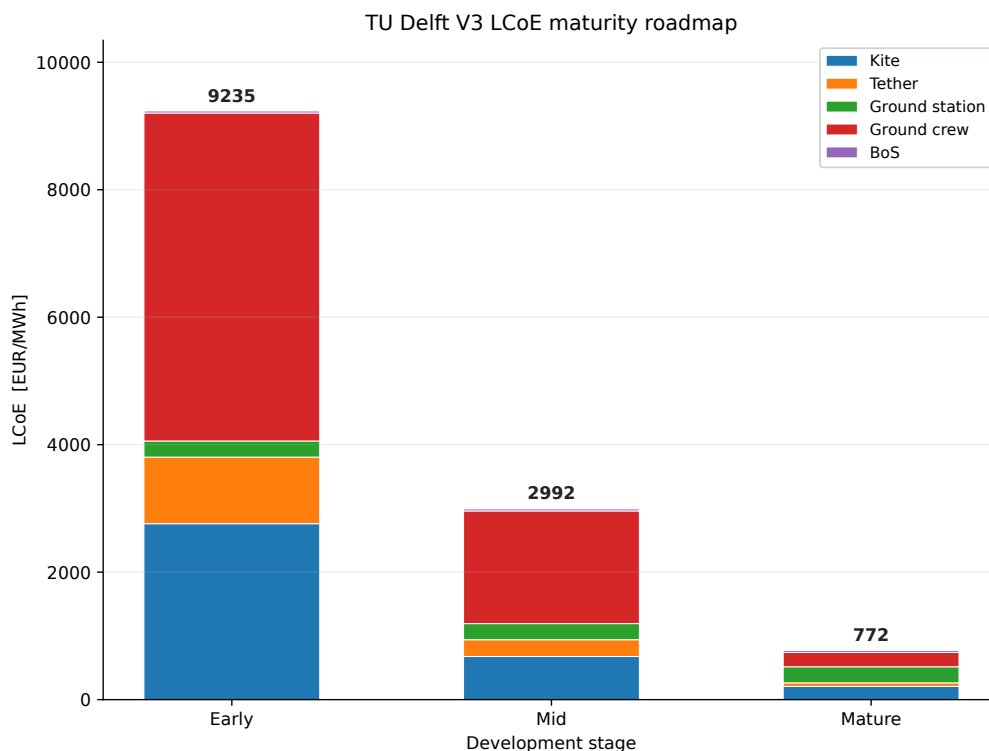


Figure 6.5: Levelised cost of energy across the Early, Mid, and Mature stages of the maturity roadmap for the fixed V3 design, decomposed into subsystem contributions.

The relative cost composition changes further in the Mature scenario. Longer canopy and tether ser-

vice lives reduce the contribution of flying-component replacement, while the remaining ground-crew requirement lowers the labour contribution. As these recurring costs decrease, the fixed ground-station investment accounts for a larger share of the LCoE. This increase in its relative contribution does not represent a higher ground-station cost, but results from the reduction of the other cost components.

The scenario comparison demonstrates that the economic performance of the fixed V3 system is highly dependent on component durability and operational autonomy. Even under the Mature assumptions, however, residual labour and replacement expenditure remain relevant. The maturity assessment therefore indicates that technological development must address recurring expenditure in addition to improving aerodynamic performance or reducing initial capital cost. The strong dependence on recurring expenditure also means that alternative designs cannot be compared through energy production and capital cost alone, because changes in wing and tether sizing affect component lifetime and replacement expenditure. These coupled effects are examined in chapter 7.

Coupled Design Optimisation and Scaling Trends

This chapter applies the coupled techno-economic framework to identify the wing area and allowable tether stress that minimise the LCoE at different generator mechanical power limits. A reference optimisation case is first examined to identify the physical and economic trade-offs governing the design in section 7.1. The resulting design and cost trends are subsequently evaluated across system scales in section 7.2, followed by a sensitivity assessment of the principal modelling assumptions in section 7.3.

7.1. Design study and the reference optimisation case

The coupled framework identifies an interior grid minimum in the wing-area and allowable-tether-stress design space. The design that achieves it follows from the framework's physical relations, rather than appearing as an unexplained optimiser output. The design study evaluates the influence of flat-wing area and allowable tether stress across five prescribed mechanical power limits for the generator. The 40 kW case is examined in detail to identify the physical and economic trade-offs governing the optimum. The resulting mechanisms provide the basis for interpreting the scaling trends across the remaining generator limits.

7.1.1. Design study definition

The design study evaluates each candidate system over a grid of the outer design variables, S and $\sigma_{t,allow}$, at five mechanical power limits for the generator, ranging from 40 to 250 (kW). Wing area governs the balance between aerodynamic loading and airborne mass. Allowable tether stress sets the tether diameter and, through it, the tether mass, aerodynamic drag, service life, and cost. The stress is sampled over 0.10 to 0.40 (GPa), chosen to bracket the optimum within the validity of the HMPE bending-fatigue relations of [35]. The wing-area range is scaled with the generator rating. The bounds used were found by testing the specific-area anchor for the reference case. A bound is widened when the lowest evaluated levelised cost occurs at that bound, so that the located optimum lies in the interior of the sampled range. The bounds and resolution of the grid are listed in Table 7.1.

Table 7.1: Design space and grid of the optimisation study. Each generator mechanical power limit is evaluated over a seven-by-five grid of wing area S and allowable tether stress $\sigma_{t,allow}$, the latter sampled at five values over 0.10 to 0.40 GPa.

P_{gen} (kW)	Wing-area range (m ²)	Grid ($S \times \sigma_{t,allow}$)
40	45-135	7×5
80	60-150	7×5
110	80-180	7×5
150	100-220	7×5
250	100-220	7×5

Each generator rating was evaluated over the full grid, thereby resolving the levelised-cost surface and the trade-offs that determine its optimum. These surfaces were computed at a reduced wind-speed resolution, because a full two-dimensional sweep at the finest resolution would be prohibitively expensive. The absolute levelised cost is sensitive to this resolution, so the reference case and the reported optima were re-evaluated at the production resolution used throughout the results. The reduced-resolution surfaces therefore establish the design trends and the stress optimum, while the production evaluations set the reported cost levels.

The tether service life is calculated from the bending-fatigue relation. So the allowable tether stress represents a trade-off between cost and performance. All design points are evaluated using the production fidelity of the quasi-steady performance model. Every node of the reference-case grid returns a feasible result, indicating that the sampled region lies within the operating envelope of the sizing and performance models.

7.1.2. Reference optimisation case

The results of the grid search are presented in a surface plot, which can be seen in Figure 7.1a. Such a plot provides useful insights not only into the found optimum value but also into the neighbouring values. The shape of the sampled surface, and not only its lowest point, indicates how sharply the optimum is defined. This distinction is decisive where a flat levelised-cost surface, or numerical variation in the operational optimisation, leaves only small differences between neighbouring candidates. The 40 kW generator mechanical power limit is selected as the reference case because its sampled LCoE surface contains an interior grid minimum in both design variables. The economic variation around this minimum is also more pronounced than at the higher generator limits.

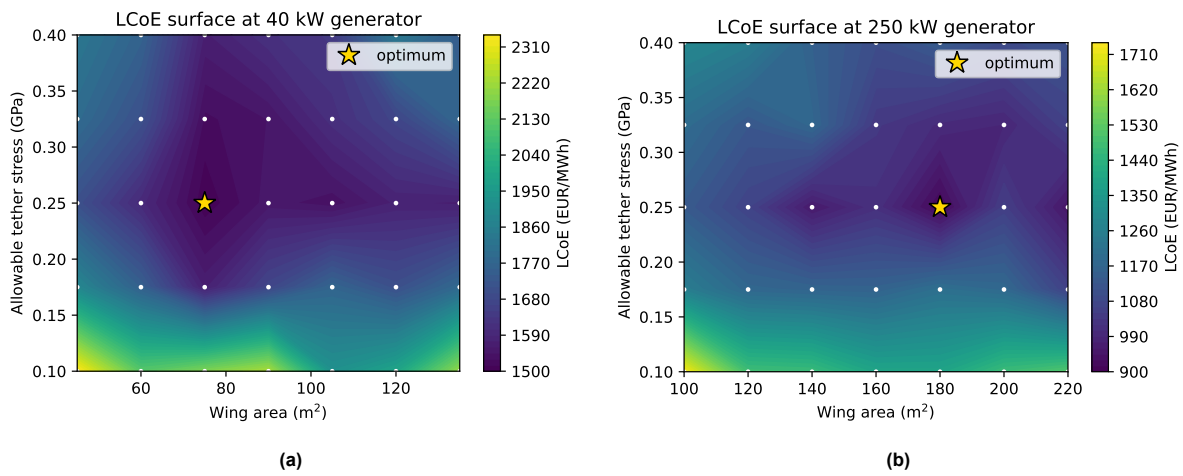


Figure 7.1: Levelised-cost surfaces over wing area S and allowable tether stress $\sigma_{t,allow}$, for (a) the 40 kW rating and (b) the 250 kW rating, with each optimum marked by a star. The 40 kW surface is a closed bowl with a sharply defined interior minimum, whereas the 250 kW surface is a broad, flat trough along the stress optimum, over which the levelised cost varies little with wing area. Both optima lie at 0.25 GPa. The colour scales differ between panels, so the panels show the shape of each surface rather than a comparison of absolute levels.

The sampled LCoE surface in Figure 7.1a contains an interior grid minimum at a wing area of 75 m² and an allowable tether stress of 0.25 GPa. The LCoE increases in both design directions around this point, indicating that the minimum is resolved within the selected bounds. The increase is more pronounced for this generator limit than for the higher generator limits considered. This can be observed at the 250 kW rating in Figure 7.1b. The surface is a broad, flat trough along the stress optimum, over which the levelised cost varies little across a wide span of wing areas. The optimum wing area is therefore weakly determined at the larger ratings, which motivates reporting it as a band in the following analysis. It should be emphasized that the color scales in the two images are not the same. The minimum points at the respective optima correspond to two different absolute values.

The generator mechanical power limit influences the operational solution but does not represent the maximum power that the drivetrain must accommodate. It constrains the reel-out and reel-in phases, but the transitions between them are not subject to the same constraint. The resulting transition peak exceeds the prescribed 40 kW limit, thereby determining the required peak capacity of the generator and drivetrain. The rated electrical power remains below the prescribed mechanical limit because it represents the maximum cycle-averaged net electrical output, including the electrical energy consumed during reel-in. The located grid minimum therefore results from the coupled balance between energy production and recurring expenditure, with the generator limit influencing the sustained pumping-cycle operation.

Table 7.2: Reference design at the 40 kW rating, taken at the optimum of the surface in Figure 7.1a. The crest factor κ is the ratio of peak mechanical power to rated electrical power, which determines the drivetrain and storage sizing.

Quantity	Symbol	Value	Unit
Flat wing area	S	75	m ²
Allowable tether stress	$\sigma_{t,allow}$	0.25	GPa
Total airborne mass	m_{air}	76.8	kg
Maximum tether force	$F_{t,max}$	24.6	kN
Tether diameter	d_t	12.1	mm
Cut-in wind speed	$v_{w,in}$	7	m s ⁻¹
Rated wind speed	$v_{w,rated}$	14.1	m s ⁻¹
Cut-out wind speed	$v_{w,out}$	25	m s ⁻¹
Capacity factor	CF	0.49	–
Crest factor	f_{crest}	3.1	–
Annual energy production	AEP	101.3	MW h
Capital expenditure	CAPEX	121	k€
Annual operating expenditure	OPEX	126	k€
Levelised cost of energy	LCoE	1502	€/MWh

The reference design is summarised in Table 7.2. It produces 101 MW h per year at a capacity factor of 0.49. These aggregate performance quantities provide the inputs required for the subsequent techno-economic assessment.

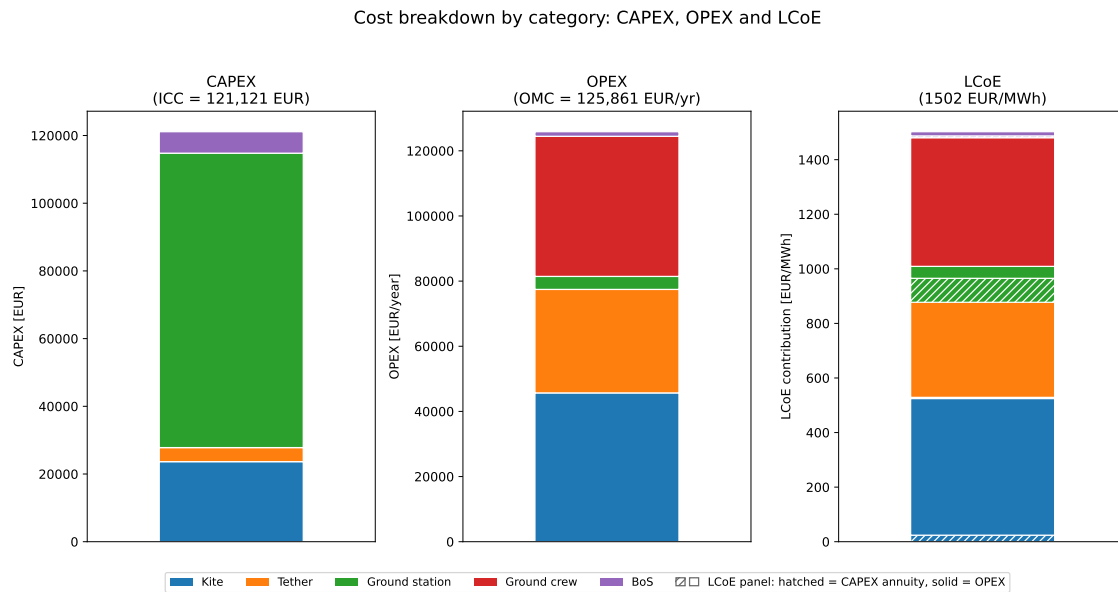


Figure 7.2: Subsystem contributions to the capital cost, operating cost, and LCoE for the optimised 40 kW configuration. The ground station is the primary driver of capital cost, while ground-crew labour and kite and tether replacement dominate both operating and levelised cost. In the levelised-cost panel, the hatched region represents the capital annuity, and the solid region represents the operating contribution.

The cost structure of the optimised design is shown in Figure 7.2. The capital cost is dominated by the ground station, which carries the drivetrain and the storage. The operating cost has a different composition, dominated by ground-crew labour together with kite and tether replacement. The annual operating cost far exceeds the annualised capital cost, so the levelised cost follows the operating-cost structure, and the ground station, which dominates the capital cost, contributes little to it. Three recurring contributions, namely labour, kite replacement, and tether replacement, account for about four-fifths of the levelised cost. Recurring labour and replacement therefore set the economics of the optimised system, as they do for the fixed V3 kite established in subsection 6.3.2.

The grid minimum results from two coupled balances. The allowable tether stress sets the tether diameter and, through it, both the material cost and the bending-fatigue life. Higher stress thins the tether and lowers its material cost, but it increases the bending stress and shortens the service life, thereby increasing replacement expenditure. These opposing effects balance at an allowable stress of 0.25 GPa. The wing area sets the second balance. A larger wing raises the annual energy production, but it also raises the airborne mass, the design tether force, and the kite and tether replacement costs, each of which scales with area. The resulting balance produces the interior grid minimum in wing area. As the system scale increases, the fixed labour contribution is progressively diluted, and the annual energy production and recurring component costs scale more similarly with wing area. The LCoE surface consequently becomes flatter.

7.2. Scaling trends across generator ratings

The minimum LCoE, wing-area selection, and pumping-cycle performance are compared across the five prescribed mechanical power limits for the generator. The lowest evaluated grid point can be identified at each limit. However, the widening wing-area bands indicate that wing area becomes weakly determined at the larger scales. The resulting fitted relation is therefore interpreted as a scaling trend rather than as a precise sizing law.

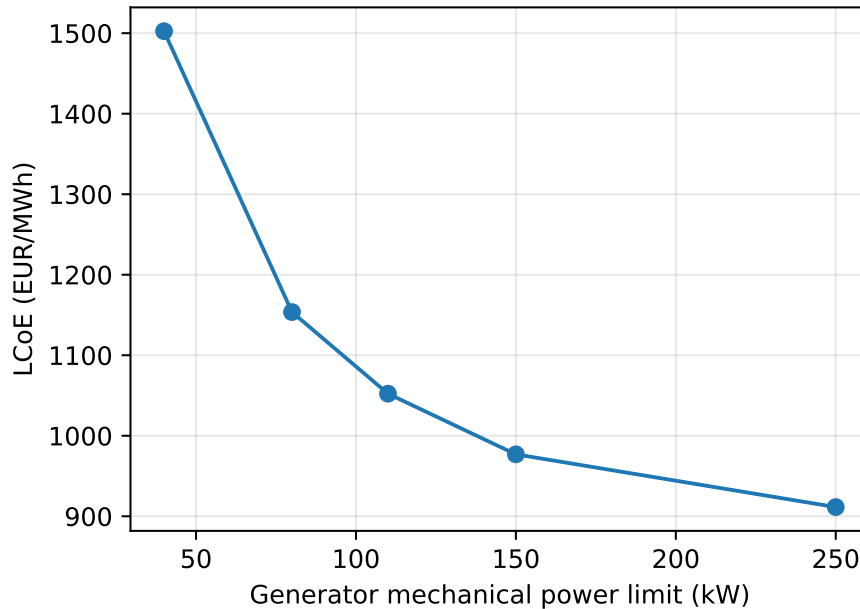


Figure 7.3: Minimum LCoE as a function of the generator's mechanical power rating. The cost decreases monotonically and in a convex manner from 1502 €/MWh to 911 €/MWh, indicating diminishing returns at the highest-rated powers.

As Figure 7.3 illustrates, the levelised cost decreases monotonically and convexly as the generator power increases. The drop is sharp for the smallest power ratings and becomes more gradual at higher ratings, indicating diminishing returns from economies of scale. This trend is consistent with the mechanism observed in the reference case. As the system grows, the fixed operating and crew expenses are spread over a greater amount of energy produced. Once this size-independent cost baseline is sufficiently diminished, the remaining expenses are mainly determined by the elements that scale with the wing. As a result, the levelised cost converges toward a lower limit. This envelope is the robust headline of this design study, as the cost differences between successive ratings are relatively large and easily outweigh any underlying variability.

The allowable tether stress optimum does not vary with scale. The design surfaces locate the optimum near 0.25 GPa at every rating, so the tether-stress trade-off between material cost and bending-fatigue life balances at the same point across the investigated range. This has been confirmed at production fidelity for all ratings. The dependence of this optimum on the assumed tether lifetime is examined in

section 7.3.

The wing-area dependence near the optimum was then characterised by a refined area sweep at this fixed stress optimum. The sweep samples the wing area more finely than the design surfaces do, thereby resolving the near-optimal region and the width of the similarity range. A corresponding refinement of the full two-dimensional grid was not performed because a fine two-dimensional sweep at every rating is computationally expensive, which is a known limitation of the study.

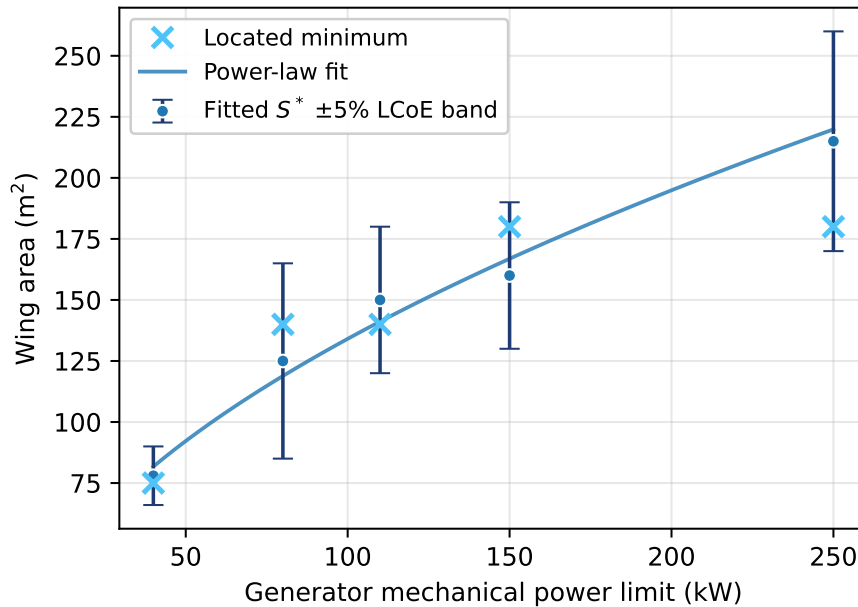


Figure 7.4: Optimised wing area against generator mechanical power limit, at an allowable tether stress of 0.25 GPa. The crosses mark the located minima, that is, the sampled designs of lowest evaluated levelised cost at each rating. These minima are weakly determined on the flat cost surface, so they scatter between neighbouring wing areas and coincide for several ratings. At each rating, the bar shows the range of wing areas within 5 % of these located minima, the round marker is the band midpoint, and the curve is a power-law fit to the midpoints. The bars widen and overlap with scale, so the wing area is not sharply determined at large ratings.

The representative wing area increases sublinearly with the generator mechanical power limit, as shown in Figure 7.4. The crosses are the located minima of the design surfaces, that is the sampled designs of lowest evaluated levelised cost, and they should not be interpreted as precisely resolved optima. The scatter between them is visible in the coincidences, where the 80 kW and 110 kW cases select the same sampled wing area, as do the 150 kW and 250 kW cases. The similarity range from the refined sweep shows the broader region within which designs remain within a comparable levelised cost, and its midpoint is used as the representative wing area. The fitted power law captures the scaling trend of these representative values, so it describes the trend rather than defining a uniquely resolved optimum or a generally applicable sizing relation.

The scatter in the identified minima originates from the inner cycle optimisation rather than from the outer grid search. The outer search evaluates a fixed set of designs and therefore introduces no numerical variation itself. The SLSQP solver used for the inner optimisation is deterministic given a fixed initial guess, but the converged solution can depend on the starting point. This sensitivity was confirmed by a direct test. When the design was held fixed and only the initial guess was varied, the results showed different converged costs. Initial guesses further from the tuned reference also caused the solver to enter its repair fallback. As neighbouring designs can differ only slightly in LCoE, this sensitivity can influence which grid point is identified as the lowest cost design. Using an initial guess tailored to each design, or obtained from a neighbouring solution, could reduce this sensitivity and is therefore identified as a direction for further work.

The similarity range widens with increasing system scale, indicating that the LCoE becomes less sensitive to wing area. At larger system scales, the relative contribution of labour costs decreases because

these costs do not scale directly with system size. At the same time, annual energy production and the dominant kite-and-tether replacement costs scale more similarly with wing area. Changes in lifetime cost and energy production therefore increasingly compensate one another. As a result, the wing area is more strongly constrained for the smallest system than for the larger systems.

The lowest evaluated LCoE occurs near an allowable tether stress of 0.25 GPa for all prescribed generator mechanical power limits. In contrast to the wing area, the preferred allowable tether stress shows no systematic variation with system scale under the reference tether lifetime relation. Its dependence on the assumed tether lifetime is examined in section 7.3.

Table 7.3: Representative optimised design at each rating, taken at the middle point of the band of the fitted matching-law wing area of Figure 7.4 with $\sigma_{t,\text{allow}} = 0.25$ GPa.

P_{gen} (kW)	40	80	110	150	250
S (m ²)	82	119	141	167	220
$\sigma_{t,\text{allow}}$ (GPa)	0.25	0.25	0.25	0.25	0.25
m_{air} (kg)	85	133	165	203	289
d_t (mm)	12.7	15.3	16.7	18.1	20.8
CF	0.51	0.48	0.45	0.44	0.41
AEP (MW h)	107	181	223	277	391
CAPEX (k€)	123	187	229	291	399
OPEX (k€)	137	183	197	220	297
LCoE (€/MWh)	1538	1229	1086	991	948

The physical quantities of the representative optimised designs are listed in Table 7.3. The derived quantities, namely the airborne mass, the tether diameter, and the annual energy production, vary monotonically with wing area, whereas the levelised cost is flat in wing area near the optimum. The located minimum area is weakly determined on this flat surface, so it coincides for two pairs, as the crosses in Figure 7.4 show. Tabulating the derived quantities at the located areas would therefore assign the same mass and tether diameter to these two ratings, and the wing area is too weakly determined to support such an equality. The derived quantities are consequently reported at the fitted trend area, which is monotonic and internally consistent across ratings. This choice changes the levelised cost only within the surface's flatness and therefore has no effect on the trends being investigated. Each column is a complete framework evaluation at the fitted wing area, so the tabulated design is computed rather than interpolated. The airborne mass and the tether diameter increase with the generator limit, because both follow the increasing wing area and maximum tether force. The capacity factor decreases across the ratings because the rated power increases as the generator limit rises, while the annual energy is bounded by the site's wind distribution. A larger system reaches its higher rated power only in stronger winds, which occur less often at the site, so the rated capacity rises faster than the captured energy.

The pumping cycle contains three distinct power quantities, as shown in Figure 7.5. The prescribed mechanical power limit for the generator constrains the reel-out and reel-in phases. The transitions between these phases are not subject to this constraint and produce a higher instantaneous peak generator shaft power. The rated electrical power remains below the prescribed mechanical limit because it represents the maximum cycle-averaged net electrical output, including the electrical energy consumed during reel-in.

The peak generator mechanical power increases from approximately 75 kW for the 40 kW limit to 369 kW for the 250 kW limit. The ratio between peak generator mechanical power and rated electrical power remains between approximately 3.1 and 3.4. The three power quantities therefore retain a similar relative separation as the system scale increases, even though the absolute difference grows substantially. Since the transition peak must be accommodated by the generator and drivetrain, sizing these components from either the rated electrical power or the prescribed mechanical limit alone would underestimate the required peak capacity. Generator and drivetrain sizing is therefore coupled to wing area through the transition peak, which increases as the larger wing produces higher tether forces. The storage is sized from the per-cycle energy swing rather than from this peak, and because

the peak-to-rated ratio stays bounded, the storage requirement grows with the wing area without becoming a limiting cost, so the optimisation towards higher annual energy incurs no disproportionate storage penalty.

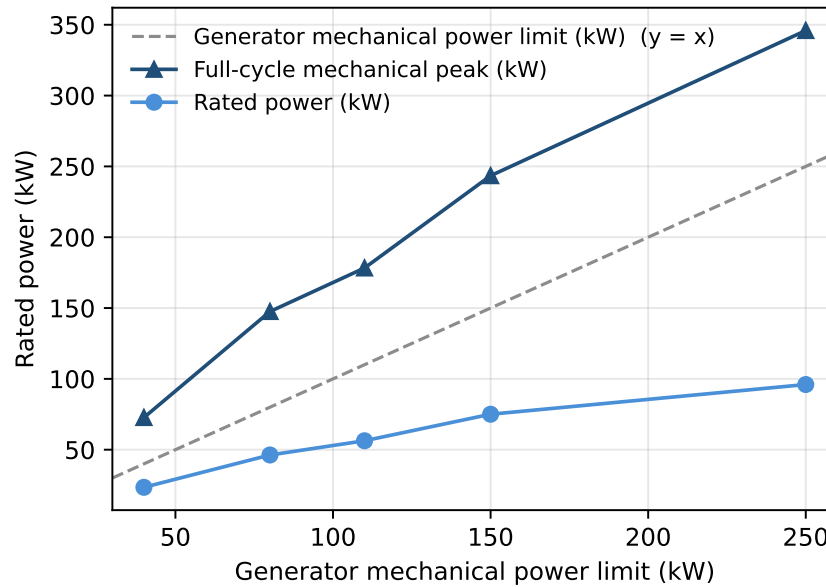


Figure 7.5: Rated electrical power, prescribed generator mechanical power limit, and peak generator mechanical power for the representative designs. The prescribed limit constrains the reel-out and reel-in phases, while the uncapped transitions produce a higher instantaneous generator shaft-power peak.

7.3. Sensitivity and robustness

The tested assumptions generally preserve the principal conclusions, except when the assumed tether life increases substantially, which shifts the preferred allowable tether stress. The analysis varies wing-mass scaling, maturity stage, and tether-life formulation individually about the reference case. The perturbations are evaluated at four generator mechanical power limits between 40 kW and 250 kW. The intermediate 110 kW limit is omitted because the adjacent limits already establish whether the scaling trend is preserved. Each variant is assessed by its effect on the LCoE, the selected design variables, the dominant cost contribution, and the direction of the scaling trend. This one-at-a-time approach tests robustness to individual assumptions but does not capture their interactions.

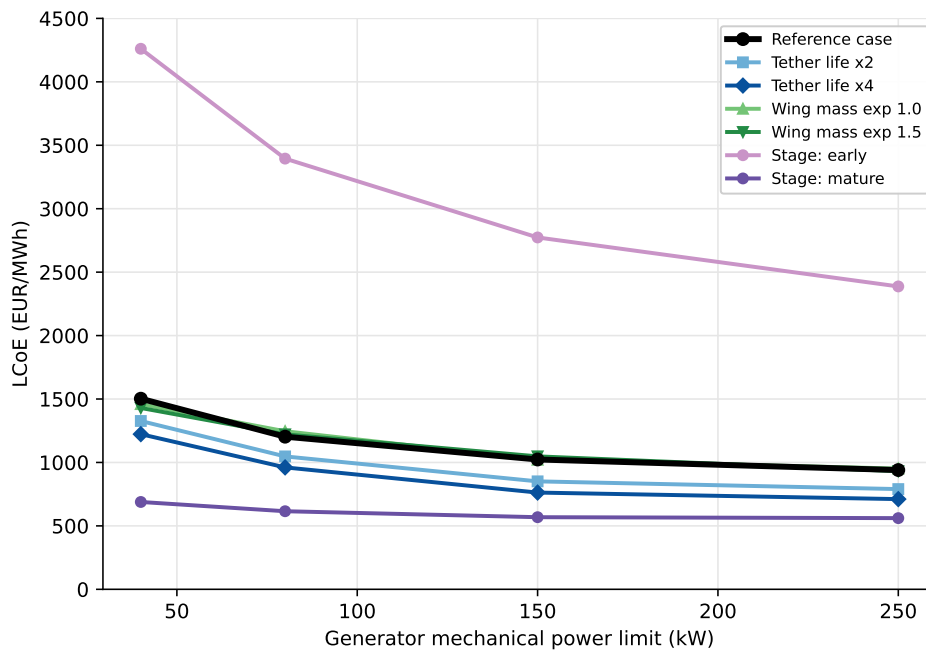


Figure 7.6: Minimum evaluated LCoE against the prescribed generator mechanical power limit, for the reference case and each sensitivity variant. Every variant preserves the decreasing LCoE trend. The maturity stage has the largest influence on the LCoE, followed by the tether-life formulation, while the wing-mass exponent has little effect.

The LCoE decreases with increasing generator mechanical power limit for every tested assumption, as shown in Figure 7.6. None of the perturbations reverses the economy-of-scale trend or eliminates its diminishing returns. The scaling conclusion remains robust within the investigated one-at-a-time perturbations. The magnitude of the sensitivity varies with scale, most notably for the maturity-stage variants. Conclusions regarding the design variables and cost composition are drawn from the optimisation results and cost breakdowns, as the overlay does not display these aspects, and are summarised in Table 7.4

Table 7.4: Effect of the investigated assumptions on the selected design and the dominant recurring cost. Both are taken from the optimisation results and the cost breakdowns. The LCoE level itself appears in Figure 7.6.

Assumption	Selected design	Dominant recurring cost
Wing-mass scaling	Reference selection retained	Reference selection retained
Maturity stage	Reference selection retained	Canopy (early), tether (mature)
Tether-life formulation	Higher allowable stress at the largest increase	Shifts to kite structure

Varying the wing-mass exponent does not materially affect the LCoE. Both variants remain close to the reference curve across the investigated limits. The optimisation results retain the wing-area band and the allowable tether stress of the reference case. The small contribution of wing mass to total expenditure, compared to ground-crew labour and kite and tether replacement, explains this insensitivity. Uncertainty in wing-mass scaling does not affect the identified design or the scaling trend.

The maturity stage also changes the dominant recurring cost. Canopy replacement dominates under the early-stage assumptions. In the 150 kW mature-stage breakdown, the ground-crew and canopy contributions decrease to $19 \text{ € MW}^{-1} \text{ h}$ and $41 \text{ € MW}^{-1} \text{ h}$, respectively. Tether replacement consequently becomes the dominant recurring cost in the LCoE. These changes in cost magnitude and composition do not materially affect the wing-area trend or the preferred allowable tether stress. The maturity assumption therefore primarily affects expected economic performance and the priority given to cost reduction, rather than design trends.

The tether-life formulation most strongly affects the selected allowable tether stress. Extending the

assumed tether life reduces LCoE by lowering replacement expenditure. A moderate increase retains the reference allowable-stress selection at all limits. The largest increase shifts the lowest evaluated LCoE towards a higher allowable stress at the smaller and intermediate limits. As tether replacement becomes less significant, the penalty for shorter service life diminishes, and the material and drag benefits of a thinner tether become more important.

The preferred allowable tether stress is therefore robust to a moderate change in tether life, but not to an increase of several times the bending-fatigue estimate. The tether-life formulation is consequently the most important investigated uncertainty for the selected tether design. Experimental calibration of tether service life would provide the greatest improvement in confidence in this design result. The maturity stage, by contrast, has the largest influence on the absolute LCoE. The two uncertainties, therefore, lead to different conclusions. Across the tested assumptions, the principal conclusions are supported by the observed trends and their underlying mechanisms, while the absolute LCoE remains indicative.

7.4. Design drivers and study comparison

The economic mechanisms identified in this study can be compared with the fixed-wing optimisation performed by Joshi et al. [12]. In the study by Joshi et al., the operating expenditure is undiscounted, and it exceeds the initial capital expenditure, this is similar to the soft-wing case. However, the optimal rating is strongly influenced by the mass penalty imposed by the rigid-wing sizing relation. Aircraft mass increases superlinearly with wing area, which raises structural cost and reduces energy performance by increasing the cut-in and rated wind speeds and lowering the capacity factor. At larger ratings, this penalty outweighs the benefit of further scaling. Storage is also a major cost contributor, while the generator's rated power is optimised through the power crest factor by balancing power-capping losses against drivetrain costs. The resulting optimum is therefore mainly shaped by aircraft mass, storage, and drivetrain sizing.

The present TU Delft V3-derived model shows a different balance. Airborne mass also increases superlinearly with wing area, but changes in the assumed wing mass scaling relation have only a limited effect on the selected designs and their LCoE. Instead, the economic result is mainly governed by recurring expenditure from canopy and tether replacement and labour requirements.

The treatment of wing service life is an important difference between the two models. Joshi et al. do not include a recurring rigid wing replacement term. The kite is included as a capital cost, while maintenance and replacement are not modelled because the required information was unavailable. In the present study, finite canopy life is explicitly linked to reel-out operating hours, which introduces canopy replacement as a recurring cost. The comparison therefore shows how the representation of component service life can affect the economic result.

Tether stress affects the same factors in both models: tether dimensions, service life, and replacement expenditure. In the fixed-wing assessment, these effects compete with more dominant other effects. Namely, aircraft mass, storage, and drivetrain sizing. In the model for soft-wing systems, tether replacement is one of the main recurring cost mechanisms and directly contributes to the trade-off between tether mass and service life.

Labour expenditure introduces another difference. The present model includes a recurring crew requirement that is approximately independent of system size. As the system becomes larger, this cost is distributed over a greater annual energy production, contributing to the reduction in LCoE. An equivalent recurring crew term is not included in Joshi et al., so part of the difference in scaling behaviour arises from different system boundaries.

Within the assumptions of both studies, the main contrast is that the optimum found by Joshi et al. is strongly influenced by rigid wing mass, storage, and drivetrain sizing, whereas the present V3-derived model is more strongly governed by canopy and tether replacement and labour expenditure. These differences depend on the assumed component lives, cost maturity, operational strategy, site conditions, and system boundaries and should therefore not be directly generalised to other systems.

Conclusion

This thesis investigated how scaling affects the performance and LCoE of soft-wing ground-generation airborne wind energy systems. To answer this question, a coupled techno-economic framework was developed that propagates design changes through system sizing and operation to annual energy production, component replacement, and lifetime cost. This allows assessment of the performance benefits of scaling alongside the associated mass and replacement penalties. The framework was applied to designs derived from the TU Delft V3 LEI kite under the wind conditions of the Bangor Erris reference site. Across prescribed generator mechanical power limits from 40 kW to 250 kW, annual energy production increased, and LCoE decreased at a diminishing rate.

Airborne mass increased more than proportionally with wing area over the investigated range. This finding followed from a component-based model in which the LEI wing and KCU were scaled separately, rather than applying a single scaling relation to the complete airborne system. The nominal LEI wing model scaled with $S^{1.22}$, while the KCU combined fixed, linear, and superlinear mass contributions. The resulting airborne mass trend is therefore component-dependent and cannot be described by one universal power law. Sensitivity cases ranging from linear to $S^{1.5}$ wing-mass scaling produced only minor changes in LCoE and did not alter the main scaling trend. Within the present framework, airborne mass affects system sizing and performance, but uncertainty in wing-mass scaling is not a leading driver of LCoE.

Component service life affected LCoE by determining how often the canopy and tether had to be replaced during the project lifetime. For the fixed TU Delft V3 reference system under the mid-maturity assumptions, the indicative calculated LCoE was approximately 2992 €/MWh. Recurring operating expenditure contributed more to this value than annualised capital expenditure. Ground crew labour formed the largest individual contribution, followed by expenditure associated with canopy and tether replacement. By contrast, the ground station dominated the initial investment but contributed comparatively little to LCoE. These results show that initial investment alone does not adequately represent the economic consequences of finite soft-wing component life.

The maturity scenarios further demonstrated the influence of operational labour and component durability. The reduction in LCoE from the Early to the Mature scenario resulted from combining fewer ground crew hours with longer canopy and tether lives. Because these assumptions changed simultaneously, this comparison alone cannot rank their individual effects. The dominant replacement contribution also changed between scenarios. Canopy replacement mattered more under short-life assumptions, while tether replacement became relatively more important as canopy life increased. The maturity comparison therefore indicates that operational autonomy and component durability are relevant development priorities, but does not establish a universal ranking between them.

Allowable tether stress introduced another trade-off between performance and service life. Increasing the allowable stress produced a thinner tether with lower aerodynamic drag and material cost, but shortened its calculated replacement interval. The combined effects on performance, initial cost, and replacement expenditure consequently determined the preferred allowable stress. This shows why component lifetime must be evaluated alongside technical performance when comparing conceptual system designs.

Across system scales, annual energy production increased while capacity factor decreased because rated power increased faster than energy production under the Bangor Erris wind conditions. The reduction in LCoE therefore came mainly from spreading recurring costs over higher energy production, rather than from improved capacity utilisation. Ground crew expenditure was assumed to remain approximately independent of system size, so its contribution per unit of energy declined as scale increased. Canopy and tether replacement expenditure increased with wing area and progressively

offset this benefit. The balance between fixed recurring expenditure and size-dependent replacement costs explains why LCoE continued to decrease, but at a diminishing rate.

The wing areas associated with the lowest evaluated LCoE increased sublinearly with the prescribed generator limit. At larger limits, LCoE also became less sensitive to the exact wing area. These systems were therefore characterised by broad regions with similar LCoE values rather than sharply resolved economic optima. The exact selected wing area is consequently less informative than the underlying balance between energy production, recurring labour, and component replacement.

The resulting scaling behaviour differs from the fixed-wing assessment used for comparison, in which LCoE initially decreases and then increases with system size. In that formulation, the selected system scale is strongly influenced by the superlinear aircraft mass penalty together with storage and drive-train sizing. In the present formulation, recurring expenditure for ground crew, canopy replacement, and tether replacement governed the change in LCoE more strongly than the investigated wing-mass relations. The comparison does not establish an inherent difference between fixed-wing and soft-wing AWE systems. The two assessments use different component-life models, labour assumptions, power-rating formulations, and system boundaries. Instead, it shows that an intermediate-scale optimum depends on which physical and economic penalties are represented and how they change with system scale.

The reduction in LCoE with scale and the sublinear increase in wing area were maintained across the individual sensitivity cases, while the preferred allowable tether stress remained within a similar range in most cases. The exact design selections were less robust because the LCoE surfaces became relatively flat at larger scales and were affected by numerical variability in the operational optimisation. The sensitivity analysis varied uncertain inputs individually and therefore did not capture interactions between them. Several mass, lifetime, and cost relations also rely on limited data or extrapolation. The numerical LCoE values and exact design selections should therefore be interpreted as indicative. The relative trends and the mechanisms causing them provide the stronger basis for interpretation.

Overall, the results show that the economic consequences of scaling cannot be inferred from airborne mass or performance alone. Within this framework, the scale benefit is driven primarily by spreading recurring labour costs over greater energy production, while increasing component replacement expenditure progressively limits this benefit. The limited sensitivity to wing-mass scaling suggests that mass reduction alone is unlikely to offer the main opportunity to lower LCoE under the present assumptions. Instead, operational autonomy and improved component durability emerge as promising directions for addressing the dominant recurring expenditures, although their individual effects should be assessed separately before establishing their relative importance. By explicitly representing finite component life and recurring replacement, the framework extends LCoE-driven conceptual assessment to soft-wing systems. Also, it supports comparing design alternatives before detailed component designs and validated commercial cost data are available.

Recommendations

The developed framework provides a basis for coupled techno-economic assessment during conceptual design. However, its numerical implementation and underlying models contain assumptions and simplifications that limit the precision and applicability of the results. Recommendations for further developments are therefore organised into five themes.

The flat LCoE surfaces and the recurring wing areas show that the numerical reliability of the individual design selection should be established before the located designs are treated as resolved. The low-cost regions should be evaluated with progressively finer grids, while selected designs should be repeated using starting points adapted to the candidate geometry. The individually windowed area sweeps can first determine whether the recurring minima follow the sampling grid, after which multiple starting points can isolate the sensitivity of the inner operational optimisation. This would separate the numerical selection of individual designs from the underlying scaling trend.

The study considers only the wing area and the allowable tether stress, while the remaining design choices are fixed or represented by simplified relations. The coupled design space should be expanded, together with the models required to represent the resulting design changes. A rating formulation based on a variable power cap and component models that relate structural mass and service life to the governing loads provides the most direct extensions. Storage capacity could subsequently be included once its influence on system operation is represented. The computational cost increases with a more extensive design space, therefore, additional variables should be introduced selectively. This would reveal further coupled trade-offs and determine whether the identified scaling trends are retained under a larger design freedom.

The aerodynamic and performance parameters are held at the reference V3 values across all scales, so the larger systems represent a scaled reference design rather than one optimised for their rating. They may therefore underperform relative to a design tailored to each scale. More advanced models should represent how the aero-structural behaviour changes with size, and the governing performance parameters should be treated as design quantities rather than fixed inputs. This would determine whether the identified scaling trends hold under a design optimised at each scale, rather than one scaled from the reference.

The framework currently represents an individual V3-derived LEI system, and it therefore cannot assess other soft-wing concepts or complete deployments. Architecture-specific models should be added to represent how other concepts are sized and how their characteristics affect performance and economics. The framework should also be extended to the farm level by linking site use, tether length, and multiple-system operation to the economic assessment. This would allow alternative system concepts and deployment strategies to be compared and determine whether the economies of scale found for an individual system remain at the deployment level.

The complete framework has not been validated against measured lifetime expenditure, and the uncertain assumptions were varied only individually. Available demonstrator and component measurements should first be used to validate the technical relations, followed by sustained operational data for the lifetime economics. A combined uncertainty assessment should also determine whether interactions between assumptions alter the principal conclusions. This would establish the conditions under which the framework provides reliable conceptual comparisons.

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